INTERNAL USE — CONTROLLED TECHNICAL REFERENCE

Gas Plant Controls & Automation Reference Manual

Revision 1.2

Allen-Bradley ControlLogix 1756 • 1794 FLEX I/O • AVEVA (Wonderware) InTouch HMI

Instrumentation • PLC Hardware and Programming • HMI • Troubleshooting • Process Control • Safety Fundamentals

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This manual provides a technical reference for gas plant automation systems, instrumentation, PLC hardware, HMI applications, troubleshooting methods, process control, and safety fundamentals.

This manual is a controlled, living document. It is reviewed and updated as plant procedures evolve. Revision 1.2 converts the earlier manual to reference-manual format, corrects technical errors identified in peer review, and adds standard ladder logic patterns (see Appendix C). Do not distribute outside the facility without authorization.

[COMPANY: Insert company name, logo, and document control number here]

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About This Manual

This manual is organized from plant-process fundamentals through PLC hardware and programming, HMI configuration, wiring standards, troubleshooting, and safety. Controlled plant documents govern wherever they differ from this manual (Section 0.1).

Navigation Guide

- **Process orientation:** Section 1 (Process Flow) and Section 2 (Instrumentation) provide the process and instrument context for every control loop.
- **Field quick reference:** Appendix E (Field Pocket Card). Desk reference: Section 11 (Quick Reference Tables).
- **Specific tasks:** Use the Table of Contents to locate the relevant subsection. Cross-references are noted inline.
- **Ladder logic:** Section 5. Section 5.4 contains the standard ladder patterns used throughout this manual.
- **Qualification and training:** Review cards, lab review questions and answer keys, the 8-week qualification plan, the competency matrix, and assessment notes are in Appendix F.
- **Troubleshooting:** Section 8.
- **Safety:** Section 0 and Section 9. Where Section 0 and a plant procedure disagree, the plant procedure governs.

Notation Conventions

Convention	Meaning
Bold text	Key term at first use, or a field label.
`Monospace text`	PLC tag name, instruction mnemonic, Neutral Text rung, or exact software text.
WARNING box	Hazard that can cause injury, death, environmental release, or loss of a protection layer.
CAUTION box	Condition that can cause equipment damage, a process upset, an unplanned trip, or use of incorrect plant data.
NOTE box	Supplementary information. Items marked "Verify on bench before field use" have not been confirmed on the plant bench controller or software version.
[COMPANY: ...]	Placeholder for plant-specific information (listed in Appendix D).
[DRAWING: ...]	Placeholder for a referenced site drawing.
Rung blocks	Ladder rungs are shown as Studio 5000 Neutral Text using v36 mnemonics, one rung per line, each preceded by a rung label. See Section 5.1.
Example values	Ranges, setpoints, IP addresses, and tags in examples are illustrative ("example only"). Use the rationalized plant values.

CAUTION — Document Ownership

Initial and date each [COMPANY: ...] placeholder entry when it is filled. Log official revisions in Appendix C (Revision Log). The accuracy of this manual depends on the accuracy of its plant-specific data.

NOTES

Section 0: Working Rules

These rules apply before work on a live loop, a project download, or an SIS cabinet. They keep BPCS examples separate from ESD/SIS terminology and standardize tag punctuation. Where a later example conflicts with these rules, these rules govern.

0.1 Plant Documents Govern

This manual is a technical reference. It is not the I/O schedule, the cause-and-effect matrix, the official tag standard, or a substitute for a P&ID or loop drawing. If a number, setpoint, IP address, wire color, or bypass rule in this manual differs from a controlled plant document, use the plant document and mark the manual placeholder for correction.

CAUTION — Example Data Until Placeholders Are Filled

Appendix D lists every [COMPANY:] and [DRAWING:] gap. Until a qualified engineer fills those items and signs Appendix C, example ranges, setpoints, and I/O slots are not live plant data. Tags such as SEP_PT_101 are example tags.

0.2 Authorization Boundaries

Technician authorization has a defined envelope. Work outside it requires a named supervisor present and a valid permit or MOC. The qualification stages referenced below are defined in the competency matrix (Appendix F.4).

Activity	Observe	Supervised practice	Independent (after sign-off)	Never without process-safety / engineering approval
Loop check on BPCS instrument	Permitted	Permitted	After Lab 1 sign-off	SIS transmitters that are SIF sensors
FLEX module hot-swap (non-SIS)	Permitted	Permitted	After Lab 2 sign-off	SIS / F&G I/O racks
BPCS ladder change / download	Permitted	Training controller only	After qualification sign-off, with MOC	Production controller without MOC
PID tuning	Read-only on live loops	Simulator / training controller	After qualification sign-off, with MOC	Any live loop without MOC
HMI faceplate / alarm priority change	Permitted	Development node	Not on the production ViewApp without MOC	ISA-18.2 priority or setpoint change without rationalization
Force, inhibit, or shelve	Permitted	Training controller	Never leave a force on a production tag	SIS override, ESD force, PSV isolation
SIS proof test / SIF bypass	Observe only	No	No	All SIS maintenance overrides
Fiscal / custody-transfer configuration	Observe metering technician	No	No	Any AGA factor, GC, or meter factor change

0.3 Tag Naming Convention

Drawings use ISA hyphens (SEP-PT-101 on a P&ID) and the PLC uses underscores (SEP_PT_101). Both forms identify the same instrument. Use the mapping below unless the plant tag standard in Section 3.3 states otherwise.

Where it appears	Form	Example	Rule
P&ID, loop drawing, work order	ISA-5.1 hyphens	SEP-PT-101	Copy the drawing exactly on forms and permits.
ControlLogix tag / alias	Underscores, no hyphens	SEP_PT_101	Logix tag names cannot contain hyphens.
UDT member	Tag.Member	SEP_PT_101.Value	One UDT instance per device or loop. Use the member the plant tag already has; never create a parallel tag for the same signal.
Raw I/O alias	…_Raw or I/O address alias	SEP_PT_101_Raw	Points at the FLEX/1756 channel; the scaled PV is SEP_PT_101.Value.
HMI / historian	Match PLC external name	SEP_PT_101	Do not invent a third spelling. Bind to the PLC tag.
Health bits	_OK suffix, 1 = healthy	STOP_OK, SEP_PAHH_OK	Named for the state that allows the process to run.
SIS / C&E documents	Plant SIS standard	Per C&E matrix	Never reuse a BPCS example tag as if it were a SIF tag. SIS status read by the BPCS is read-only (e.g., SIS_ESD_OK).

Discrete fail-safe convention

This convention applies to every lab and example in this manual unless a plant drawing shows the opposite and a supervisor has confirmed it.

Item	Field wiring	DI when healthy / not pressed	Ladder instruction on the run/health path	What de-energizes the output
Stop, overload, trip input	NC contact (opens on demand or on cable break)	1 (24 VDC present)	XIC of a health bit (e.g., `XIC(STOP_OK)`)	Bit goes 0; XIC opens; OTE drops
E-stop / MCR	NC contacts in the hardwired coil circuit or safety relay; auxiliary NC contact to a DI	E_STOP_OK = 1 when released and intact	`XIC(E_STOP_OK)` in the run path (monitoring only)	The hardwired circuit removes coil power directly; the PLC command also drops
Start pushbutton	NO momentary	0 at rest; 1 while pressed	`XIC(START_PB)` in parallel with the seal-in	Releasing START_PB after the seal-in is established does not drop the output
Run feedback / limit switch	As drawn (usually NO)	1 = asserted condition	XIC of the asserted bit	Device leaves the asserted state

A health bit is named for the state that is TRUE when the process is allowed to run (STOP_OK, OL_OK, E_STOP_OK, SEP_PAHH_OK). A health bit is never named STOP_PB. Every example and lab in this manual follows this convention. XIO of an _OK bit appears only in trip-detect and first-out rungs.

> **NOTE — Pair the rung with the loop drawing**
> Before writing or testing a discrete bit, read the loop drawing: contact type (NO/NC), powered state, and whether the card is sinking or sourcing. The instruction matches the bit meaning, not the marking on the pushbutton.

0.4 BPCS vs SIS Terminology

The Basic Process Control System (BPCS) is the ControlLogix / FLEX / InTouch system that runs the plant day to day. The Safety Instrumented System (SIS) is a separate, SIL-rated system with dedicated sensors, logic solver, and final elements. Snippet 6 and Lab 8 in this manual are BPCS trip-and-permissive examples. They are not SIS logic and they are not an ESD SIF.

Layer	What it is	Technician scope before qualification sign-off	Terms to use	Terms not to use
BPCS	Process control, permissives, process alarms, equipment protection that is not SIL-rated	Training controller practice; observe live systems; no production download	Trip, permissive, interlock, process alarm	ESD SIF, SIL shutdown, proof-test bypass
Alarm system	ISA-18.2 operator alarms on the HMI	Rationalize on paper; no live setpoint edits	Alarm, shelve, suppress, priority	Trip, SIF, ESD
SIS / SIF	Independent safety function per IEC 61511	Observe a proof test; never operate overrides	SIF, SIS, proof test, MOC, bypass permit	Anything that implies the BPCS is the safety layer
PSV / mechanical	Last-line mechanical relief	Never gag or isolate without an authorized isolation procedure	PSV, MAWP, flare	Soft interlock
F&G	Fire and gas detection / voted output	Observe; do not inhibit detectors	F&G zone, voted alarm	BPCS voting substitutes

> **WARNING — Do Not Copy BPCS Trip Logic into a Production ESD**
> Never paste Snippet 6, Lab 8, or any BPCS bypass pattern into a live ESD or SIS routine. Unauthorized bypass of a safety function violates OSHA PSM requirements and can be fatal. SIS changes are performed only under the plant functional-safety procedure.

0.5 Work Control Flow

Use this sequence on every job that touches live equipment or configuration. Skip a box only when the work order explicitly says the step does not apply.

Step	Action	Who	Done when
1	Read the work order, P&ID, and loop drawing. Define the exact tag and the as-left condition.	Technician	Scope is one sentence and one tag list
2	Decide if MOC is required (configuration, setpoint, logic, SIS, bypass, new device).	Technician and Supervisor	MOC number or written "MOC not required"
3	Permits: LOTO, hot work, line break, confined space, bypass, as applicable.	Authorized issuer	Permits in hand
4	Notify the board operator. Agree alarm suppression and expected bad-quality time.	Technician and Operations Personnel	Operator repeats the tag back
5	Do the work. No extra forces, no extra inhibits, no unscoped changes.	Technician	As-found recorded
6	Restore: wiring, covers, forces cleared, inhibits cleared, shelves expired or removed.	Technician	As-left recorded
7	Operator confirms live values and alarm state. Close permits. File the form.	Technician and Operations Personnel	Permit closed, form signed

0.6 Forces, Inhibits, and Leftover Software Traps

Leftover forces and inhibited I/O connections are a standing hazard. Complete this checklist at the start and end of every job. Lab 11 practices it on the training controller.

Check	Where to look	Pass condition
I/O forces	Studio 5000 Force table / controller status	Zero forces. If a force must remain, it is on the permit and the HMI shows it.
Program forces / toggles	Search for Force / EnableForce tags	No leftover test bits TRUE
Inhibited connections	I/O tree — inhibited modules	No unexpected inhibits
MSG / comms inhibit	Message configuration / program bits	Production reads healthy
HMI alarm shelve	Alarm summary shelf list	No critical shelf past expiry
BPCS bypass timer	SEP_Byp.TON / bypass permit	Timer expired or permit still open and visible to operator
Downloads	Offline vs online compare	Official project path updated; no laptop-only copy

0.7 Backup, Download, and OT Hygiene

Most serious control-system incidents are not 4–20 mA math errors. They are the wrong .ACD file downloaded to the wrong controller, a USB stick on a process machine, or a force left in place.

Rule	Practice
Official copy	The controlled Studio 5000 / InTouch project lives in the plant repository — not only on a laptop Desktop.
Compare before download	Online/offline compare. If unexpected diffs exist, stop and call the owner of the system.
Download window	Production downloads require a planned window, operator notification, and a named approver.
Training vs production	Training controller IPs and project names must be visually distinct. Never browse production from a lab project.
Removable media	No personal USB on engineering workstations or HMI nodes. Use the approved transfer method only.
Accounts	Use an individual login. Do not share HMI operator or engineering passwords. Lock the station when unattended.
After a controller replace	Load the official project, confirm firmware, confirm redundancy SYNCHRONIZED, confirm I/O tree green, then loop-check critical tags.

0.8 Software Stack at Time of Writing

Fill this table from the plant engineering workstation. Menus and dialogs described in this manual are written for the versions listed and may differ in other major versions.

Component	Plant standard (fill in)	Examples in this manual assume
ControlLogix CPU	[COMPANY: catalog + firmware]	1756-L8x (ControlLogix 5580) family in production; bench exercises use a 1769-L33ER (CompactLogix 5370) at Studio 5000 Logix Designer v36
Comm module	[COMPANY: EN2T / EN3TR / DLR]	EtherNet/IP, 1756-EN2T
Remote I/O	[COMPANY: 1794 and/or 5094]	1794-AENTR + FLEX I/O
HMI	[COMPANY: InTouch Classic / System Platform / other]	AVEVA InTouch HMI 2020 R2 (WindowMaker / WindowViewer)
Linx / OPC	[COMPANY: RSLinx Classic / FT Linx / KEPServer]	CIP path + OPC DA/UA
Historian	[COMPANY: product]	Plant historian for SOE / Lab 12
SIS platform	[COMPANY: product — view only, do not edit]	Independent of BPCS examples

0.9 Illustrative Calculations

The dew-point Structured Text in Section 5.3 and the simplified orifice formula in Appendix B illustrate syntax and structure only. They are not plant correlations and are not used for hydrate decisions, custody tickets, or MOC justifications. Use the plant-approved calculation, flow computer, or vendor method.

NOTES

Section 1: Gas Plant Process Flow Overview

1.1 Natural Gas Processing Plant Overview

A **natural gas processing plant** receives raw wellhead gas — a mixture of methane (CH_4), heavier hydrocarbons (ethane, propane, butane, pentane+), water vapor, hydrogen sulfide (H_2S), carbon dioxide (CO_2), and trace contaminants — and transforms it into pipeline-quality "sales gas" plus valuable liquid products. The processing train follows a predictable sequence of unit operations, each with its own instrumentation, control loops, and safety systems.

This sequence is fundamental to automation maintenance and troubleshooting. Every control loop, HMI screen, alarm, and interlock exists to serve a specific step in this chain. The six primary stages are summarized below.

Stage 1 — Inlet Separation

Raw wellhead gas enters the plant through inlet metering skids and flows into **inlet separators** (also called slug catchers on wet-gas systems). These vessels use gravity, baffles, and sometimes coalescing elements to separate three phases: gas (vapor), condensate (liquid hydrocarbons), and produced water. Key controlled variables are separator pressure and liquid levels. A pressure control valve (PCV) regulates gas outlet pressure; level control valves (LCV) drain liquids to downstream treating.

Stage 2 — Compression

After separation, the gas may need **compression** to raise it to pipeline or treating pressures. Reciprocating and centrifugal compressors are common. Critical instrumented variables include suction/discharge pressures and temperatures, vibration (on rotating equipment), lube-oil pressure, and surge control on centrifugal compressors. Anti-surge control loops and high-high pressure shutdowns (PAHH) are essential safety functions here.

Stage 3 — Dehydration (Glycol)

Glycol dehydration removes water vapor to prevent hydrate formation and meet pipeline water dew-point specifications (typically <7 lb H_2O/MMSCFD or <4°C dew point). Tri-ethylene glycol (TEG) absorbs water from the gas in a contactor (absorber column) and is regenerated in a reboiler. Controlled variables include contactor pressure, reboiler temperature (typically 195–205°C), and glycol circulation rate (flow control).

Stage 4 — Amine Treating (H_2S / CO_2 Removal)

Amine treating (also called sweetening) removes acid gases — primarily H_2S and CO_2 — using amine solvents (MEA, DEA, MDEA, or blends). Gas contacts lean amine in an absorber; the rich amine is regenerated in a stripper (regenerator) column. Pipeline specifications limit H_2S to ≤4 ppm and CO_2 to ≤2%. Analyzer transmitters (AT) continuously monitor treated gas H_2S content. Critical controlled variables: absorber pressure, regenerator temperature (typically 115–125°C), lean amine flow rate, and reboiler duty.

Stage 5 — NGL Fractionation

Natural Gas Liquids (NGL) fractionation separates the heavier hydrocarbon liquids (C2+) into marketable products: ethane, propane, butane, and natural gasoline (pentane+). This is accomplished using a series of distillation columns (deethanizer, depropanizer, debutanizer). Each column has precise temperature and pressure control profiles, with multiple tray or packed sections. Product purity is monitored by analyzer transmitters (chromatographs or on-line BTU analyzers).

Stage 6 — Metering / Custody Transfer

Custody transfer metering measures the final sales-gas volume and energy content for commercial billing. Turbine meters, ultrasonic meters, or orifice meters with AGA-7 or AGA-3 calculations are typical. These meters require high accuracy (±0.25% or better), temperature/pressure compensation, and chromatograph integration for BTU content. Regulatory compliance (FERC, RRC Texas) governs documentation requirements.

> [COMPANY: Insert site-specific P&ID reference numbers for each stage here — e.g., P&ID-001 Inlet Separation, P&ID-010 Compression, P&ID-020 Dehydration, etc.]

1.2 Key Process Streams Table

Stream Name	Source	Destination	Key Parameter	Typical Range
Wellhead Inlet Gas	Field gathering header	Inlet Separator	Pressure	200–1,200 PSIG
Inlet Separator Gas Outlet	Inlet Separator	Compression suction	Pressure, H_2O content	100–900 PSIG
Inlet Separator Condensate	Inlet Separator (liquid boot)	Condensate stabilizer	Level, flow	0–100% level
Inlet Separator Produced Water	Inlet Separator (water boot)	Water treatment / disposal	Level, flow	0–100% level

Stream Name	Source	Destination	Key Parameter	Typical Range
Compressor Discharge Gas	Compressor discharge KO drum	Glycol contactor	Pressure, Temperature	900–1,440 PSIG, 100–160°F
Dehydrated Sweet Gas	Glycol contactor outlet	Amine absorber or sales	Water dew point	<4°C dew point
Rich Amine	Amine absorber sump	Amine regenerator	H_2S loading, flow	0.4–0.5 mol H_2S/mol amine
Lean Amine	Amine regenerator reboiler	Amine absorber top	Temperature, flow	120–135°F, 10–50 GPM
Acid Gas (Tail Gas)	Amine regenerator overhead	Sulfur recovery / incinerator	H_2S concentration	30–80 mol% H_2S
Sweet Pipeline Gas	Amine absorber outlet	NGL fractionation / sales	H_2S, CO_2, BTU	<4 ppm H_2S, 950–1,100 BTU/SCF
NGL Mixed Stream	Deethanizer bottoms	Depropanizer	C2/C3 ratio, pressure	150–300 PSIG
Propane Product	Depropanizer overhead	Storage / pipeline	Purity, pressure	>95 vol% C3, 180–220 PSIG
Natural Gasoline	Debutanizer bottoms	Storage tanks	RVP, flow	10–14 psi RVP
Sales Gas (Custody Transfer)	Custody transfer meter run	Downstream pipeline operator	Volume, BTU content	Per contract spec

1.3 Critical Control Loops at Each Stage

Loop Tag	Stage	Process Variable	Setpoint Range	Control Action
SEP-PIC-101	Inlet Separation	Separator operating pressure	150–250 PSIG (example; transmitter 0–300 PSIG, MAWP 285 PSIG)	PCV throttles gas outlet
SEP-LIC-101	Inlet Separation	Separator oil/gas interface level	40–60% of range	LCV drains condensate
SEP-LIC-102	Inlet Separation	Separator water level	30–50% of range	LCV drains produced water
COMP-PIC-201	Compression	Suction header pressure	Varies by stage	Speed control / recycle valve
COMP-TIC-201	Compression	Interstage cooler outlet temp	100–120°F	Air fin cooler fan pitch or bypass
COMP-FIC-201	Compression	Anti-surge recycle flow	Anti-surge curve	Anti-surge control valve (ACV)
DEHYD-PIC-301	Dehydration	Glycol contactor pressure	900–1,440 PSIG	PCV back-pressure valve
DEHYD-TIC-301	Dehydration	TEG reboiler temperature	195–205°C	Fuel gas control valve to reboiler
DEHYD-FIC-301	Dehydration	Lean TEG circulation rate	2–5 GPM/MMSCFD	Glycol pump stroke / VFD speed
AMINE-PIC-401	Amine Treating	Absorber operating pressure	800–1,200 PSIG	PCV outlet control valve
AMINE-TIC-401	Amine Treating	Regenerator reboiler temperature	115–125°C	Steam or fuel gas control valve
AMINE-LIC-401	Amine Treating	Absorber sump level	45–55%	Rich amine pump discharge LCV
FRAC-TIC-501	NGL Fractionation	Deethanizer overhead temp	−30 to −10°F	Reflux flow control
FRAC-PIC-501	NGL Fractionation	Deethanizer operating pressure	200–400 PSIG	Condenser duty or overhead PCV
METER-FIC-601	Custody Transfer	Gas sales flow	Per contract	Monitor only — check meter

1.4 Safety Interlock Overview

Every stage of the gas plant is protected by a layered safety architecture. Technicians must understand three fundamental safety device categories before working on any loop.

Emergency Shutdown (ESD)

An **Emergency Shutdown (ESD)** system is a dedicated logic solver (or SIS controller — see Section 9) that monitors critical process variables (pressure, level, temperature, fire/gas) and, upon detecting an unsafe condition, drives the process to a safe state by closing isolation valves, tripping compressors, and opening blowdown valves. ESD systems are independent of the Basic Process Control System (BPCS/PLC). ESD logic is typically implemented as de-energize-to-trip (fail-safe) — the valve is held open by power; loss of signal closes it.

Pressure Safety Valves (PSV)

A Pressure Safety Valve (PSV) is a purely mechanical last line of defense. It is set to open at or below the vessel Maximum Allowable Working Pressure (MAWP) and relieves excess pressure directly to the flare system without any electronic signal. Alarm and trip setpoints are ordered PAH < PAHH < PSV set pressure ≤ MAWP. PSVs require periodic testing and must never be blocked in without proper isolation permits. A PSV is never defeated without authorization.

PAHH / PALL / LAHH / LALL Descriptions

Acronym	Full Name	Trigger Condition	Typical Action
PAHH	Pressure Alarm High-High	Process pressure exceeds highest safe limit	BPCS or SIS trip (per C&E matrix) — shutdown or blowdown
PAH	Pressure Alarm High	Process pressure exceeds normal operating limit	Operator alert; manual intervention
PAL	Pressure Alarm Low	Process pressure drops below normal operating limit	Operator alert; check for leaks
PALL	Pressure Alarm Low-Low	Process pressure at critically low level	BPCS or SIS trip (per C&E matrix) — compressor suction protection
LAHH	Level Alarm High-High	Vessel level at maximum safe limit	BPCS or SIS trip (per C&E matrix) — close inlet, open drain
LALL	Level Alarm Low-Low	Vessel level dangerously low (pump runout)	BPCS or SIS trip (per C&E matrix) — trip pump, close outlet
TAHH	Temperature Alarm High-High	Temperature at maximum safe limit	BPCS or SIS trip (per C&E matrix) — trip heater or compressor
FSLL	Flow Switch Low-Low	Flow drops to unsafe minimum (e.g., lube-oil)	BPCS or SIS trip (per C&E matrix) — trip rotating equipment

WARNING — Safety Interlock Bypass

Never bypass an ESD input, PSV, or PAHH setpoint without a written Management of Change (MOC) authorization, a valid bypass permit, a temporary alarm or operator notification in place, and approval from a process safety engineer. Unauthorized bypasses are a violation of OSHA PSM regulations (29 CFR 1910.119) and may have fatal consequences.

NOTES

Section 2: Instrumentation Fundamentals

2.1 The 4–20 mA Signal Standard

The **4–20 mA current loop** is the dominant analog signal standard in process control. It represents a process variable as a proportional current: 4 mA = 0% of range (live zero), 20 mA = 100% of range. The signal is a current (not a voltage), which makes it inherently immune to line resistance and EMI noise over long cable runs — both critical advantages in an outdoor gas plant environment.

Why a "Live Zero" at 4 mA?

A healthy, powered transmitter outputs at least 4 mA even when the process is at its minimum value. If the signal drops below 3.6 mA, the DCS/PLC can detect it as a **broken wire or transmitter power failure** — a "below live zero" fault — and annunciate an alarm. A signal of 0 mA unambiguously indicates an open circuit.

Signal Conversion Formula

```
(* Convert mA signal to % of span *)
Percent (%) = (mA - 4) / 16 × 100

(* Convert % to engineering units *)
EU Value = PV_Min + (Percent / 100) × (PV_Max - PV_Min)

(* Combined single-step conversion *)
EU Value = PV_Min + ((mA - 4) / 16) × (PV_Max - PV_Min)

(* Reverse: EU to expected mA output *)
mA = ((EU Value - PV_Min) / (PV_Max - PV_Min)) × 16 + 4

(* Example: PT range 0–300 PSIG, reading 12 mA *)
EU = 0 + ((12 - 4) / 16) × (300 - 0) = 0.5 × 300 = 150 PSIG
```

Wiring Types

Wiring Type	Description	Power Source	Typical Use
2-Wire (Loop-Powered)	Transmitter powered by the loop supply (24 VDC); signal and power share same two wires	Control room barrier / PLC card	Pressure, temperature transmitters — most common in gas plant field instruments
3-Wire	Separate power supply wire (+ power, − common, + signal)	Local external 24 VDC supply	Sensors requiring separate power (some flow meters, RTDs)
4-Wire (Line-Powered)	Separate power pair (L1/N for AC, or + /− for DC) plus separate signal pair	AC line power or external DC	Analyzers, high-power transmitters, multivariable transmitters

2.2 Instrument Types

Transmitter Type	ISA Tag	Measurement	Common Range	Notes
Pressure Transmitter	PT	Gauge (PSIG), absolute (PSIA), or differential pressure (ΔP in inH_2O)	0–300 PSIG (separator); 0–1,500 PSIG (high-pressure)	Rosemount 3051; Endress+Hauser Cerabar; uses capacitive or silicon sensor
Temperature Transmitter	TT	Process fluid temperature (°F or °C)	−50 to 500°F typical	Usually RTD (Pt100) or thermocouple (J, K type) plus 4-wire head-mount transmitter (e.g., Rosemount 644)
Flow Transmitter	FT	Volumetric or mass flow rate	Varies — 0–200 MMSCFD gas; 0–500 BPD liquid	Types: orifice plate + DP cell, Coriolis (mass), magnetic (liquid), ultrasonic, vortex
Level Transmitter	LT	Liquid level in vessel or tank	0–100% (0–120 inches typical)	Types: DP level (displacer), guided wave radar (GWR), non-contacting radar, buoyancy; 4-20mA output
Analyzer Transmitter	AT	Gas/liquid composition or property	0–20 ppm H_2S; 0–100% LEL; 950–1,150 BTU/SCF	Types: H_2S electrochemical, O_2, chromatograph, BTU analyzer, pH; 4-20mA or digital (Modbus) output
Vibration Transmitter	VT	Shaft vibration on rotating equipment (mils peak-to-peak or mm/s RMS)	0–10 mils (reciprocating); 0–2 in/s (centrifugal)	Proximity probes (Bently Nevada), accelerometers; often direct 4-20mA or EDDL to machine monitoring system

2.3 HART Protocol Basics

HART (Highway Addressable Remote Transducer) is a bi-directional digital communication protocol superimposed on the standard 4–20 mA signal. It uses frequency-shift keying (FSK) to transmit digital data at 1,200 baud without disturbing the analog signal.

- **Primary Variable (PV):** The main measurement — represented by the 4–20 mA signal and also transmitted digitally. Example: process pressure in PSIG.
- **Secondary Variable (SV):** A secondary measurement transmitted digitally only. Example: sensor temperature on a pressure transmitter.
- **Tertiary / Quaternary Variables (TV, QV):** Additional digital variables on multivariable transmitters (e.g., temperature-compensated flow from a multivariable DP transmitter).
- **Polling Address:** Each HART device has a polling address: 0–15 for HART 5 devices and 0–63 for HART 6/7 devices. In point-to-point mode (the most common), the polling address is 0 and the 4–20 mA analog signal is active. In multidrop mode, the analog signal is fixed (typically 4 mA) and devices use non-zero addresses.
- **HART Communicator:** A handheld device (e.g., Emerson 475 or AMS Trex) that connects across the loop (at the terminals or at the barrier) to read/write configuration, trim the instrument, run diagnostics, and perform calibration.

> **NOTE — HART in the Field**
> Connect the HART communicator across a loop resistance of 230–600 Ω (typically 250 Ω), either at the instrument terminals or at the field junction box, not at the PLC card terminals. If total loop resistance is below 230 Ω, the HART communicator will not establish communication; add a HART resistor.

2.4 Instrument Calibration

Zero and Span

Zero is the adjustment that sets the transmitter's output to exactly 4.000 mA at the Lower Range Value (LRV — 0% of input range). **Span** is the adjustment that sets the output to exactly 20.000 mA at the Upper Range Value (URV — 100% of input range). Always adjust zero first, then span, then re-check zero (they can interact).

5-Point Calibration Procedure

A **5-point calibration** verifies transmitter accuracy across five input values: 0%, 25%, 50%, 75%, and 100% of span. For an upscale check (ascending) and a downscale check (descending), this produces 10 data points — sufficient to detect hysteresis and linearity errors.

Calibration Record Template

Tag	Description	Range	0% mA (4.00)	25% mA (8.00)	50% mA (12.00)	75% mA (16.00)	100% mA (20.00)	As-Found Error	As-Left Error	Technician	Date
SEP-PT-101	Inlet Sep Pressure	0–300 PSIG									_____
[Add additional rows as needed — COMPANY: attach to site calibration management system record]											

2.5 Loop Check Procedure (Step-by-Step)

A **loop check** verifies that a complete instrument loop — from field sensor through wiring, junction box, cable, IS barrier, PLC AI card, to HMI display — functions correctly before commissioning or after maintenance.

1. **Obtain authorization.** Confirm MOC or loop check work order. Notify the control room operator that the loop is being tested and may produce false readings or alarms — request alarm suppression if needed.
2. **Gather documentation.** Retrieve the loop drawing, P&ID, instrument datasheet, and calibration record for the instrument under test.
3. **Inspect field wiring.** At the field instrument, check terminal connections are tight, wire labels match the loop drawing, no corrosion or moisture intrusion, and cable gland is sealed.
4. **Measure loop current at field instrument.** Connect a calibrated milliammeter (or use the HART communicator's mA measurement function) in series with the loop. Verify minimum current (≥3.6 mA) is present — indicates loop is powered and complete.

5. **Apply 0% input.** Inject 4.000 mA using a precision mA source (loop calibrator, e.g., Beamex MC6 or Fluke 789). Confirm PLC AI card reads 0% of engineering units. Confirm HMI displays correct 0% value.
6. **Apply 25%, 50%, 75%, 100% inputs.** For each point, confirm PLC and HMI values track the injected signal within acceptable tolerance (typically ±0.1% of span for most process instruments).
7. **Verify alarms.** Inject a value above the PAH setpoint — confirm alarm annunciates on HMI with correct tag, description, and priority. Test PAHH if applicable.
8. **Check fail-safe behavior.** Open the loop (disconnect one wire). Confirm the PLC flags bad quality through the channel under-range/fault status and that a bad-quality alarm annunciates on the HMI.
9. **Restore wiring.** Reconnect all wires, verify connections are tight, replace covers and glands.
10. **Clear alarm suppression.** Notify control room that loop check is complete and loop is returned to service. Confirm operator restores normal alarm state.
11. **Document results.** Record all readings on the loop check form or calibration record. Note any deviations, corrective actions taken, and the as-left condition. Sign and date.

2.6 Intrinsic Safety Basics

Intrinsic Safety (IS) is a method of explosion protection for instruments in hazardous (classified) locations. It limits the electrical energy (voltage, current, and capacitance/inductance) in the hazardous area circuit to levels incapable of igniting a flammable atmosphere — even under fault conditions. IS circuits are approved for use in Class I, Div. 1 (or ATEX/IECEx Zone 0/1) hazardous locations.

Zener Barriers

A **Zener barrier** (or simply "barrier") is a passive IS device mounted in the safe area (control room). It contains Zener diodes that clamp voltage to a safe level and resistors that limit fault current. Zener barriers require a certified intrinsically safe earth bond (≤1Ω ground connection) to function correctly. They are simple and inexpensive but sensitive to ground quality.

Galvanic Isolators

A **galvanic isolator** provides IS protection using transformer or optical isolation — no ground bond required. It converts the hazardous-area signal to the safe area with complete galvanic isolation. Galvanic isolators are preferred when ground loop issues exist or when a reliable IS earth is not available. They are more expensive than Zener barriers but easier to install correctly.

IS vs. Non-IS Wiring Rules

Rule	IS Circuits	Non-IS Circuits
Cable type	IS-approved cable (blue sheath per NEC common practice)	Standard instrumentation cable
Conduit/tray separation	Must be segregated from non-IS wiring (separate tray or conduit)	Standard routing per plant practice
Labeling	All IS terminals and cables labeled "INTRINSICALLY SAFE"	Standard I&E labels
Connection in hazardous area	Permitted live (Class I, Div. 1 if IS certified)	Requires hot work permit or area deenergization
Ground bond (Zener barrier)	Required — ≤1Ω to certified IS earth	Standard panel ground
Documentation	Entity concept parameters must be verified (Voc, Isc, Ca, La of barrier vs. transmitter)	Standard loop drawing

2.7 Common Instrument Faults and Symptoms

Fault	Symptom	Likely Cause	Corrective Action
Open circuit (broken wire)	mA = 0; HMI shows bad quality / low fault alarm	Broken wire, loose terminal, corroded connection	Trace loop wiring; check all terminal connections with milliammeter
Below live zero	mA ≤ 3.6 (NAMUR NE43 low fault); channel under-range status set; HMI shows bad quality	Partial short, power supply issue, failing transmitter, transmitter in low fault mode	Check loop supply voltage; measure current at transmitter terminals; read HART status; replace transmitter if power is correct
High fault / over-range	mA ≥ 21 (NAMUR NE43 high fault); over-range status set	Over-range input, shorted signal lead, sensor failure, transmitter in high fault mode	Verify the process is not actually at the URV; isolate transmitter; test sensor element

Fault	Symptom	Likely Cause	Corrective Action
Noisy / oscillating signal	PV value jumps rapidly; trend shows spikes	Inadequate shielding, ground loop, EMI from VFD/motor near cable, loose connection	Check shield grounding (single-point ground); re-route cable away from power; add damping in transmitter
Offset / span error after maintenance	PV reads consistently high or low vs. reference gauge	Transmitter zero shifted, process connection not equalized, impulse line blocked	Re-zero transmitter at reference; check for plugged impulse lines; recalibrate
Transmitter drift over time	Slow creep in reading; calibration record shows increasing as-found error	Electronic aging, process ingress into sensor, thermal cycling	Increase calibration frequency; consider replacement; protect against thermal shock
Positioner not responding to command	Valve position % does not match output command from PLC	Air supply failure, positioner tuning, I/P transducer failure, mechanical linkage slip	Check instrument air supply (min 20 PSIG clean dry air); stroke valve manually; check positioner feedback linkage; retune positioner
HART communication fails	HART communicator shows "No device found"	Loop resistance below 230 Ω, wrong polling address, IS barrier limiting current	Ensure 230–600 Ω loop resistance (add a 250 Ω HART resistor if needed); verify polling address; check barrier entity parameters

NOTES

Section 3: Allen-Bradley ControlLogix 1756

3.1 Hardware Architecture

The **Allen-Bradley ControlLogix 1756** platform is a modular, rack-based PLC system manufactured by Rockwell Automation. It is the industry standard for large-scale process control in oil and gas, chemical, and power generation facilities. Each system consists of one or more chassis (racks) populated with interchangeable I/O, communication, and CPU modules connected via a high-speed backplane.

Chassis Sizes

Part Number	Slots	Typical Use
1756-A4	4 slots	Small remote I/O panel or satellite rack
1756-A7	7 slots	Medium remote rack or small main rack
1756-A10	10 slots	Standard main rack — most common in gas plant control panels
1756-A13	13 slots	Large main rack with extensive I/O
1756-A17	17 slots	Maximum density rack — used for high-channel-count applications

Common 1756 Modules

Part Number	Description	Channels	Signal Type	Notes
1756-L85E	ControlLogix 5580 CPU (Logix 5585)	N/A	Processing	Up to 40 MB user memory; one embedded EtherNet/IP port; SD card slot; may occupy any slot (slot number is used in communication paths)
1756-EN2T	EtherNet/IP Communication Module	N/A	100 Mbps Ethernet	Connects rack to plant EtherNet/IP network; supports CIP messaging and I/O connections
1756-IB16	Digital Input — 24 VDC Sink/Source	16 DI	24 VDC discrete	8mA input current; 1ms filter (adjustable); screw terminal IFM required (1756-TBNH)
1756-OB16	Digital Output — 24 VDC Source	16 DO	24 VDC discrete	0.5A/pt, 8A/module; fused groups; check external fusing for inductive loads
1756-IF8	Analog Input — 8-Channel	8 AI	4–20 mA, 0–20 mA, or voltage	Integer mode: 0–20.58 mA = −32768…32767 counts (4 mA ≈ −20030); floating-point mode scales in the module; 249 Ω sense resistor (current mode wiring per 1756-UM009)
1756-OF8	Analog Output — 8-Channel	8 AO	4–20 mA or 0–10 V	Output fault and program-mode states are configurable (hold last state or a user-defined value)
1756-RM2	Redundancy Module	N/A	Fiber crosslink	Used in pairs; connects primary and secondary controller chassis; handles crossload and switchover
1756-PA75	AC Power Supply	N/A	120/240 VAC input, 5VDC/24VDC backplane output	75W output; always in power supply slot (left end of chassis); use 1756-PB75 for 24 VDC input

Backplane Communication

The **backplane** is the internal bus connecting all modules in a chassis. Data is transferred between the CPU and I/O modules at a rate governed by the **Requested Packet Interval (RPI)** — a configurable parameter (in milliseconds) that specifies how often the CPU reads input data from or writes output data to each module. A smaller RPI means faster updates but higher backplane and network load.

Parameter	Typical Value	Notes
Local I/O RPI	2–20 ms	Local chassis I/O; faster than remote I/O is practical
Remote I/O RPI (FLEX)	10–100 ms	Set based on process speed and network loading
Backplane data rate	Gigabit internal	Not user-configurable; managed by firmware
Slot number	0 to (chassis size − 1)	The CPU may occupy any slot; the slot number is used in communication paths. Modules are numbered left to right from the power supply end.

Redundancy: Primary / Secondary Controller

Redundancy in ControlLogix uses two identical chassis (primary and secondary) connected via 1756-RM2 modules and a fiber optic crosslink cable. The primary controller is active (scanning, controlling); the secondary is in a synchronized warm-standby state. The secondary receives periodic **crossloads** — transfers of the tag database, I/O image, and program state — so it can take over with zero data loss. **Switchover triggers** include: CPU fault, power loss on primary, communication loss, or user-commanded switchover. Switchover time is typically <100 ms for a warm-redundant system.

> **NOTE — Redundancy Maintenance**
> Always confirm the secondary controller is in SYNCHRONIZED state before performing any work on the primary controller. A non-synchronized secondary cannot take over cleanly. Check the redundancy status in RSLinx or Studio 5000 Redundancy Management before any planned maintenance that could cause a primary fault.

3.2 Studio 5000 / RSLogix 5000 Basics

Studio 5000 Logix Designer (formerly RSLogix 5000) is the programming environment for all ControlLogix systems. Technicians use it to create, edit, download, and monitor PLC programs.

Project Tree Structure

Node	Description	Key Settings
Controller	Root node; defines CPU type, firmware rev, and controller-scoped tags	Controller properties, redundancy settings, safety settings
Tasks	Container for all scan tasks; defines execution scheduling	Task type, period (for Periodic), priority
Programs	Logical grouping within a Task; each has its own local tag database	Program-scoped tags; schedule inhibit options
Routines	Executable code unit within a Program; written in Ladder, ST, FBD, or SFC	Routine type; Main Routine assigned in Program properties
Tags	Variable database; controller-scoped or program-scoped	Data type, scope, alias, external access (read/write from HMI)
I/O Configuration	Hardware tree mirroring physical rack and module layout	Module properties, RPI, inhibit, fault actions
Add-On Instructions	User-defined or library reusable instruction blocks	AOI definition, parameters, local tags, logic

Task Types

Task Type	When to Use	Typical Period / Priority
Continuous	Main process control scan; executes repeatedly without interruption. Only one continuous task per controller.	Runs at background priority; period = scan time
Periodic	Timed interrupts — high-speed loops, sequential logic at fixed intervals (e.g., every 10 ms for a fast pressure control loop)	1–2,000 ms; priority 1–15 (1 = highest)
Event	Triggered by a specific hardware event (module input change, motion complete, DIO event) — use for time-critical responses	Triggered; priority 1–15; max execution time configurable

Program / Routine Hierarchy

Within each Task, Programs execute in order (top to bottom in the task list). Within each Program, the **Main Routine** executes first; it calls sub-routines using JSR (Jump to Subroutine) instructions. Organizing complex programs into sub-routines (e.g., SEP_STARTUP, SEP_NORMAL_OP, SEP_SHUTDOWN, SEP_ALARM_SCAN) improves readability and maintenance. Fault Routine and Power-Up Routine are optional special routines executed only on fault or restart events.

Tag Database and Scope

- **Controller-scoped tags:** Accessible from any Program or Routine in the project, and from external devices (HMI, MSG instructions). Use for tags shared across programs or monitored by the HMI.
- **Program-scoped tags:** Local to one Program only. Use for internal calculation variables, timers, and counters not needed elsewhere. Cannot be accessed by name from outside the Program.

Data Types Table

Data Type	Size	Range / Description	Typical Use
BOOL	1 bit	0 or 1	Discrete I/O, flags, alarm bits, permissives
SINT	8 bits	−128 to 127	Small integer counters, status codes
INT	16 bits	−32,768 to 32,767	Medium-range counts; rarely used — prefer DINT
DINT	32 bits	−2,147,483,648 to 2,147,483,647	Counters, state machines, integer math
REAL	32 bits (IEEE 754 float)	±3.4 × 10^{38}; ~7 significant digits	Analog values, PV, SP, engineering units calculations, PID
STRING	88 bytes (DINT LEN + 82 SINT DATA)	Up to 82 ASCII characters	Equipment names, alarm messages, recipe strings
TIMER (TON/TOF/RTO)	Structure	.EN, .TT, .DN, .ACC, .PRE	Timing delays, pulse generation
COUNTER (CTU/CTD)	Structure	.CU, .CD, .DN, .OV, .UN, .ACC, .PRE	Counting events, cycles, batches
UDT	User-defined	Combination of any data types	PID loops, valve faceplates, equipment modules (see below)
AOI	User-defined instruction	Custom instruction with parameters and internal logic	Reusable valve, motor, PID, or alarm blocks

UDT Example — PID Loop Structure

A User-Defined Type (UDT) for a PID loop can organize the loop parameters under a single tag name (e.g., SEP_PIC_101). The example below is illustrative; the native PID instruction uses its own PID structure (Section 5.2, Snippet 4).

```
(* UDT: PID_LOOP_TYPE — illustrative member definitions *)
PV          : REAL    (* Process variable, engineering units (PSIG) *)
SP          : REAL    (* Setpoint, engineering units (PSIG) *)
OUT         : REAL    (* Controller output 0.0–100.0 % *)
Kp          : REAL    (* Proportional gain *)
Ki          : REAL    (* Integral: 1/s when PID.PE = 0; Ti in min/repeat when PE = 1 *)
Kd          : REAL    (* Derivative: s when PID.PE = 0; Td in min when PE = 1 *)
OutMin      : REAL    (* Output minimum clamp (%) *)
OutMax      : REAL    (* Output maximum clamp (%) *)
Mode        : DINT    (* 0 = Manual, 1 = Auto, 2 = Cascade *)
ManOut      : REAL    (* Manual output value when Mode = 0 *)
AutoMode    : BOOL    (* TRUE when in auto *)
InAuto      : BOOL    (* Permissive for auto mode entry *)
Hi_Alarm    : BOOL    (* PV high alarm active *)
HiHi_Alarm  : BOOL    (* PV high-high alarm active *)
Lo_Alarm    : BOOL    (* PV low alarm active *)
LoLo_Alarm  : BOOL    (* PV low-low alarm active *)
PAH_SP      : REAL    (* High alarm setpoint *)
PAHH_SP     : REAL    (* High-high alarm setpoint *)
PAL_SP      : REAL    (* Low alarm setpoint *)
PALL_SP     : REAL    (* Low-low alarm setpoint *)
Fault       : BOOL    (* Instrument fault — bad AI signal *)
```

Add-On Instructions (AOIs)

An Add-On Instruction (AOI) is a custom, reusable instruction block with its own logic, parameters, and local tags. It appears in the ladder rung like a native instruction. Rockwell Automation and equipment vendors (e.g., Fisher, Emerson) provide AOI libraries for valve control, PID, alarm management, and motor control. Use existing, tested AOIs in preference to rebuilding equivalent logic.

3.3 Tag Naming Convention

Consistent tag naming is critical for HMI integration, alarm management, historian configuration, and long-term maintainability. Follow Section 0.3: hyphens on P&IDs, underscores in Logix, one spelling in HMI. The recommended convention follows ISA-5.1 functional identification augmented with plant area codes. When the official plant standard is pasted into the [COMPANY:] box below, that standard wins.

Format: [AREA]-[FunctionCode][Number].[Attribute]

- **AREA:** Plant area abbreviation (e.g., SEP = Inlet Separator, COMP = Compressor, DEHYD = Dehydration, AMINE = Amine Unit, FRAC = Fractionation, METER = Metering)
- **FunctionCode:** ISA instrument function code (PT, TT, FT, LT, AT, XV, PCV, LCV, FCV, MOT)
- **Number:** Loop/device number (3 digits, sequential within area, e.g., 101, 102...)
- **Attribute:** Sub-element of a UDT or structure (e.g., .Value, .SP, .OUT, .Mode, .Hi_Alarm)

Examples Table

Tag	Description	Data Type	Unit
SEP_PT_101.Value	Inlet Separator pressure transmitter PV	REAL	PSIG
SEP_PT_101.Hi_Alarm	Inlet Separator pressure high alarm	BOOL	—
SEP_LIC_101.SP	Inlet Separator level controller setpoint	REAL	%
SEP_LIC_101.OUT	Inlet Separator level controller output	REAL	%
COMP_XV_201	Compressor isolation valve — discrete state	BOOL	0=Closed, 1=Open
COMP_MTR_201.Running	Compressor motor run feedback	BOOL	—
SIS_ESD_OK	SIS ESD healthy status (read-only in BPCS)	BOOL	1 = healthy, 0 = SIS tripped
AMINE_AT_301.Value	Amine unit outlet H_2S analyzer PV	REAL	ppm
FRAC_TIC_401.Kp	Fractionator temperature controller — proportional gain	REAL	—
METER_FT_501.Value	Custody transfer metering flow transmitter PV	REAL	MMSCFD

[COMPANY: Insert the plant's official tag naming standard here — include approved area codes, numbering series, and any deviations from ISA-5.1 convention. Reference document number: _____]

3.4 Networking

EtherNet/IP Overview

EtherNet/IP (Ethernet Industrial Protocol) is the Rockwell Automation standard network for ControlLogix communication. It uses standard IEEE 802.3 Ethernet hardware and the **CIP (Common Industrial Protocol)** application layer. Two messaging types are fundamental:

- **Implicit Messaging (I/O messaging):** Real-time, cyclic data exchange between controller and I/O devices. The connection is established at startup and data is transferred automatically at the RPI rate. Used for: local and remote FLEX I/O data, drive I/O, safety I/O.
- **Explicit Messaging (MSG instruction):** On-demand, request-response data exchange. The controller initiates a read or write to another device upon logic triggering. Used for: reading data from another PLC, writing recipe setpoints, communication with HMI OPC server, diagnostics.

Device Level Ring (DLR) Topology

A Device Level Ring (DLR) is a high-availability Ethernet topology in which devices are connected in a ring rather than a star. The ring supervisor sends beacon frames (typically every 400 µs) to detect a cable break. On a fault, the ring supervisor reconfigures the network to a linear topology within about 3 ms, restoring communication without a managed switch at every node. DLR is common in FLEX I/O banks, VFD networks, and remote I/O panels.

CIP Basics

CIP (Common Industrial Protocol) is the application-layer protocol used by EtherNet/IP (and also DeviceNet and ControlNet). It defines objects, services, and a consistent device model. Key CIP services used in plant work:

- **CIP Data Table Read/Write:** Used in MSG instructions to read or write a specific tag in another controller by name.
- **CIP Generic:** Used for manufacturer-specific operations, drive parameter reads, and module configuration.
- **Connected Explicit:** Persistent connection for repeated MSG reads (more efficient than unconnected when polling frequently).

Network Address Layout Template

Device	IP Address	Subnet	Role
CTRL-1 (Primary ControlLogix)	[COMPANY: e.g., 10.10.1.10]	255.255.255.0	Primary process controller
CTRL-1S (Secondary ControlLogix)	[COMPANY: e.g., 10.10.1.11]	255.255.255.0	Redundant secondary controller
FLEX-SEP-01 (1794-AENTR, Separator Bank)	[COMPANY: e.g., 10.10.1.20]	255.255.255.0	Remote FLEX I/O — inlet separator
FLEX-COMP-01 (1794-AENTR, Compressor Bank)	[COMPANY: e.g., 10.10.1.21]	255.255.255.0	Remote FLEX I/O — compressor area
COMP-PLC-1 (Compressor PLC)	[COMPANY: e.g., 10.10.1.30]	255.255.255.0	Compressor package OEM PLC — MSG target
WMGR-01 (Wonderware Application Server)	[COMPANY: e.g., 10.10.1.50]	255.255.255.0	SCADA/HMI server
OPC-SVR-01 (OPC DA/UA Server)	[COMPANY: e.g., 10.10.1.51]	255.255.255.0	OPC server — HMI to PLC data bridge
ENG-WS-01 (Engineering Workstation)	[COMPANY: e.g., 10.10.1.100]	255.255.255.0	Studio 5000 / InTouch WindowMaker development
HMI-OP-01 (Operator Console #1)	[COMPANY: e.g., 10.10.1.110]	255.255.255.0	InTouch WindowViewer — DCS workstation

NOTES

Section 4: 1794 FLEX I/O

4.1 Hardware Overview

The **Allen-Bradley 1794 FLEX I/O** system is a distributed, modular I/O platform designed for remote installation close to field devices — reducing field wiring runs and control panel space. Each FLEX I/O bank consists of a communication adapter followed by up to eight I/O modules, each snapping onto a terminal base mounted on a DIN rail. The system connects to the ControlLogix controller over EtherNet/IP or DeviceNet.

Adapter Types

Part Number	Network	Description	Key Difference
1794-AENTR	EtherNet/IP	Dual-port EtherNet/IP adapter with DLR support	Preferred for modern installations; supports Device Level Ring for redundant network path; configures as I/O connection in Studio 5000
1794-ADN	DeviceNet	DeviceNet adapter (older installations)	Limited to 500 kbps DeviceNet speed; requires 1756-DNB DeviceNet scanner in main rack; found in legacy gas plant systems

Terminal Base Types

Part Number	Description	Notes
1794-TB3	Standard screw-clamp terminal base (3-wire field terminals)	Most common; accepts all FLEX I/O modules; hand-screw clamp terminals
1794-TB3S	Spring-clamp terminal base (3-wire)	Spring clamp requires no screwdriver — push-in and release; faster wiring, vibration-resistant; preferred for high-vibration locations
1794-TBN	Narrow-form terminal base	Reduced footprint for space-constrained panels; fewer field terminals per base

Hot-Swap Capability

A critical advantage of FLEX I/O is **hot-swap** — the ability to remove and replace an individual I/O module while the adapter and other modules in the bank remain powered and communicating. The module is mechanically separable from its terminal base: the module (which contains the electronics) lifts off, while the terminal base (which retains field wiring) stays mounted. This means a failed analog input module can be replaced in the field without de-energizing the entire panel or disconnecting field wiring. Refer to Lab 2 (Section 10) for the full hot-swap procedure.

Module Rail Assembly Rules

1. The **adapter module (1794-AENTR or 1794-ADN) is always first** on the left — the system cannot operate without it as the first module.
2. Up to **8 I/O modules** can follow the adapter on a single bank (total 9 positions including adapter).
3. Modules are connected mechanically and electrically by the terminal base expansion cable — each base has a right-side connector that plugs into the left-side connector of the next base.
4. The last terminal base must have a **terminator cap (1794-END)** on the right-side expansion connector to seal the bus and prevent communication errors.
5. Power wiring: 24 VDC I/O power enters through the adapter's power terminals and distributes across the bank. AC modules (1794-IM8, 1794-OM8) require separate, isolated AC power to their terminal base.

4.2 Common FLEX I/O Modules

Part Number	Description	Points	Signal Range	Notes
1794-IB16	Digital Input — 24 VDC Sink/Source	16 DI	24 VDC (10.8–26.4V operating)	Filter time configurable (0–8ms); suitable for limit switches, proximity sensors, valve position feedback
1794-OB16	Digital Output — 24 VDC Source	16 DO	24 VDC, 0.5A/point	Electronic short-circuit protection; suitable for solenoid valves, pilot lights, relay coils
1794-IE8	Analog Input — 4–20 mA or 1–5 V	8 AI	4–20 mA or 1–5 V; 4–20 mA mode: 4 mA = 0 counts, 20 mA = 30840 counts (module-scaled, left-justified 12-bit)	Individual channel configuration; reads 0 at or below 4 mA, so use the channel under-range status for wire-break detection

Part Number	Description	Points	Signal Range	Notes
1794-OE4	Analog Output — 4–20 mA or 1–5 V	4 AO	4–20 mA; 4 mA = 0 counts, 20 mA = 30840 counts	Configurable fault state (hold, low clamp, high clamp); used for control valve positioners, VFD speed references
1794-IM8	Digital Input — 120/240 VAC	8 DI	85–265 VAC	Used where field switches are AC-powered (older installations, motor starters with AC auxiliary contacts)
1794-OM8	Digital Output — 120/240 VAC	8 DO	85–265 VAC, 0.5A/point	Isolated relay or triac outputs; used for AC-powered solenoids, contactors in legacy panels

4.3 I/O Mapping Table — Gas Plant Example

The I/O Mapping Table is one of the most important maintenance documents. It links every physical terminal in the FLEX I/O bank to a specific field device, PLC tag, and P&ID reference. Keep this table updated whenever field devices are added, removed, or relocated.

[COMPANY: This table is a template — replace example rows with actual plant I/O assignment. Reference plant I/O schedule drawing number: _____]

Slot	Ch	Tag Name	Description	Signal Type	Range	P&ID Ref	Device Type	Wire Color	Terminal	Notes
1	0	SEP_PT_101_Raw	Inlet Sep Pressure Xmtr	AI 4–20mA	0–300 PSIG	P&ID-001	Rosemount 3051CD	Blue/White	TB1-1/2	IS barrier reqd
1	1	SEP_LT_101_Raw	Inlet Sep Level Xmtr	AI 4–20mA	0–100%	P&ID-001	Magnetrol GWR	Red/White	TB1-3/4	2-wire loop power
1	2	SEP_LT_102_Raw	Inlet Sep Water Level Xmtr	AI 4–20mA	0–100%	P&ID-001	Rosemount 3301 GWR	Red/White	TB1-5/6	2-wire loop power
1	3	SEP_TT_101_Raw	Inlet Sep Temp Xmtr	AI 4–20mA	−50–250°F	P&ID-001	Rosemount 644H RTD	Green/White	TB1-7/8	4-wire RTD head transmitter
1	4	SEP_FT_101_Raw	Inlet Sep Gas Outlet Flow	AI 4–20mA	0–50 MMSCFD	P&ID-001	Emerson 3051S MV DP	Blue/White	TB1-9/10	Compensated DP flow
1	5	COMP_PT_201_Raw	Compressor Suction Press Xmtr	AI 4–20mA	0–600 PSIG	P&ID-010	Rosemount 3051CD	Blue/White	TB1-11/12	IS barrier reqd
1	6	COMP_TT_201_Raw	Compressor Suction Temp Xmtr	AI 4–20mA	−50–350°F	P&ID-010	Rosemount 644H TC-K	Green/White	TB1-13/14	Thermocouple type K
1	7	COMP_PT_202_Raw	Compressor Discharge Press Xmtr	AI 4–20mA	0–1500 PSIG	P&ID-010	Rosemount 3051CD	Blue/White	TB1-15/16	High-pressure range
2	0	SEP_PCV_101_Raw	Sep Pressure Ctrl Valve Position	AI 4–20mA	0–100%	P&ID-001	Fisher positioner fdbk	Yellow/White	TB2-1/2	Valve position feedback
2	1	DEHYD_TT_301_Raw	TEG Reboiler Temperature Xmtr	AI 4–20mA	100–500°F	P&ID-020	Rosemount 644H RTD	Green/White	TB2-3/4	High-temp process
2	2	AMINE_AT_301_Raw	Amine Outlet H2S Analyzer	AI 4–20mA	0–20 ppm	P&ID-030	Ametek Model 101 H2S	Orange/White	TB2-5/6	4-wire analyzer; check alignment monthly
2	3	FRAC_PT_401_Raw	Deethanizer Press Xmtr	AI 4–20mA	0–500 PSIG	P&ID-040	Rosemount 3051CD	Blue/White	TB2-7/8	IS barrier reqd
3	0	SEP_PCV_101_CMD	Sep Pressure Ctrl Valve Command	AO 4–20mA	0–100%	P&ID-001	Fisher DVC6200 positioner	Yellow/White	TB3-1/2	Fail-open valve (air-to-close)

Slot	Ch	Tag Name	Description	Signal Type	Range	P&ID Ref	Device Type	Wire Color	Terminal	Notes
3	1	SEP_LCV_101_CMD	Sep Level Ctrl Valve Command	AO 4–20mA	0–100%	P&ID-001	Fisher DVC6200 positioner	Yellow/White	TB3-3/4	Fail-closed valve (air-to-open)
3	2	DEHYD_FCV_301_CMD	TEG Circulation Flow Ctrl Cmd	AO 4–20mA	0–100%	P&ID-020	Masoneilan positioner	Yellow/White	TB3-5/6	TEG pump control valve
3	3	COMP_SV_201_CMD	Compressor Speed Reference	AO 4–20mA	0–100%	P&ID-010	ABB ACS880 VFD speed ref	Yellow/Blk	TB3-7/8	0-100% = 0-3600 RPM
4	0	COMP_XV_201_Open	Comp Inlet XV-201 Open Limit	DI 24VDC	0/1	P&ID-010	Limit switch (NO contact)	White/Blk	TB4-1	1=valve fully open
4	1	COMP_XV_201_Closed	Comp Inlet XV-201 Closed Limit	DI 24VDC	0/1	P&ID-010	Limit switch (NO contact)	White/Blk	TB4-2	1=valve fully closed
4	2	SIS_ESD_OK	SIS ESD healthy status (read-only)	DI 24VDC	0/1	P&ID-ESD-001	SIS logic solver output relay	Red/Blk	TB4-3	1 = SIS healthy; 0 = SIS tripped or bypassed. BPCS reads only; never written by BPCS logic
4	3	FG_Z3_OK	F&G Detector Zone 3 Vote	DI 24VDC	0/1	P&ID-FG-003	F&G System voted output relay	Red/Blk	TB4-4	1 = healthy; 0 = fire/gas detected (fail-safe); bypass requires MOC
[COMPANY: Add additional I/O rows here — complete all fields before commissioning. Verify wire colors against site cable schedule.]										

4.4 FLEX I/O Configuration in Studio 5000 (Step-by-Step)

1. **Open the project in Studio 5000.** In the Controller Organizer, right-click **I/O Configuration → New Module**.
2. **Select the adapter module.** Search for "1794-AENTR" and select it. Enter the adapter's IP address (e.g., 10.10.1.20), name (e.g., FLEX_SEP_01), and confirm chassis/slot assignment. Set the RPI for the adapter — this governs how frequently the ControlLogix polls the entire FLEX bank. Typical value: 20 ms.
3. **Add I/O modules under the adapter.** Right-click the newly created adapter in the I/O tree → **New Module**. Add each I/O module in slot order (Slot 1 = first I/O module after adapter, Slot 2 = second, etc.). Select the correct part number (e.g., 1794-IE8 for AI), configure its signal type (mA or voltage), and set any module-specific options (filter time for DI, fault state for AO).
4. **Verify connection status online.** Go online (Comm → Who Active → connect to controller). In the I/O Configuration tree, modules in a healthy state show a green checkmark. A red "X" indicates a fault — check physical connections, IP address, and module type match.
5. **Map tags.** In the Controller Tags, verify that auto-generated I/O tags (e.g., FLEX_SEP_01:1:I.Ch0Data) exist. Create alias tags with engineering names (e.g., SEP_PT_101_Raw) pointing to these I/O tags for use in logic and HMI.

Common FLEX I/O Faults

Fault	Code / Indicator	Likely Cause	Action
Module Fault	16#0116 (keying/revision mismatch) or module fault status; red module LED	Module type mismatch with configuration, module hardware failure, incorrect firmware	Verify module part number matches configuration; replace module; check firmware revision compatibility
Comms Loss to Adapter	16#0203 (connection timed out) or 16#0204 (connection request error / unconnected timeout); adapter NS LED flashing red (connection timeout) or solid red (duplicate IP)	Ethernet cable disconnected, IP address conflict, wrong subnet, adapter powered off	Ping adapter IP; check cable and switch port; verify IP address in adapter (use RSLinx Browse or BOOTP-DHCP server)

Fault	Code / Indicator	Likely Cause	Action
Wrong RPI	Task overrun; adapter connection fault if RPI < network latency	RPI set too fast for network capacity or number of connections	Increase RPI (e.g., from 5ms to 20ms) for non-critical loops; separate high-speed and low-speed I/O onto different banks
Channel Fault	Individual channel fault bit in I/O tag; HMI shows bad quality on that tag	Open-circuit 4-20mA loop (broken wire), over-range input, short circuit on DO	Measure loop current at field terminals; check wiring for breaks; replace field device if necessary

NOTES

Section 5: Ladder Logic Fundamentals

This section uses Studio 5000 Logix Designer v36 instruction names. Rungs are written in Neutral Text, the text form that Logix Designer exports and accepts for pasting. Each rung block lists a rung label followed by one rung per line. Branches are written as `[path1 ,path2 ]` and each rung ends with a semicolon.

5.1 Core Instructions Reference

The working set below covers all logic in this manual. Legacy mnemonics used in older projects and literature are shown in the second column.

v36 name	Legacy name	Function	Neutral Text example
XIC	—	Examine If Closed: rung continues when the bit is 1.	`XIC(STOP_OK)`
XIO	—	Examine If Open: rung continues when the bit is 0 (bit must be 0 to proceed).	`XIO(SEP_PT_101.Bad)`
OTE	—	Output Energize: bit follows rung state; cleared at prescan.	`OTE(Motor_Run)`
OTL / OTU	—	Output Latch / Unlatch: bit retained until the opposite instruction executes; retained through prescan and power cycle.	`OTL(SEP_Trip_OK)`
ONS	—	One Shot: rung-out true for one scan on a false-to-true transition. Storage bit is set at prescan, so no pulse occurs on the first scan.	`ONS(Area.Reset_ONS)`
TON	—	Timer On Delay: .DN sets after .PRE ms of continuous true rung; reset when rung goes false.	`TON(Delay_TON,5000,0)`
TOF	—	Timer Off Delay: .DN is 1 while rung is true and stays 1 for .PRE ms after rung goes false.	`TOF(Fan_Rundown,60000,0)`
RTO	—	Retentive Timer On: accumulates while true; holds .ACC when false; cleared only by RES.	`RTO(Run_Hours_RTO,3600000,0)`
CTU / CTD	—	Count Up / Down on the rung false-to-true transition (edge tracked by .CU / .CD). .DN = .ACC ≥ .PRE.	`CTU(Comp_Cycle_CTU,1000,0)`
RES	—	Reset a timer or counter.	`RES(Comp_Cycle_CTU)`
EQ / NE	EQU / NEQ	Equal / Not Equal.	`EQ(Comp_State,1)`
GT / LT	GRT / LES	Greater Than / Less Than.	`GT(SEP_PT_101.Value,SEP_PT_101.PAH_SP)`
GE / LE	GEQ / LEQ	Greater Than or Equal / Less Than or Equal.	`LE(Ctr.ACC,0)`
LIMIT	LIM	Limit test: true when Low ≤ Test ≤ High.	`LIMIT(0,SEP_PT_101.Value,300)`
MOVE	MOV	Copy Source to Destination.	`MOVE(4,Comp_State)`
ADD / SUB / MUL / DIV	—	Arithmetic: Source A op Source B → Destination.	`MUL(Raw,Gain,Calc)`
SQRT	SQR	Square root (DP flow).	`SQRT(FT_DP,FT_Root)`
TRUNC	TRN	Truncate REAL to integer value.	`TRUNC(Calc,Whole)`
JSR / SBR / RET	—	Jump to subroutine with optional parameters; SBR receives, RET returns.	`JSR(SEP_LEVEL_CONTROL,0)`
MSG	—	Explicit message to another device (CIP Data Table Read/Write).	`MSG(Comp_MSG_Read)`
GSV / SSV	—	Get / Set System Value (controller and module object attributes). Use GSV for system data only; MSG status is read directly from MESSAGE members (.ER, .ERR, .EXERR).	`GSV(Program,THIS,MajorFaultRecord,FaultRec[0])`
PID	—	Ladder PID: PID(PID,PV,Tieback,CV,PIDMasterLoop,InHoldBit,InHoldValue). Executes only while EnableIn is true.	`PID(SEP_PIC_101,SEP_PT_101.Value,0,SEP_PCV_101_CMD,0,0,0)`
ALMA	—	Analog alarm with HH/H/L/LL limits, deadband, and on-delay (native alternative to compare-and-timer alarms).	Configured in the ALMA instruction dialog

NOTE
Studio 5000 v36 displays the v36 mnemonics. Acceptance of legacy mnemonics (GRT, MOV, etc.) when pasting Neutral Text is not confirmed for the plant version. Verify on bench before field use.

Low-use instructions (CMP, OSR, OSF, FAL, FSC, JMP, LBL) are listed in Appendix G. They are not used in interlocks.

Rules applied to every rung in this manual

- **One OTE per tag.** A tag written by an OTE is written in exactly one place. Use the Cross Reference tool to confirm before adding a rung.
- **Never write an input.** Physical inputs and SIS status bits (e.g., SIS_ESD_OK) are read-only in BPCS logic.
- **Health bits.** Run and trip paths use XIC of an _OK bit (1 = healthy). XIO of an _OK bit appears only in trip-detect and first-out rungs (Section 0.3).
- **Never drive a discrete output directly from an OTL.** Latched bits survive prescan and power cycles; the output OTE must re-evaluate its permissives every scan.
- **Gate remote and HMI data.** Data from MSG reads or HMI writes is used only behind a communications-healthy condition (Snippet 8, Pattern A8).

Timer status bits

Bit	TON	TOF	RTO
.EN	Rung-in is true	Rung-in is true	Rung-in is true
.TT	Timing (rung true, .ACC < .PRE)	Timing (rung false, .ACC < .PRE)	Timing (rung true, .ACC < .PRE)
.DN	.ACC ≥ .PRE with rung still true	1 while rung true; clears when .ACC reaches .PRE after rung goes false	.ACC ≥ .PRE; stays set until RES
Rung false	.ACC and .DN reset	Timing starts	.ACC held

CAUTION — Prescan, Power-Up, and Communications Loss
Prescan runs on every Program-to-Run transition and power-up. During prescan:
- OTE bits are cleared.
- OTL/OTU bits keep their last value.
- ONS storage bits are set (no first-scan pulse).
- TON timers are reset; RTO status bits are cleared but .ACC is kept.
- DINT and REAL values are kept.

On HMI communications loss, HMI-written request bits stay at their last value. On MSG failure, destination data goes stale and keeps its last value. Clear HMI requests in logic after use (Pattern A1/A7) and gate MSG data on communications health (Snippet 8). Review the analog output fault and program-mode states for each loop.

5.2 Annotated Ladder Examples

Each example lists its tags, the rungs in Neutral Text, and the operating notes. Tag names follow Section 0.3. Setpoints are examples only.

Snippet 1: Motor Start/Stop with Seal-In (Fail-Safe Health Bits)

Tag	Type	Meaning
START_PB	BOOL input	NO momentary start pushbutton; 1 while pressed
STOP_OK	BOOL input	NC stop circuit; 1 = not pressed and intact
OL_OK	BOOL input	Overload relay NC contact; 1 = healthy
E_STOP_OK	BOOL input	Auxiliary NC contact of the hardwired E-stop circuit; 1 = released
Motor_Run	BOOL output	Starter coil command

Rung 0 – Run with seal-in
```
[XIC(START_PB) ,XIC(Motor_Run) ]XIC(E_STOP_OK)XIC(STOP_OK)XIC(OL_OK)OTE(Motor_Run);
```

A broken stop wire, a tripped overload, or a pressed E-stop sets its _OK bit to 0 and drops the output. The E-stop is wired into the hardwired coil circuit or safety relay (Section 5.4); the PLC input only monitors it. On a Program-to-Run transition the OTE is cleared and the motor does not restart until START_PB is pressed again.

Snippet 2: TON Timer — Pump Delay Start

Rung 0 – 5 s start delay (permissives inside the timed path)
```
XIC(AutoMode)XIC(Pump_Cmd)XIC(P101.STOP_OK)XIC(P101.OL_OK)TON(Delay_TON,5000,0);
```
Rung 1 – Start output (permissives repeated on the output)
```
XIC(Delay_TON.DN)XIC(P101.STOP_OK)XIC(P101.OL_OK)OTE(Pump_Start_Out);
```

Loss of any permissive resets the timer and drops the output in the same scan. PRE is in milliseconds (5000 = 5 s).

Snippet 3: CTU Counter — Compressor Cycle Count

Rung 0 – Count run starts (CTU counts on the rung false-to-true transition, tracked by .CU)
```
XIC(Comp_Run_FB)CTU(Comp_Cycle_CTU,1000,0);
```
Rung 1 – Maintenance due
```
XIC(Comp_Cycle_CTU.DN)OTE(Comp_Maint_Due);
```
Rung 2 – Reset from HMI (only when due)
```
XIC(HMI_Maint_Reset)XIC(Comp_Maint_Due)RES(Comp_Cycle_CTU);
```
Rung 3 – Clear the HMI request
```
XIC(HMI_Maint_Reset)OTU(HMI_Maint_Reset);
```

Debounce Comp_Run_FB (for example with a short TON) if the feedback contact chatters; each chatter would otherwise count as a start. Counter .ACC is retained through power cycles.

Snippet 4: PID Loop — Separator Pressure Control

The PID instruction executes only while its rung is true and computes correctly only when called at a fixed interval. Place it unconditionally in a periodic task (100 ms in this example) with .UPD equal to the task period (0.1 s). Use .SWM (software manual) with .SO (software output) for manual mode; do not condition the PID rung to stop it.

Member	Example value	Meaning
.SP	200.0 PSIG (example)	Setpoint in engineering units
.MAXI / .MINI	300.0 / 0.0	Unscaled PV range (PV already in PSIG)
.MAXS / .MINS	300.0 / 0.0	Engineering-unit scaling range
.MAXO / .MINO	100.0 / 0.0	Output limits (%)
.MAXCV / .MINCV	100.0 / 0.0	CV range written to the output tag
.UPD	0.1	Loop update time (s) = periodic task period
.PE	0	0 = independent gains (Kp unitless, Ki in 1/s, Kd in s); 1 = dependent (KP = Kc, KI = Ti in min/repeat, KD = Td in min)
.CA	Per valve action	0 = reverse (E = SP − PV); 1 = direct (E = PV − SP). For an outlet PCV where a higher command opens the valve, PV above SP must raise the output (.CA = 1).
.SWM / .SO	—	Software manual mode / manual output (%)

Rung 0 – Output tracking while in auto (bumpless entry to manual)
```
XIO(SEP_PIC_101.SWM)MOVE(SEP_PIC_101.OUT,SEP_PIC_101.SO);
```
Rung 1 – Manual request from HMI
```
XIC(HMI_PIC101_ManReq)OTL(SEP_PIC_101.SWM);
```
Rung 2 – Auto request from HMI (only with a good PV)
```
XIC(HMI_PIC101_AutoReq)XIO(SEP_PT_101.Bad)OTU(SEP_PIC_101.SWM);
```
Rung 3 – Force manual on bad PV
```
XIC(SEP_PT_101.Bad)OTL(SEP_PIC_101.SWM);
```
Rung 4 – Clear HMI requests
```
[XIC(HMI_PIC101_ManReq) ,XIC(HMI_PIC101_AutoReq) ][OTU(HMI_PIC101_ManReq) ,OTU(HMI_PIC101_AutoReq) ];
```
Rung 5 – PID (unconditional)
```
PID(SEP_PIC_101,SEP_PT_101.Value,0,SEP_PCV_101_CMD,0,0,0);
```

Software manual mode is bumpless toward manual because .SO tracks .OUT while in auto; on return to auto the PID resumes from the current output. Setpoint tracking (PV tracking) in manual is configured on the PID setup tab where required by the loop philosophy.

NOTE
PID scaling members, .CA selection, and the bumpless transfer behavior shown here. Verify on bench before field use.

Snippet 5: GT Compare and Alarm — High Pressure (On-Delay and Deadband)

Example values: SEP_PT_101.PAH_SP = 250 PSIG, PAH_DB = 5 PSIG, PAH on-delay 5 s. The PAHH trip setpoint (270 PSIG) is handled by the trip string in Snippet 6 and Pattern A3. See Pattern A2 for the general form; ALMA is the native alternative.

Rung 0 – Clear point = SP − deadband
```
SUB(SEP_PT_101.PAH_SP,SEP_PT_101.PAH_DB,SEP_PT_101.PAH_Clr);
```
Rung 1 – On-delay above setpoint (good PV only)
```
XIO(SEP_PT_101.Bad)GT(SEP_PT_101.Value,SEP_PT_101.PAH_SP)TON(SEP_PT_101.PAH_On,5000,0);
```
Rung 2 – Alarm with deadband seal-in
```
[XIC(SEP_PT_101.PAH_On.DN) ,XIC(SEP_PT_101.PAH)GT(SEP_PT_101.Value,SEP_PT_101.PAH_Clr) ]XIO(SEP_PT_101
.Bad)OTE(SEP_PT_101.PAH);
```

The alarm is not latched in the PLC; acknowledgment is handled by the HMI alarm system (ISA-18.2). A bad PV suppresses the process alarm and raises the bad-quality alarm instead (Pattern A3).

Snippet 6: BPCS Trip / Permissive Logic (Not a SIL ESD)

WARNING — BPCS Only
This is BPCS equipment-protection logic. It is not an SIS function and is never copied into an ESD or SIS controller (Section 0.4). SIS status is read as SIS_ESD_OK and never written.

Rung 0 – Trip causes healthy (all _OK bits, no PAHH trip)
```
XIC(SEP_PAHH_OK)XIC(COMP_PAHH_OK)XIC(FG_Z3_OK)XIC(MAN_TRIP_OK)XIO(SEP_PT_101.PAHH_Trip)OTE(SEP_Causes_
OK);
```
Rung 1 – Trip (also on first scan: the system comes up tripped)
```
[XIO(SEP_Causes_OK) ,XIC(S:FS) ]OTU(SEP_Trip_OK);
```
Rung 2 – Reset only with all causes healthy
```
XIC(Area.Reset_Pulse)XIC(SEP_Causes_OK)OTL(SEP_Trip_OK);
```
Rung 3 – Trip cancels the open request
```
XIO(SEP_Trip_OK)OTU(SEP_XV_101.Open_Req);
```
Rung 4 – Solenoid (energize to open, de-energize to close)
```
XIC(SEP_Trip_OK)XIC(SEP_XV_101.Open_Req)OTE(SEP_XV_101.SOL);
```
Rung 5 – Trip alarm
```
XIO(SEP_Trip_OK)OTE(SEP_Trip_Alm);
```

After a reset the valve does not re-open automatically; the operator issues a new open request. Area.Reset_Pulse is generated by Pattern A1. The first-out capture in Pattern A6 is placed before Rung 1.

Snippet 7: Analog Scaling — 4–20 mA to Engineering Units

For a 1794-IE8 channel in 4–20 mA mode, the module reports 0 counts at 4 mA and 30840 counts at 20 mA. For a 0–300 PSIG transmitter, Gain = span / 30840.0 = 300.0 / 30840.0 and EU_Min = 0.0.

Rung 0 – Scale (good signal only; value holds last good reading when Bad)
```
XIO(SEP_PT_101.Bad)MUL(SEP_PT_101_Raw,SEP_PT_101.Gain,SEP_PT_101.Calc)ADD(SEP_PT_101.Calc,SEP_PT_101.E
U_Min,SEP_PT_101.Value);
```

- **1794-IE8:** reads 0 at or below 4 mA. A wire break cannot be detected from the count value; use the channel under-range status bit for bad quality (Pattern A3).
- **1756-IF8:** use floating-point mode and configure the channel scaling in the module so the tag is already in engineering units. In integer mode, 0–20.58 mA maps to −32768…32767 counts (4 mA ≈ −20030 counts).
- **SCL** is available only in Function Block Diagram and Structured Text. **SCP** is not a native ControlLogix instruction. Ladder scaling uses MUL/ADD as shown, or a scaling AOI.

NOTE
IE8 under-range status member names (taken from the module profile in the I/O tree). Verify on bench before field use.

Snippet 8: MSG Instruction — Read Tag from Remote Compressor PLC

Configure Comp_MSG_Read as CIP Data Table Read. Add the peer controller to the I/O tree and browse to it for the communication path. Comp_Speed_Remote is the destination tag.

Rung 0 – Poll timer (1 s, only when no message is in progress)
```
XIO(Comp_MSG_Read.EN)TON(Comp_MSG_Poll,1000,0);
```
Rung 1 – Send
```
XIC(Comp_MSG_Poll.DN)MSG(Comp_MSG_Read);
```
Rung 2 – Stale-data timer (resets each time .DN is seen)
```
XIO(Comp_MSG_Read.DN)TON(Comp_MSG_Stale,5000,0);
```
Rung 3 – Communications fault
```
XIC(Comp_MSG_Stale.DN)OTE(Comp_Comms_Fault);
```
Rung 4 – Use remote data only when healthy
```
XIO(Comp_Comms_Fault)MOVE(Comp_Speed_Remote,Comp_Speed);
```

Every use of the remote data is gated with XIO(Comp_Comms_Fault). Error information is read from the MESSAGE members .ER, .ERR, and .EXERR (no GSV needed). Common codes: 16#0001 connection failure (extended 16#0204 unconnected timeout; 16#0203 connection timeout), 16#0004 IOI syntax error, 16#0005 destination unknown, 16#0006 packet space. Default .UnconnectedTimeout is 30 s; connected messages time out at .ConnectionRate (7.5 s) × .TimeoutMultiplier (4).

NOTE

MSG path and error codes on the 1769-L33ER bench controller. Verify on bench before field use.

5.3 Advanced Ladder Patterns

JSR/SBR/RET Parameter Passing

MainRoutine – Rung 0: call with two inputs and one return
```
XIC(SEP_Running)JSR(SEP_LEVEL_CONTROL,2,SEP_LT_101.Value,SEP_LIC_101.SP,SEP_LIC_101.OUT);
```
SEP_LEVEL_CONTROL – Rung 0: receive inputs
```
SBR(PV_Local,SP_Local);
```
SEP_LEVEL_CONTROL – last rung: return result
```
RET(OUT_Local);
```

Parameter passing is shown in simplified form. Encapsulating the logic in one routine means the algorithm is edited in one place. For new reusable logic, an AOI is generally preferred.

NOTE

JSR/SBR/RET parameter syntax in Neutral Text. Verify on bench before field use.

State Machine Pattern — Compressor Sequencer

States: 0 = Idle, 1 = Purge, 2 = Startup, 3 = Running, 4 = Shutdown. Transition rungs write Comp_Next; the last transition rung copies Comp_Next to Comp_State, so each state executes for a full scan before the next transition. Abort conditions are evaluated after the normal transitions so they take priority.

Rung 0 – First scan: Idle
```
XIC(S:FS)[MOVE(0,Comp_State) ,MOVE(0,Comp_Next) ];
```
Rung 1 – Idle → Purge
```
EQ(Comp_State,0)XIC(Comp_Start_Req)XIC(Comp_Perm_OK)XIC(COMP_Trip_OK)MOVE(1,Comp_Next);
```
Rung 2 – Purge timer (60 s)
```
EQ(Comp_State,1)TON(Comp_Purge_TON,60000,0);
```
Rung 3 – Purge → Startup
```
EQ(Comp_State,1)XIC(Comp_Purge_TON.DN)MOVE(2,Comp_Next);
```
Rung 4 – Startup timeout timer (30 s)
```
EQ(Comp_State,2)TON(Comp_Start_TON,30000,0);
```
Rung 5 – Startup → Running at speed
```
EQ(Comp_State,2)GE(Comp_Speed,Comp_Speed_SP)MOVE(3,Comp_Next);
```
Rung 6 – Startup timeout → Shutdown
```
EQ(Comp_State,2)XIC(Comp_Start_TON.DN)LT(Comp_Speed,Comp_Speed_SP)MOVE(4,Comp_Next);
```
Rung 7 – Shutdown timer (30 s)
```
EQ(Comp_State,4)TON(Comp_Shutdn_TON,30000,0);
```
Rung 8 – Shutdown → Idle
```
EQ(Comp_State,4)XIC(Comp_Shutdn_TON.DN)MOVE(0,Comp_Next);
```
Rung 9 – Abort: trip or stop request in states 1–3 → Shutdown
```
[XIO(COMP_Trip_OK) ,XIC(Comp_Stop_Req) ]GT(Comp_State,0)LT(Comp_State,4)MOVE(4,Comp_Next);
```
Rung 10 – Commit transition
```
MOVE(Comp_Next,Comp_State);
```
Rung 11 – Action: purge command
```
EQ(Comp_State,1)XIC(COMP_Trip_OK)OTE(Comp_Purge_Cmd);
```
Rung 12 – Action: run command in Startup and Running
```
[EQ(Comp_State,2) ,EQ(Comp_State,3) ]XIC(COMP_Trip_OK)OTE(Comp_Run_Cmd);
```

Each output is written by one action rung only. Comp_Start_Req and Comp_Stop_Req are one-scan request bits cleared after use (Pattern A1). State-based TON timers reset automatically when the state is left.

Structured Text Example — Dew Point Calculation (Structure Only)

CAUTION — Placeholder Formula

The correlation in this routine is a placeholder formula, not for use. It demonstrates Structured Text syntax only (Section 0.9). Use the plant-approved correlation.

```
(* Calc_DewPoint — Structured Text structure example
   Tags (REAL): T_Gas_F, P_Gas_PSIA, W_Content, Dew_Point_C, Dew_Point_F,
                Dew_Point_HiAlarm_SP   (BOOL): Calc_Fault, Dew_Point_Hi_Alarm *)

Calc_Fault := 0;                        (* Clear status every execution *)
Dew_Point_Hi_Alarm := 0;

T_Gas_C := (T_Gas_F - 32.0) * 5.0 / 9.0;

IF (W_Content > 0.0) AND (P_Gas_PSIA > 0.0) THEN
    (* PLACEHOLDER FORMULA — NOT FOR USE *)
    Dew_Point_C := -40.0 + (W_Content / P_Gas_PSIA * 9500.0);
    IF Dew_Point_C > T_Gas_C THEN      (* Cannot exceed gas temperature *)
        Dew_Point_C := T_Gas_C;
    END_IF;
    IF Dew_Point_C  Dew_Point_HiAlarm_SP THEN
        Dew_Point_Hi_Alarm := 1;
    END_IF;
ELSE
    Calc_Fault := 1;                   (* No valid measurement; Dew_Point_F holds last value *)
END_IF;
```

Function Block Diagram (FBD) — PID Loop Description

In Function Block Diagram (FBD), the PID loop appears as graphical blocks with wires connecting them, similar to a P&ID. A typical FBD pressure loop uses:

- **Input reference (IREF):** the 1756-IF8 channel in floating-point mode (already in PSIG), or raw counts into an **SCL** block for scaling → SEP_PT_101.Value.
- **ALMA:** analog alarm block comparing the PV to HH/H/L/LL limits with deadband and on-delay → alarm bits.

- **PIDE:** enhanced PID block (FBD/ST only) receiving PV and SP → CV (0–100 %).
- **FGEN (optional):** function generator to characterize the control valve → linearized command.
- **Output reference (OREF):** writes the command to the 1756-OF8 channel (floating-point mode scales 0–100 % to 4–20 mA) → SEP_PCV_101 positioner.

5.4 Standard Ladder Patterns

The patterns below are the standard building blocks referenced throughout this manual. Tag names are examples; members such as SEP_PT_101.Bad are UDT members.

Pattern A1: HMI Reset Pulse (One Scan)

HMI pushbuttons write a request bit. The PLC converts it to a one-scan pulse and clears the request so a lost HMI connection cannot leave it set.

Rung 0 – One-scan reset pulse
```
XIC(Area.HMI_Reset)ONS(Area.Reset_ONS)OTE(Area.Reset_Pulse);
```
Rung 1 – Clear the HMI request
```
XIC(Area.HMI_Reset)OTU(Area.HMI_Reset);
```

Pattern A2: Analog Alarm with On-Delay and Deadband

The alarm activates after the PV stays above the setpoint for the on-delay and clears only when the PV falls below SP − deadband. ALMA is the native alternative.

Rung 0 – Clear point
```
SUB(SEP_PT_101.PAH_SP,SEP_PT_101.PAH_DB,SEP_PT_101.PAH_Clr);
```
Rung 1 – On-delay
```
XIO(SEP_PT_101.Bad)GT(SEP_PT_101.Value,SEP_PT_101.PAH_SP)TON(SEP_PT_101.PAH_On,5000,0);
```
Rung 2 – Alarm with deadband
```
[XIC(SEP_PT_101.PAH_On.DN) ,XIC(SEP_PT_101.PAH)GT(SEP_PT_101.Value,SEP_PT_101.PAH_Clr) ]XIO(SEP_PT_101
.Bad)OTE(SEP_PT_101.PAH);
```

Pattern A3: Bad-Quality Detection and Trip Behavior

Bad quality is derived from the channel status (under-range, over-range, connection fault) with a short filter. Each trip input is configured to trip or not trip on bad quality (BadTrips), per the C&E philosophy. For a 1794-IE8, OverRange can be taken as GT(SEP_PT_101_Raw,31804) (≈ 20.5 mA).

Rung 0 – Filter channel faults (2 s)
```
[XIC(SEP_PT_101.UnderRange) ,XIC(SEP_PT_101.OverRange) ,XIC(SEP_PT_101.Conn_Fault) ]TON(SEP_PT_101.Bad
_On,2000,0);
```
Rung 1 – Bad quality (connection fault is immediate)
```
[XIC(SEP_PT_101.Bad_On.DN) ,XIC(SEP_PT_101.Conn_Fault) ]OTE(SEP_PT_101.Bad);
```
Rung 2 – PAHH on-delay (good PV)
```
XIO(SEP_PT_101.Bad)GT(SEP_PT_101.Value,SEP_PT_101.PAHH_SP)TON(SEP_PT_101.PAHH_On,1000,0);
```
Rung 3 – PAHH trip (or bad quality when configured to trip)
```
[XIC(SEP_PT_101.PAHH_On.DN) ,XIC(SEP_PT_101.Bad)XIC(SEP_PT_101.BadTrips) ]OTE(SEP_PT_101.PAHH_Trip);
```

Pattern A4: Valve Position Discrepancy (Fail to Open / Fail to Close)

PRE = full stroke time × 1.5. The same pattern detects a motor fail-to-start using XIC(Run_Cmd)XIO(Run_FB).

Rung 0 – At commanded position
```
[XIC(XV_201.Open_Cmd)XIC(XV_201.ZSO)XIO(XV_201.ZSC) ,XIO(XV_201.Open_Cmd)XIC(XV_201.ZSC)XIO(XV_201.ZSO
) ]OTE(XV_201.AtPos);
```
Rung 1 – Discrepancy timer (15 s example)
```
XIO(XV_201.AtPos)TON(XV_201.Disc_TON,15000,0);
```
Rung 2 – Fail to open
```
XIC(XV_201.Disc_TON.DN)XIC(XV_201.Open_Cmd)OTL(XV_201.FTO_Alm);
```
Rung 3 – Fail to close
```
XIC(XV_201.Disc_TON.DN)XIO(XV_201.Open_Cmd)OTL(XV_201.FTC_Alm);
```
Rung 4 – Reset when at position
```
XIC(Area.Reset_Pulse)XIC(XV_201.AtPos)[OTU(XV_201.FTO_Alm) ,OTU(XV_201.FTC_Alm) ];
```

Pattern A5: Start Permissives vs Running Trips

Permissives are required only to start. Trips are required to start and to keep running.

Rung 0 – Start permissives
```
XIC(P101.Suct_Valve_Open)XIC(P101.Seal_Flush_OK)GT(SEP_LT_101.Value,P101.Min_Lvl)OTE(P101.Perm_OK);
```
Rung 1 – Running trips
```
XIC(P101.STOP_OK)XIC(P101.OL_OK)XIC(SEP_Trip_OK)OTE(P101.Trip_OK);
```
Rung 2 – Run command
```
[XIC(P101.Start_Pulse)XIC(P101.Perm_OK) ,XIC(P101.Run_Cmd) ]XIO(P101.Stop_Pulse)XIC(P101.Trip_OK)OTE(P
101.Run_Cmd);
```

Pattern A6: First-Out Capture

Records which cause tripped first. Place these rungs before the trip latch (Snippet 6, Rung 1).

Rung 0 – PAHH first
```
XIO(SEP_PAHH_OK)XIO(SEP_FO.Captured)[OTL(SEP_FO.PAHH) ,OTL(SEP_FO.Captured) ];
```
Rung 1 – LAHH first
```
XIO(SEP_LAHH_OK)XIO(SEP_FO.Captured)[OTL(SEP_FO.LAHH) ,OTL(SEP_FO.Captured) ];
```
Rung 2 – Manual trip first
```
XIO(MAN_TRIP_OK)XIO(SEP_FO.Captured)[OTL(SEP_FO.MAN) ,OTL(SEP_FO.Captured) ];
```
Rung 3 – Reset with causes healthy
```
XIC(Area.Reset_Pulse)XIC(SEP_Causes_OK)
[OTU(SEP_FO.PAHH) ,OTU(SEP_FO.LAHH) ,OTU(SEP_FO.MAN) ,OTU(SEP_FO.Captured) ];
```

Pattern A7: Auto / Manual / Off Mode Handling with HOA

HMI mode and start/stop requests are one-scan requests cleared by the last rung. The field HOA switch in AUTO is required for any PLC command. On first scan the pump is placed in Off.

Rung 0 – Auto request
```
XIC(P101.HMI_Auto)[OTL(P101.Auto) ,OTU(P101.Man) ];
```
Rung 1 – Manual request
```
XIC(P101.HMI_Man)[OTU(P101.Auto) ,OTL(P101.Man) ];
```
Rung 2 – Off request or first scan
```
[XIC(P101.HMI_Off) ,XIC(S:FS) ][OTU(P101.Auto) ,OTU(P101.Man) ];
```
Rung 3 – Run command
```
[XIC(P101.Auto)XIC(P101.Auto_Req) ,XIC(P101.Man)
[XIC(P101.HMI_Start)XIC(P101.Perm_OK) ,XIC(P101.Run_Cmd) ]XIO(P101.HMI_Stop) ]XIC(P101.Trip_OK)XIC(P10
1.HOA_Auto)OTE(P101.Run_Cmd);
```
Rung 4 – Clear HMI requests
```
[XIC(P101.HMI_Auto) ,XIC(P101.HMI_Man) ,XIC(P101.HMI_Off) ,XIC(P101.HMI_Start) ,XIC(P101.HMI_Stop) ]
[OTU(P101.HMI_Auto) ,OTU(P101.HMI_Man) ,OTU(P101.HMI_Off) ,OTU(P101.HMI_Start) ,OTU(P101.HMI_Stop) ];
```

Pattern A8: Communications Heartbeat

Each controller increments a counter every second; the peer counter must change within 5 s or a communications fault is declared.

Rung 0 – 1 s tick
```
XIO(HB.Tick_TON.DN)TON(HB.Tick_TON,1000,0);
```
Rung 1 – Increment local heartbeat
```
XIC(HB.Tick_TON.DN)ADD(HB.PLC_Count,1,HB.PLC_Count);
```
Rung 2 – Detect peer change
```
NE(HB.Peer_Count,HB.Peer_Last)MOVE(HB.Peer_Count,HB.Peer_Last)OTE(HB.Peer_Changed);
```
Rung 3 – Stale timer
```
XIO(HB.Peer_Changed)TON(HB.Peer_Stale,5000,0);
```
Rung 4 – Heartbeat fault
```
XIC(HB.Peer_Stale.DN)OTE(HB.Peer_Fault);
```

Pattern A9: Level Start/Stop with Hysteresis

The pump auto request latches at the high level and unlatches at the low level or on bad quality, so the pump does not cycle around a single setpoint.

Rung 0 – Request on at On_SP
```
GE(SEP_LT_101.Value,SEP_LT_101.On_SP)XIO(SEP_LT_101.Bad)OTL(P101.Auto_Req);
```
Rung 1 – Request off at Off_SP or bad PV
```
[LE(SEP_LT_101.Value,SEP_LT_101.Off_SP) ,XIC(SEP_LT_101.Bad) ]OTU(P101.Auto_Req);
```

NOTE

Conn_Fault and channel status member names in Patterns A3 depend on the module profile. Verify on bench before field use.

E-Stop and Master Control Relay Wiring

Element	Wiring	PLC role
E-stop pushbuttons	NC contacts in series in the hardwired coil circuit or safety relay input	None in the trip path; the hardwired circuit removes power directly
Safety relay / MCR	Monitored safety relay or master control relay removes output power	Optional status input
Auxiliary contact	NC auxiliary contact to a DI (E_STOP_OK)	Monitoring, alarm, and run-command drop: `XIC(E_STOP_OK)` in the run path
Reset	Hardwired reset on the safety relay	PLC reset (Pattern A1) does not bypass the hardwired reset

Run path with E-stop monitoring
```
[XIC(START_PB) ,XIC(Motor_Run) ]XIC(E_STOP_OK)XIC(STOP_OK)XIC(OL_OK)OTE(Motor_Run);
```

UDT Templates

UDT	Member	Type	Meaning
MTR_BASIC	Run_Cmd	BOOL	Run command (one OTE)
	Run_FB	BOOL	Run feedback from starter auxiliary contact
	STOP_OK / OL_OK	BOOL	Health inputs (1 = healthy)
	Perm_OK / Trip_OK	BOOL	Start permissives / running trips (Pattern A5)
	Auto / Man / HOA_Auto	BOOL	Mode state and field HOA in AUTO (Pattern A7)
	HMI_Start / HMI_Stop / HMI_Auto / HMI_Man / HMI_Off	BOOL	HMI one-scan requests, cleared in logic
	FTS_TON / FTS_Alm	TIMER / BOOL	Fail-to-start discrepancy (Pattern A4 form)
XV_BASIC	Open_Req / Open_Cmd	BOOL	Open request and resulting command
	SOL	BOOL	Solenoid output (de-energize to close)
	ZSO / ZSC	BOOL	Open / closed limit switches
	AtPos	BOOL	At commanded position
	Disc_TON	TIMER	Discrepancy timer (PRE = stroke time × 1.5)
	FTO_Alm / FTC_Alm	BOOL	Fail-to-open / fail-to-close alarms

NOTES

Section 6: AVEVA (Wonderware) InTouch HMI

6.1 AVEVA (Wonderware) InTouch Overview

Wonderware InTouch (now AVEVA InTouch HMI) is one of the most widely deployed HMI platforms in the oil and gas industry. It is encountered in two primary forms at the gas plant:

> **NOTE**
> AVEVA Plant SCADA (formerly Citect SCADA) is a separate AVEVA product and is not covered in this manual.

InTouch (Classic / Stand-Alone) vs. System Platform (OMI)

Feature	InTouch Classic	AVEVA System Platform (OMI)
Architecture	Single-node or peer-to-peer; .application file-based	Distributed; Galaxy repository; server-client architecture
Tag database	Local to each InTouch application; .dbf file	Centralized in Galaxy (SQL Server); all nodes share one tag database
Redundancy	Manual failover or I/O server redundancy only	Native server redundancy; Application Server failover
Scalability	Up to ~60,000 I/O tags per node	Virtually unlimited; enterprise-scale
Best use	Small-to-medium plants; standalone panels; legacy upgrades	Large, multi-unit plants; enterprise integration; historian integration
Development tool	WindowMaker	ArchestrA IDE (Galaxy Repository browser)
Runtime environment	WindowViewer	InTouchViewApp or OMI viewer

WindowMaker vs. WindowViewer

- **WindowMaker:** The development environment for InTouch Classic. Used to create graphic displays, define tagnames, write QuickScripts, configure animation links, and set up alarm groups. Development must be done offline or on a development workstation (not while WindowViewer is running on the same PC, in most configurations).
- **WindowViewer:** The runtime environment for InTouch. Operators use WindowViewer; it executes graphic displays, communicates with PLCs via I/O servers (OPC, DDE, SuiteLink), and runs all scripts and animations. WindowViewer is always running on operator consoles.

Galaxy Repository (System Platform)

The **Galaxy** is the central database of all configuration objects in a System Platform installation. All templates, instances, scripts, graphic symbols, and communication configuration are stored in a SQL Server database called the Galaxy repository. Changes made in the ArchestrA IDE are deployed to target nodes (Application Server, ViewApp nodes). This single-source-of-truth model eliminates the inconsistency of maintaining separate tag databases on multiple HMI nodes.

ArchestrA Object Model Key Objects

Object Type	Description	Purpose
Area	Logical grouping object (e.g., SEP_Area, COMP_Area)	Organizes equipment objects by plant area; owns child objects
ApplicationServer (AppEngine)	Runs on a server PC; hosts all instrument and equipment objects	Executes tag data acquisition, alarming, scripting, historian logging
WinPlatform	Platform object for a Windows-based HMI node	Hosts ViewApp or InTouchViewApp instances on operator consoles
ViewApp	Graphics display host object	Manages startup, resolution, and navigation for HMI screens on a node
InTouchViewApp	InTouch-based ViewApp for legacy InTouch applications	Allows classic InTouch apps to run within System Platform framework

6.2 HMI Tag List Template

The HMI tag list defines every process variable visible to operators: its source PLC tag, engineering range, alarm setpoints, and deadband. Verify this list against the PLC tag database after any modification.

HMI Tag	Tag Type	PLC Tag Source	Description	Units	Min	Max	Alarm LoLo	Alarm Lo	Alarm Hi	Alarm HiHi	Deadband	Notes
SEP_PT_101	Analog	SEP_PT_101.Value	Inlet Sep Pressure	PSIG	0	300	100	150	250	270	5.0	PAHH = BPCS trip (example); SIS PAHH is separate
COMP_PT_201	Analog	COMP_PT_201.Value	Compressor Suction Press	PSIG	0	600	50	80	520	550	8.0	PALL trips compressor
DEHYD_PT_301	Analog	DEHYD_PT_301.Value	Glycol Contactor Press	PSIG	800	1440	820	840	1400	1430	10.0	Monitor only
FRAC_PT_401	Analog	FRAC_PT_401.Value	Deethanizer Press	PSIG	150	500	160	170	480	495	5.0	PSV at 500 PSIG
METER_PT_501	Analog	METER_PT_501.Value	Custody Transfer Meter Press	PSIG	0	1200	—	—	—	—	2.0	Metering — alarm separate
SEP_TT_101	Analog	SEP_TT_101.Value	Inlet Sep Temperature	°F	−50	250	30	40	180	200	2.0	Hydrate risk if low
COMP_TT_201	Analog	COMP_TT_201.Value	Compressor Discharge Temp	°F	50	400	—	—	350	380	3.0	TAHH trips compressor
AMINE_TT_301	Analog	AMINE_TT_301.Value	Amine Regen Reboiler Temp	°F	200	300	220	230	270	280	2.0	Optimal 246–257°F
FRAC_TT_401	Analog	FRAC_TT_401.Value	Deethanizer Overhead Temp	°F	−50	100	−35	−30	−5	0	1.5	Product purity indicator
SEP_LT_101	Analog	SEP_LT_101.Value	Inlet Sep Liquid Level	%	0	100	10	15	80	90	2.0	LAHH closes inlet, opens drain
AMINE_LT_201	Analog	AMINE_LT_201.Value	Amine Absorber Sump Level	%	0	100	15	20	75	85	2.0	Critical for pump protection
GLYCOL_LT_301	Analog	GLYCOL_LT_301.Value	TEG Storage Tank Level	%	0	100	20	25	90	95	2.0	Inventory management
FRAC_LT_401	Analog	FRAC_LT_401.Value	Deethanizer Sump Level	%	0	100	10	15	80	90	2.0	Controls bottom product flow
SEP_FT_101	Analog	SEP_FT_101.Value	Inlet Sep Gas Flow	MMSCFD	0	50	2	3	45	48	0.5	Total plant throughput indicator
COMP_FT_201	Analog	COMP_FT_201.Value	Compressor Recycle Flow	MMSCFD	0	20	—	—	15	18	0.2	Anti-surge monitor
METER_FT_501	Analog	METER_FT_501.Value	Custody Transfer Sales Flow	MMSCFD	0	60	—	—	—	—	0.1	Fiscal metering — RRC regulated
SEP_XV_101	Discrete	SEP_XV_101	Sep Inlet Block Valve	Open/Closed	0	1	—	—	—	—	—	Closes on BPCS trip
COMP_XV_201	Discrete	COMP_XV_201	Compressor Suction Block Valve	Open/Closed	0	1	—	—	—	—	—	Permissive for compressor start
ESD_XV_001	Discrete	ESD_XV_001	Main Plant ESD Block Valve	Open/Closed	0	1	—	—	—	—	—	SIS final element — fail-closed
SEP_PCV_101	Analog	SEP_PCV_101_CMD	Sep Pressure Ctrl Valve Position	%	0	100	—	—	—	—	1.0	Modulating — PID output
AMINE_LCV_201	Analog	AMINE_LCV_201_CMD	Amine Rich Amine LCV Position	%	0	100	—	—	—	—	1.0	Level control output

HMI Tag	Tag Type	PLC Tag Source	Description	Units	Min	Max	Alarm LoLo	Alarm Lo	Alarm Hi	Alarm HiHi	Deadband	Notes
COMP_MTR_201	Discrete	COMP_MTR_201.Running	Compressor Motor Status	Run/Stop	0	1	—	—	—	—	—	Feedback from MCC
PUMP_MTR_101	Discrete	PUMP_MTR_101.Running	Transfer Pump Motor Status	Run/Stop	0	1	—	—	—	—	—	Feedback from MCC starter
AMINE_AT_301	Analog	AMINE_AT_301.Value	Amine Outlet H2S Conc	ppm	0	20	—	—	3	4	0.1	Pipeline spec: <4 ppm
GAS_AT_401	Analog	GAS_AT_401.Value	Sales Gas BTU Content	BTU/SCF	900	1150	940	950	1120	1130	1.0	Pipeline spec: 967–1100 BTU
SIS_ESD_OK	Discrete	SIS_ESD_OK	SIS ESD Healthy Status (read-only)	Healthy/Tripped	0	1	—	—	—	—	—	0 = SIS has tripped — critical
FG_Z3_OK	Discrete	FG_Z3_OK	F&G Zone 3 Alarm	Normal/Alarm	0	1	—	—	—	—	—	0 = alarm active (fail-safe)
COMP_PAHH_OK	Discrete	COMP_PAHH_OK	Compressor PAHH Status	Normal/Trip	0	1	—	—	—	—	—	Hardwired input; 1 = healthy
Comp_Efficiency	Analog	Comp_Efficiency	Compressor Isentropic Efficiency	%	0	100	—	—	—	—	1.0	Calculated tag — PLC math routine
Sep_Retention_Time	Analog	Sep_Retention_Time	Inlet Sep Estimated Retention Time	min	0	60	2	3	45	55	0.5	Calculated from level and flow

6.3 Alarm Management

ISA-18.2 Alarm Philosophy Summary

ISA-18.2 (Management of Alarm Systems for the Process Industries) governs alarm system design, rationalization, monitoring, and lifecycle management. Its core philosophy: every alarm must require operator action, must be unique, and must not exceed the operator response capacity. Benchmark rates per operator console: about 6 alarms per hour (1 per 10 minutes) is very likely acceptable; about 12 per hour (2 per 10 minutes) is the maximum manageable. An alarm flood is more than 10 alarms in 10 minutes.

Alarm Priority Levels

Priority	Definition	Response Time	Annunciation	Example
Critical (Priority 1)	Immediate hazard to people, environment, or major equipment — ESD-level event	<5 minutes	Audible (distinct tone), full-screen popup, red flashing on HMI	PAHH trip, fire detection, H2S >10 ppm
High (Priority 2)	Significant process exceedance — product quality, equipment protection, regulatory limit	5–15 minutes	Audible, red steady on HMI alarm banner	PAH, TAHH, LAHH, compressor vibration high
Medium (Priority 3)	Process deviation outside normal operating range — corrective action required	15–60 minutes	Audible (lower priority tone), yellow on alarm banner	PAL, TAL, analyzer high, pump seal leak
Low (Priority 4)	Advisory — process approaching a limit; operator should monitor	Next available opportunity	No audible; displayed only in alarm list; cyan or white on HMI	Instrument calibration due, maintenance timer expired

Alarm Rationalization Worksheet Template

Tag	Description	Cause	Consequence	Safeguards	Priority	Setpoint	Deadband	Operator Action
SEP-PT-101 PAHH	Inlet Sep High-High Pressure	Control valve fails closed; blockage downstream	Vessel over-pressure; BPCS trip and loss of throughput	PSV-101 (285 PSIG set = MAWP); BPCS trip closes XV-101; SIS PAHH (separate)	Critical (1)	270 PSIG (example)	5 PSIG	Investigate PCV position immediately; check downstream; prepare for BPCS trip
AMINE-AT-301 PAH	H2S in Treated Gas High	Amine degradation; low amine flow; high inlet H2S	Off-spec gas to pipeline; regulatory violation; pipeline tariff penalty	Analyzer redundancy; manual grab sample; operator response	High (2)	3 ppm	0.2 ppm	Check amine circulation rate, reboiler temp; notify gas control; take grab sample
[Tag]	[Description]	[Root cause of upset]	[What happens if operator does not respond]	[Independent protection layers already in place]	[1–4]	[value + units]	[value + units]	[specific, actionable response]

Common Bad Actor Alarms and Suppression Strategies

- **Chattering alarms:** Alarm activates and clears repeatedly within a short period (e.g., a level alarm oscillating ±2% around a setpoint). Solution: increase deadband, move setpoint further from normal operation, or add time delay filter.
- **Stale alarms:** Alarm that has been active for days or weeks without operator action. Solution: shelve with expiry time, investigate root cause, redesign setpoint or process.
- **Nuisance alarms during startup/shutdown:** Normal process transients exceed alarm limits during non-steady-state. Solution: implement state-based alarming (alarms active only in applicable operating mode).
- **Flood alarms:** A single process upset triggers dozens of alarms simultaneously (consequence alarms). Solution: rationalize to first-out alarm design; use alarm suppression for downstream alarms when root cause is identified.

6.4 HMI Screen Design Guidelines

Navigation Hierarchy

Effective HMI navigation follows a **four-level hierarchy** that allows operators to quickly drill from a plant-wide overview to a specific instrument detail without confusion:

1. **Level 1 — Plant Overview:** Single-screen birds-eye view of all major process units with key KPIs (throughput, pressures, alarms). Always accessible from any screen.
2. **Level 2 — Area Overview:** One screen per major process area (Separation, Compression, Dehydration, Amine, Fractionation, Metering). Shows all loops, equipment statuses, and active alarms in that area.
3. **Level 3 — Unit/Equipment:** Detailed P&ID-style mimic for one unit (e.g., Inlet Separator). Shows all instruments with live values, control valve positions, motor statuses, and setpoints.
4. **Level 4 — Faceplate / Detail:** Pop-up controller faceplate for a single loop — PV, SP, mode, output %, trend, alarm setpoints. Accessed by clicking on any instrument on a Level 3 screen.

ISA 101 Color Standards (Simplified)

Element	Display treatment	Meaning
Background	Light or medium gray (low saturation)	Neutral background; color is reserved for abnormal conditions
Process lines and vessels	Dark gray or black lines; no color coding by stream in normal state	Gas, liquid, and utility streams distinguished by labels and line style
Equipment normal (running, stopped, open, closed)	Shades of gray, filled vs. outlined shapes, and text state	Normal states are not colored
Alarm — Critical / HiHi	Red, with alarm shape and priority indicator; flashing until acknowledged	Immediate action required
Alarm — High priority	Distinct high-priority color (per site style guide), flashing until acknowledged	Prompt action required
Alarm — Medium / Low priority	Distinct priority colors (per site style guide)	Action required within the response time

Element	Display treatment	Meaning
Acknowledged alarm	Same priority color, steady (not flashing)	Operator has seen the alarm; condition still active
Return to normal, unacknowledged	Per site style guide (e.g., outlined priority indicator)	Condition cleared; awaiting acknowledgment

NOTE
Alarm colors are used only for alarm states and are not reused for any other purpose (for example, yellow is not used for both an alarm priority and equipment status). Confirm all colors with the site HMI style guide.

Recommended Screen List

Screen Name	Purpose	Key Objects	Navigation Links
PLANT_OVERVIEW	Full plant bird's-eye KPI dashboard	Total flow, main pressures, ESD status, active alarm count, equipment status summaries	→ All Area screens
SEP_OVERVIEW	Inlet separation area	Inlet separator graphic, PT-101, LT-101, LT-102, TT-101, XV-101, PCV-101, FT-101	← Plant; → SEP_DETAIL; → COMP_OVERVIEW
COMP_OVERVIEW	Compression area	Compressor graphic, PT-201, TT-201, VT-201, MTR-201 status, anti-surge loop, XV-201	← SEP; → DEHYD; → COMP_FACEPLATE
DEHYD_OVERVIEW	Glycol dehydration	Contactor, reboiler, TEG pump, PT-301, TT-301, LT-301, FCV-301	← COMP; → AMINE_OVERVIEW
AMINE_OVERVIEW	Amine treating unit	Absorber, regenerator, amine pump, AT-301 H2S, TT-301, LT-201, LCV-201	← DEHYD; → FRAC_OVERVIEW
FRAC_OVERVIEW	NGL fractionation	Deethanizer, depropanizer columns, PT-401, TT-401, LT-401, product quality AT	← AMINE; → METER_OVERVIEW
METER_OVERVIEW	Custody transfer metering	Meter runs, FT-501, PT-501, BTU analyzer AT-401, daily totals, contract limits	← FRAC; → PLANT_OVERVIEW
ESD_STATUS	ESD system status and bypass management	Cause and effect matrix display, bypass status, SIS I/O status, PAHH/LAHH/TAHH indicators	← Any screen via ESD button
ALARM_SUMMARY	Active alarm list — all areas	Sorted by priority; timestamp, tag, description, current value, setpoint, acknowledge button	← Any screen via Alarm button
TREND_OVERVIEW	Multi-trend historical and live data	Configurable 8-pen trend for any selected tags; time range selector	← Any faceplate via Trend button

6.5 InTouch Script Examples

QuickScript: Acknowledge All Active Alarms

```
' InTouch QuickScript: Acknowledge All Active Alarms
' Trigger: Action script on the "Acknowledge All" pushbutton

IF $OperatorLevel >= 2 THEN
    ' Log the acknowledgment with operator name and time
    LogMessage("All alarms acknowledged by: " + $OperatorName +
               " at " + Text($DateTime, "MM/DD/YY HH:MM:SS"));
    ' Acknowledge all alarms in all groups
    AlarmAckAll("$System", 0);
    MessageBox("All active alarms have been acknowledged.", "Alarm Acknowledgment", 0);
ELSE
    MessageBox("Insufficient access level to acknowledge all alarms.", "Access Denied", 0);
ENDIF;
```

NOTE
The AlarmAckAll function name and arguments differ between InTouch versions and alarm providers. Verify on bench before field use.

QuickScript: PID Mode Request (Auto / Manual)

The HMI does not write the PID mode or setpoint directly. It sets a one-scan request bit; the PLC performs the mode change, validates PV quality, and clears the request (Section 5.2, Snippet 4). No script delay is needed.

```
' InTouch QuickScript: PID Mode Request — SEP_PIC_101
' Trigger: Action scripts on the MAN and AUTO buttons of the PID faceplate
' Tag bindings (I/O tags to the PLC):
'   PIC101_ManReq  -> HMI_PIC101_ManReq   (discrete, PLC clears after use)
'   PIC101_AutoReq -> HMI_PIC101_AutoReq  (discrete, PLC clears after use)

' MAN button
IF $OperatorLevel >= 1 THEN
    PIC101_ManReq = 1;
    LogMessage("SEP_PIC_101 MAN requested by " + $OperatorName);
ENDIF;

' AUTO button
IF $OperatorLevel >= 1 THEN
    PIC101_AutoReq = 1;
    LogMessage("SEP_PIC_101 AUTO requested by " + $OperatorName);
ENDIF;
```

NOTE

A discrete pushbutton with a Set action can replace the script. Verify on bench before field use.

NOTES

Section 7: Wiring Placeholders and Field Drawings

7.1 Control Panel Wiring Standards

Wire Color Standards

Signal Type	Color	AWG	Notes
120 VAC Line (Hot)	Black	12 or 14	NEC Article 310; use 12 AWG for circuits >15A
120 VAC Neutral	White	12 or 14	Never switch the neutral; bond to ground at service entrance only
120 VAC Ground / Safety Earth	Green or bare	12 or 14	Always connected to equipment chassis and panel ground bus
24 VDC Positive (+)	Red	16 or 18	ISA 5.4 practice; tie to DC positive bus via fuse
24 VDC Negative / Common (−)	Blue or Black	16 or 18	ISA 5.4 practice; tie to DC common bus; do NOT ground unless required
4–20 mA Analog Signal (+)	White	18 or 20	Shielded twisted pair (STP); shield drain to single-point ground at panel
4–20 mA Analog Signal (−)	Black	18 or 20	Paired with white; do not confuse with AC hot wiring
Intrinsically Safe (IS) Circuits	Light Blue (preferred)	18 or 20	Common ISA/IEC practice; mandatory physical segregation from non-IS wiring
Thermocouple Extension (Type K+)	Yellow	20 (TC ext wire)	Must use thermocouple extension wire (not copper) from sensor to terminal block; maintain polarity
Thermocouple Extension (Type K−)	Red	20 (TC ext wire)	Red is always negative for USA TC extension cables
Control (Discrete Signal)	Orange	18	Distinguishes discrete control from power; common in motor control wiring

Terminal Block Numbering Convention

Terminal blocks must be numbered sequentially with a logical, documented system. The recommended convention:

- **Format:** [Panel ID]-TB[Group]-[Terminal Number] — example: MCC-1-TB2-14
- Power distribution terminals: group TB1 (AC power), TB2 (24 VDC), TB3 (24 VDC common)
- Analog signal terminals: group TB10–TB19 (one group per FLEX I/O bank or AI card)
- Digital input terminals: group TB20–TB29
- Digital output terminals: group TB30–TB39
- IS barrier terminals: group TB50–TB59 — physically separated from all other terminals
- Earth/ground terminals: group TBG (green terminal blocks, always grounded)

Cable Separation Rules

- **Minimum 12 inches (305 mm) of physical separation** between signal cables (4–20 mA, discrete I/O, network) and power cables (480 VAC motor, VFD output, 120 VAC control power) for parallel runs exceeding 10 feet.
- Where cable trays must cross, the crossing shall be at **90 degrees** to minimize coupling.
- VFD output cables (motor leads) are a strong source of EMI and must be in **dedicated, separate conduit** — never share conduit or tray with signal cables.
- **Never route analog signal cables through conduit shared with power cables**, even if only passing through for a short distance.

Shield Grounding

- Each shielded instrument cable must have its drain wire (shield) connected to the **instrument signal ground bus** in the control panel — one end only (panel end).
- The **field end of the shield must be left floating** (not connected to the instrument housing, junction box, or conduit). Connecting both ends creates a ground loop, which introduces noise.
- The instrument signal ground bus must be bonded to the **main panel earth ground** with a single, low-impedance connection.
- IS circuit drain wires must connect to the **IS earth bar** — not the same bus as non-IS ground.

7.2 Wiring Diagram Placeholders

NOTE — Drawing Placeholders
The following subsections describe the content of required field wiring drawings. Obtain the actual drawings from the plant engineering department and insert them here, or reference the drawing number from the plant drawing management system.

[DRAWING: Elementary Wiring Diagram — Motor Control Circuit (MCC)]
Description: This drawing shows the complete motor control circuit for a typical motor starter at the plant MCC. Components shown include: main disconnect switch, primary fuses, motor contactor coil circuit (typically 120 VAC coil), overload (OL) relay with NC auxiliary contact wired in series with contactor coil, HOA (Hand-Off-Auto) selector switch, PLC Digital Output contact wired to the "Auto" position of HOA, pilot lights (Red=Run, Green=Stop), auxiliary contacts for PLC feedback (DI), and motor nameplate data. Drawing number: [COMPANY: _____]

[DRAWING: Loop Wiring Diagram — 4–20 mA Pressure Transmitter to PLC AI Card]
Description: This drawing shows the complete loop wiring for one 4-20mA pressure transmitter (e.g., SEP-PT-101) from the field connection through to the PLC analog input card. Components shown in order: transmitter field terminals (+/− loop), field junction box terminals, multi-pair instrument cable (shield, +, −), IS Zener barrier or galvanic isolator (if applicable) mounted in control room, terminal blocks in control panel, 250Ω HART resistor (if HART is used), 1756-IF8 or 1794-IE8 AI card terminal. Drawing number: [COMPANY: _____]

[DRAWING: Digital Input Wiring — Proximity Switch / Limit Switch to DI Card]
Description: This drawing shows wiring for a 2-wire proximity sensor or a limit switch connecting to a 1794-IB16 or 1756-IB16 digital input card. Components shown: 24 VDC sensor power supply from DI card or separate source, sensor signal wire, cable shield grounded at panel end, DI card input terminal, and common (0V) return. For IS applications, IS barriers are shown. Drawing number: [COMPANY: _____]

[DRAWING: Digital Output Wiring — PLC DO Card to Solenoid Valve]
Description: This drawing shows a 1794-OB16 or 1756-OB16 digital output driving a 24 VDC solenoid valve coil. Components shown: DO terminal (positive switched output), external 24 VDC fuse (0.5A–1A per output group), solenoid valve coil terminals, flyback suppression diode across solenoid coil (cathode to positive), 24 VDC common return to DO card. Drawing number: [COMPANY: _____]

[DRAWING: ESD Loop Wiring — SIS Solenoid Valve Wiring]
Description: This drawing shows the wiring for a SIL-rated ESD solenoid valve in a de-energize-to-trip configuration. Components shown: SIS logic solver output relay (fail-safe output), SIL-rated solenoid valve (spring-return, fail-closed), proof test terminal strip (to allow loop testing without de-pressurizing valve), 24 VDC dedicated SIS power supply (separate from BPCS power), dual redundancy configuration if applicable (1oo2 output), cable shielding and IS termination per SIS manufacturer requirements. Drawing number: [COMPANY: _____]

7.3 Grounding and Shielding Reference

- **Control Room Ground Bus:** A bare copper or tinned copper bus bar runs the length of the control room cable tray, bonded to the building steel and the electrical system safety earth. All panel ground wires, cable shields, and IS earth bars bond to this bus.
- **Panel Bonding:** Every metal panel, enclosure, cable tray section, and conduit run must be bonded to the plant ground grid with a minimum #6 AWG green insulated conductor. Jumper bonds (braided strap or #6 AWG) are required at cable tray joints where paint may interrupt conductivity.
- **EMI Mitigation for VFD Installations:**
- Install VFD in a metal enclosure with proper EMI gasket sealing.
- Use shielded motor cable (VFD-rated cable, all three phases + shield + PE) from VFD output to motor; ground both ends of the shield for VFD motor cable.
- Install output dV/dt filters or sine wave filters between VFD and motor if cable length exceeds 100 feet or if motor insulation is not inverter-rated.
- Keep VFD input power wiring, VFD output (motor) wiring, and signal wiring in separate conduits and trays.
- Do not run communication cables (Ethernet, DeviceNet) in the same conduit as VFD motor leads.

NOTES

Section 8: Troubleshooting Guide

8.1 Systematic Troubleshooting Methodology

Effective troubleshooting is a disciplined, sequential process. Random probing wastes time, introduces new problems, and can create safety incidents. Technicians should apply this 7-step process every time, regardless of urgency.

1. **Define the Problem:** State clearly what is NOT working. "PV not reading" is vague. "SEP_PT_101 reads 0 PSIG on the HMI but the field pressure gauge reads 205 PSIG" is precise. A well-defined problem is half solved.
2. **Observe Symptoms:** Gather all observable evidence before touching anything. Check: HMI tag quality, alarm history, PLC online diagnostics, fault LEDs on modules, related tags, when the problem started, what changed just before it started.
3. **Identify Possible Causes:** Generate a list of all plausible causes based on symptoms. Use logic: if the problem is isolated to one tag, the cause is in that specific loop. If multiple loops on the same module are bad, the module or its power is suspect.
4. **Test Hypotheses (Least Invasive First):** Start with checks that require no interaction with live equipment — visual inspection, software diagnostics, HMI configuration review. Then move to measurements that require minimal interaction (check voltage at terminals without disconnecting wires). Avoid disconnecting wires or removing modules until simpler checks are exhausted.
5. **Isolate the Fault:** Using test results, narrow the fault to a specific component, cable segment, or software element. Example: signal is correct at field junction box but wrong at panel terminal → fault is in the cable between junction box and panel.
6. **Repair or Replace:** Apply the correct fix. Do not bypass the root cause; address it directly. Obtain required permits (LOTO, hot work, entry) before physical work. Document the work performed.
7. **Verify and Document:** Confirm normal operation is restored (loop check, HMI verification, alarm test). Complete the work order or maintenance record. Update the loop drawing if any wiring was changed. Notify the control room operator that the loop is back in service.

WARNING — Before Any Troubleshooting

LOTO (Lockout/Tagout): Isolate and de-energize equipment before touching live terminals or removing modules, unless procedures explicitly permit live maintenance and a JSA (Job Safety Analysis) has been completed. Never assume a circuit is de-energized; verify with a calibrated meter. In Class I, Division 1 areas, obtain a hot work permit before using any non-IS tools or equipment. When in doubt, consult the supervisor.

8.2 Troubleshooting Tables

Table A: PLC / Control System Faults

Symptom	Possible Cause	Diagnostic Steps	Solution
I/O module fault LED on (solid or flashing red)	Module type mismatch; failed module; configuration error; blown fuse on output module	Check module fault code in Studio 5000 → I/O Config → Module Properties → Fault/Status. Note 16# fault code. Check module part number matches configuration.	Correct module type/config; replace failed module; check output fuses for DO modules
Loss of comms to FLEX I/O bank	Ethernet cable disconnected; IP address conflict; adapter powered off; wrong RPI	Ping adapter IP address. Check network switch port LED. Check adapter NS/MS LEDs. Verify IP address in adapter via RSLinx Browse → BOOTP.	Re-seat cable; resolve IP conflict; restore power; adjust RPI if excessive packet loss
Controller in FAULT mode (RUN LED off, FAULT LED on)	Major fault: unhandled exception, task overrun, I/O connection timeout, corrupted program	Studio 5000 → Controller Diagnostics → Fault Log. Note fault type and code. Check if fault is recoverable (minor) or requires clear.	Clear minor fault (GSV/SSV or online "Clear Faults"). For major fault: investigate cause, fix program, download and restart. Contact Rockwell TAC if code is unknown.
Task overrun alarm	Program scan time exceeds periodic task period; MSG instructions adding latency; too many I/O modules with short RPI	Studio 5000 → Online → Controller Properties → Tasks → view actual execution time vs. period. Identify the program with the longest scan time.	Increase task period; optimize program (reduce unnecessary calculations per scan); offload low-priority logic to a slower periodic task; increase RPI for non-critical I/O

Symptom	Possible Cause	Diagnostic Steps	Solution
Tag value stuck at minimum or maximum count	AI card channel fault (open circuit = under-range / minimum count; over-range = maximum count); scaling error; I/O connection inhibited	Check raw I/O tag value in Studio 5000 online. Measure actual mA at AI card terminals. Check module channel fault bits.	Fix field instrument or wiring issue; correct scaling; un-inhibit I/O connection
PID output saturated at 0 or 100%	Integral windup (PV cannot reach SP due to process or mechanical limitation); PID in wrong mode; output limits wrong; process disturbance too large	Check: Is process variable moving? Is control valve actually moving? Check output limits in PID block. Check for valve or actuator failure.	Fix process issue first; add anti-windup (output clamping); adjust output limits; switch to manual and observe process; check valve for mechanical failure
MSG instruction faulting (.ER=1)	Route path incorrect; target tag not published (wrong external access); connection timeout; target controller in fault	Check MSG.ERR code. Verify route path IP address. In target PLC, confirm tag is controller-scoped with "Read/Write" external access. Ping target IP.	Correct route path; fix external access on target tag; increase MSG timeout; investigate and fix target controller fault

Table B: Instrument / Field Device Faults

Symptom	Possible Cause	Diagnostic Steps	Solution
4-20 mA signal reading 0 mA (open circuit)	Broken wire, corroded terminal, failed transmitter power supply, open IS barrier fuse	Measure mA at transmitter terminals — if <1 mA there too, measure loop supply voltage. Check IS barrier fuse if applicable. Trace wiring back to panel.	Repair broken wire; clean corroded terminals; replace barrier fuse; replace transmitter if supply voltage is correct but output is zero
Signal reading ≤ 3.6 mA (below live zero — NAMUR NE43 low fault)	Transmitter powered but in fault state; sensor failure; partial short in wiring	Connect HART communicator — read transmitter status. Check if transmitter has a configured low fault output (NAMUR NE43 low = 3.6 mA). Inspect sensor element.	Replace failed sensor (RTD, thermocouple, diaphragm); reconfigure transmitter fault mode; repair wiring partial short
Signal at or above 21 mA (NAMUR NE43 high fault) or over-range	Process over-range; shorted signal wire; transmitter configured for high fault output (NAMUR 21 mA)	Verify process is not actually at or above instrument URV. Check for shorted wiring. Connect HART and read status — does transmitter report over-range?	If process is in range: replace transmitter. Fix shorted wire. Reconfigure transmitter fault output direction if desired.
Noisy / oscillating signal	Ground loop (shield grounded at both ends), EMI from VFD or motor, loose connection, inadequate shielding	Check trend in historian for noise pattern (random vs. periodic). Periodic noise often = VFD frequency. Remove one end of shield ground — if noise drops, ground loop confirmed. Wiggle wiring to find loose connection.	Ground shield at panel end only; re-route cable away from power source; add ferrite core on cable near noise source; increase transmitter damping; tighten loose connections
Transmitter reading offset error after maintenance	Impulse lines not properly vented/equalized after isolation; transmitter zero not re-established; process connection different from before	Close process isolation valve and equalize manifold. Note transmitter reading with zero DP applied — should read 4.000 mA exactly. If offset, re-zero transmitter.	Re-zero transmitter with equalized manifold valves using HART communicator or pushbutton; vent all impulse lines; retighten process fittings
Positioner not responding to command signal	No instrument air supply; positioner I/P converter failure; feedback linkage slipped; positioner needs re-tuning; valve mechanically seized	Check instrument air supply pressure (min 20 PSIG clean dry air). Stroke valve manually from positioner pushbutton if available. Measure AO signal (mA) at positioner input terminals — is command getting there?	Restore air supply; replace I/P transducer; re-align and tighten feedback linkage; retune positioner (autotune if available); free mechanical valve if seized (contact mechanical team)

Table C: HMI / SCADA Faults

Symptom	Possible Cause	Diagnostic Steps	Solution
Tag showing "???" in InTouch	I/O server disconnected from PLC; tag not found in server; PLC comms failure; InTouch not licensed	Check InTouch I/O server (e.g., RSLinx, FactoryTalk Linx) — is it running? Is the PLC topic connected (green)? Check WindowViewer → Special → System Information for comms status.	Restart I/O server; verify PLC is online and accessible; check firewall; re-map tag if server topic changed; verify InTouch license

Symptom	Possible Cause	Diagnostic Steps	Solution
Alarm not annunciating on HMI	Alarm suppressed; alarm not configured in InTouch alarm db; wrong alarm group; alarm deadband too wide; tag in bad quality (will not alarm)	Check InTouch Tag Dictionary — is alarm limit configured? Is alarm suppressed? Check alarm group configuration. Manually force tag to alarm value and observe.	Remove suppression; configure alarm limits; correct alarm group; adjust deadband; fix underlying bad quality issue
Historical trend data missing	Historian not running; disk full; logging disabled for tag; trend group not configured; historian comms offline	Check Wonderware Historian (or eDNA, OSIsoft PI) service status. Check disk space on historian server. Verify tag logging configuration (is logging enabled? scan class assigned?)	Restart historian service; free disk space; enable logging on tag; verify historian license and configuration
OPC / DDE communication timeout	OPC server lost connection to PLC; network congestion; OPC server software crashed; PLC went offline	Check OPC server application (e.g., RSLinx Enterprise, KEPServerEX) for connection status. Ping PLC IP address. Check event log for OPC server errors.	Restart OPC server; verify PLC is online; check network switch; re-configure OPC topic if connection parameters changed
InTouch I/O server disconnected	RSLinx Classic/Enterprise service stopped; license expired; Windows update restarted service; network path to PLC changed	Services.msc — is RSLinx service running? Check RSLinx diagnostics for active topic connections (green = connected; red = faulted).	Start RSLinx service; renew license; restore network path; reconfigure EtherNet/IP driver in RSLinx if PLC IP changed

Table D: Network Faults

Symptom	Possible Cause	Diagnostic Steps	Solution
EtherNet/IP adapter (1794-AENTR) red status LED	No EtherNet/IP connection from controller; IP not set; ring break in DLR; wrong RPI	Check adapter MS LED (module status) and NS LED (network status). NS flashing green = IP set, no connection; NS flashing red = connection timeout; NS solid red = duplicate IP. Ping adapter IP. In Studio 5000, check I/O connection status.	Verify controller is online and I/O connection is not inhibited; re-set adapter IP address via BOOTP if needed; check DLR ring continuity
Duplicate IP address conflict	Another device on the network assigned the same IP address (DHCP conflict, misconfigured device)	Ping the IP address — if a response is received that does not match the expected MAC address, there is a conflict. Use ARP table (arp -a) to identify MAC addresses.	Identify conflicting device (check switch MAC address table); change IP of the non-critical device; use BOOTP/DHCP server with MAC reservations to prevent recurrence
High CIP packet loss	Network switch overloaded; excessive broadcast traffic; VFD EMI coupling into Ethernet cable; cable damage; duplex mismatch	Check switch port statistics for error counters, discards, and utilization. Use RSLinx diagnostic to view I/O connection status and packet counters. Inspect cable and connectors.	Upgrade network switch; isolate plant network from office network (VLANs); use shielded Ethernet cable in EMI-heavy environments; fix duplex mismatch (set to 100 Mbps / Full Duplex)
DLR ring break detected	Cable cut, connector failure, or device failure in the DLR ring; switch port failure	Studio 5000 → I/O Config → FLEX Adapter → Module Properties → Ring Diagnostics. Note which node reported the break. Physical inspection of cable at that node.	Repair or replace damaged cable; replace failed network switch port; restore ring connection; DLR supervisor will automatically reconfigure the network while break is being repaired — I/O communication continues on linear topology

8.3 Quick Diagnostic Checklists

Checklist: When a Control Loop Won't Respond

1. Verify the loop is NOT in bypass or maintenance mode — check bypass key switch and HMI status.
2. Check PLC controller status — is it in RUN mode (not FAULT or PROGRAM mode)?
3. Check I/O module status LEDs and Studio 5000 I/O tree — any faults on the AI or AO module?
4. Verify the PID rung executes unconditionally in a periodic task and that .UPD equals the task period.
5. Check PID mode: is .SWM (software manual) or .MO (operator manual) set? If so, the output is under manual control.
6. Verify the process variable (PV) is valid: is the AI reading a realistic value, stuck at a limit, or flagged Bad?

7. Measure mA signal at the AI card terminals — does it match the expected process value?
8. Go to the field instrument — does it have power? Is its local indicator reading correctly?
9. For AO (output): measure mA at the AO card output terminals — is the PLC actually sending the commanded signal?
10. Check the field actuator (control valve): is air supply present? Is positioner receiving the mA signal?
11. Stroke the valve manually from the positioner — does it move freely? If not, call mechanical team.
12. Check for any pending task overrun or program fault that may have halted the control routine.
13. Review the PID tuning — is integral gain so high that the loop oscillates and appears unresponsive?
14. Check for a latched BPCS trip (SEP_Trip_OK = 0) that may be overriding the control valve command.
15. Review loop drawing for any manual bypass valves or isolation valves that may have been left closed.

Checklist: Before Calling the Engineer

1. ☐ All fault LEDs, alarm messages, and error codes have been observed, recorded, and photographed where permitted.
2. ☐ Exact tag names, module slot numbers, and I/O addresses involved have been noted.
3. ☐ Alarm history and trends have been checked for when the problem first appeared.
4. ☐ The problem has been confirmed NOT to be a normal process upset (check the P&ID, ask the operator).
5. ☐ The most obvious causes (power, wiring, physical inspection) have been tested.
6. ☐ Measurements (mA readings, voltage readings, LED states) have been documented.
7. ☐ Changes to the system in the last 24 hours (maintenance, modifications, configuration changes) have been identified.
8. ☐ The maintenance history (CMMS) has been checked for recent work on this loop or nearby equipment.
9. ☐ Any corrective action attempted, and its result, has been documented.
10. Is this issue affecting plant safety, production rate, or regulatory compliance? (Determines urgency of escalation.)

Checklist: After Any Control System Change (MOC / Post-Change Verification)

1. Complete and close the Management of Change (MOC) form — get required signatures before going live.
2. Download the project to the controller and verify no download errors or I/O module faults appear.
3. Perform a loop check on all affected tags — verify field-to-HMI signal integrity (Section 2.5).
4. Test all modified or new ladder logic rungs using forced I/O (with proper LOTO and operator awareness).
5. Verify all alarm setpoints are correctly configured in both PLC and HMI.
6. Test alarm annunciation — confirm alarms appear on HMI with correct priority, description, and audible annunciation.
7. Test ESD and safety interlock logic if any changes were made — function test in accordance with proof test procedure.
8. Verify PLC program backup has been saved to the project file server (not just to the controller flash).
9. Update all affected loop drawings, I/O schedules, and cause and effect matrix to reflect the change.
10. Notify all affected operators and document the change in the Control Room Logbook.
11. Update the HMI tag list and historian configuration if new tags were added.
12. Close out all work orders, permits (LOTO, hot work), and document "as-left" condition of all affected instruments.

NOTES

Section 9: Safety and SIS Basics

9.1 Process Safety Fundamentals

HAZOP (Hazard and Operability Study)

A **HAZOP** is a structured, systematic review of a process design to identify potential hazards and operability problems. A multidisciplinary team (process engineer, instrument engineer, operator, safety specialist, and often an independent facilitator) reviews every process node using standardized **guidewords** applied to process parameters (flow, pressure, temperature, composition, level).

Guideword	Meaning	Example Applied to Flow
No / None	Complete negation of the design intent	No flow when flow is required (blocked line, pump trip)
More	Quantitative increase above design	More flow than expected (control valve fails open)
Less	Quantitative decrease below design	Less flow than required (partial blockage, pump degraded)
Reverse	Opposite direction from design intent	Reverse flow (check valve failure, backpressure event)
Other Than	Complete substitution — wrong material, phase, or condition	Liquid carry-over in gas stream (separator performance failure)
As Well As	Additional (impurity or phase) present	Water in product hydrocarbon (separator malfunction)
Part Of	Only part of the design intent is achieved	Partial blockage in one branch of a multi-branch system

Layers of Protection Analysis (LOPA)

LOPA is a semi-quantitative method for evaluating whether existing and proposed protection layers provide sufficient risk reduction for a specific accident scenario. The risk is evaluated against a company risk tolerance criterion (typically expressed as a maximum acceptable frequency of a serious event — e.g., 1×10^{-4} per year for a fatality).

The layers of protection, from innermost to outermost, are:

1. **BPCS (Basic Process Control System):** The PLC/DCS control system maintaining normal operating conditions. NOT an independent protection layer if it is also the initiating cause.
2. **Operator Alarm and Response:** An alarm that an operator detects and acts upon within a required response time. Typically assigned a Probability of Failure on Demand (PFD) of 0.1 (1 in 10 chance of failure) per occurrence.
3. **Safety Instrumented System (SIS):** An independent logic solver with dedicated sensors and final elements, designed per IEC 61511 using devices certified to IEC 61508 or justified by prior use. Provides the most reliable automated protection.
4. **Relief Devices (PSV, rupture disk):** Mechanical devices requiring no logic — open automatically at set pressure. Highly reliable but require periodic testing.
5. **Physical Protection (Dikes, Blast Walls):** Passive protection limiting consequences after a release. Does not prevent the event, only mitigates consequences.

SIS vs. BPCS — Key Differences

Feature	BPCS (ControlLogix PLC)	SIS (Safety Logic Solver)
Purpose	Normal process control — maintain setpoints	Detect unsafe conditions — drive to safe state
Independence	Shares I/O, sensors, and HMI with BPCS	Fully independent sensors, logic solver, final elements
Failure mode	Fail to any state (not necessarily safe)	Designed to fail to a defined safe state (de-energize to trip)
Certification	General industrial control; not SIL-rated (IEC 61511 requires BPCS independence from the SIS)	Designed per IEC 61511; devices certified to IEC 61508 or justified by prior use; SIL-rated for the safety function
Modification	Routine MOC process	Formal functional safety change — requires re-validation and proof test
Testing	Functional testing as part of commissioning and routine maintenance	Mandatory periodic proof tests (typically annual or biennial)

IEC 61511 Safety Lifecycle Overview

IEC 61511 is the international standard for the functional safety lifecycle of Safety Instrumented Systems in the process industry. Devices used in a SIS are certified to IEC 61508 or justified by prior use. The lifecycle includes:

- **Phase 1:** Hazard and risk assessment (HAZOP, LOPA) — identify SIFs and required SIL
- **Phase 2:** SIS design — architecture, technology selection, SIL verification

- **Phase 3:** Installation, commissioning, and validation testing
- **Phase 4:** Operations and maintenance — proof tests, bypasses, management of change
- **Phase 5:** Decommissioning

9.2 Safety Integrity Level (SIL) Basics

SIL Level	PFD Range (Demand Mode)	Risk Reduction Factor	Typical Application
SIL 1	10^{-1} to 10^{-2} (1-in-10 to 1-in-100 failure probability)	10 to 100	High-high pressure shutdown on a non-critical vessel; most common SIL level in gas processing
SIL 2	10^{-2} to 10^{-3} (1-in-100 to 1-in-1,000)	100 to 1,000	Compressor emergency shutdown; ESD on amine absorber; high H2S shutdowns
SIL 3	10^{-3} to 10^{-4} (1-in-1,000 to 1-in-10,000)	1,000 to 10,000	High-consequence scenarios (LNG, hydrogen, high-toxicity — rare in standard gas plants)
SIL 4	10^{-4} to 10^{-5} (1-in-10,000 to 1-in-100,000)	10,000 to 100,000	Nuclear industry; extremely rare in hydrocarbon processing

The **Risk Reduction Factor (RRF)** is the inverse of the PFD: RRF = 1/PFD. A SIL 2 function with PFD = 0.005 has an RRF = 200, meaning it reduces the frequency of the hazardous event by a factor of 200 compared to having no SIF.

9.3 Safety Instrumented Function (SIF)

SIF Architecture

Every Safety Instrumented Function consists of three subsystems: **Sensors** (detect the hazardous condition) → **Logic Solver** (evaluates sensor inputs against the safety function requirements) → **Final Elements** (act to drive the process to a safe state). Each subsystem must be independently specified, verified, and maintained.

Voting Logic

Voting Logic	Description	Reliability Advantage	Availability Disadvantage	Use Case
1oo1 (1-out-of-1)	Single sensor/element; one device must trip to initiate SIF	High availability (trips on demand)	Single point of failure — spurious trip if sensor fails high; dangerous failure if fails safe	SIL 1 simple applications; low consequence of spurious trip
1oo2 (1-out-of-2)	Two sensors; EITHER one can initiate trip	Lower PFD (dangerous failure less likely — both must fail)	Higher spurious trip rate — either sensor failure trips system	Where process cannot tolerate dangerous failure; safety favored
2oo2 (2-out-of-2)	Two sensors; BOTH must agree to initiate trip	Very low spurious trip rate (both sensors must fail)	Higher PFD — if one sensor fails dangerously, SIF won't trip	Where spurious trips cause major production loss; typically 2oo2 used only when maintenance team can respond quickly
2oo3 (2-out-of-3)	Three sensors; ANY TWO must agree to initiate trip	Balanced — good reliability AND good availability; best of both	Higher cost (three instruments, three field loops)	SIL 2 and SIL 3 applications; compressor ESD; critical process vessels

Proof Testing

A proof test is a periodic functional test of the entire SIF (sensor → logic solver → final element) to detect dangerous undetected (DU) failures, which would prevent the SIF from responding on demand. The proof test interval (typically annual or biennial) is a key parameter in the SIL verification calculation; longer intervals increase the average PFD. A proof test is never performed on a live process without a written proof test procedure, an authorized bypass in place, and communication with the control room operator.

9.4 Gas Plant Specific Safety Topics

H_2S Hazard

Hydrogen sulfide (H_2S) is a colorless, flammable, extremely toxic gas with a characteristic "rotten egg" odor at low concentrations. The sense of smell is lost at higher concentrations, which makes H_2S uniquely dangerous. Exposure limits and typical plant action levels:

Concentration	Basis / Effect	Action Required
0.01–0.3 ppm	Odor detection threshold	Odor is not a reliable warning (olfactory fatigue)
1 ppm	ACGIH TLV-TWA (8 h)	Monitor; investigate sustained readings
5 ppm	ACGIH STEL (15 min)	Investigate source; verify ventilation; SCBA ready
10 ppm	NIOSH REL (10-min ceiling); OSHA construction PEL (TWA); common plant alarm level	Don SCBA or supplied-air respirator; evacuate non-essential personnel; follow plant H_2S response plan
20 ppm	OSHA general industry acceptable ceiling	Evacuate unless in supplied-air respiratory protection
50 ppm	OSHA general industry peak (max 10 min, once per 8-h shift, if no other exposure)	SCBA required; evacuate; notify emergency response team
100 ppm	IDLH (NIOSH)	SCBA required; full emergency response; evacuation
500–1000 ppm	Rapid unconsciousness; potentially fatal	Full emergency; do NOT enter without SCBA and buddy system

NOTE

OSHA does not define an action level for H_2S. Plant alarm and evacuation levels are set by the plant H_2S contingency plan and govern.

PPE requirements for H_2S: fixed H_2S detectors (electrochemical), personal H_2S monitors worn by all personnel in gas-handling areas, SCBA at muster points and with the emergency response team, buddy system mandatory in areas above the plant H_2S alarm level (commonly 10 ppm), and wind direction awareness at all times.

Flammable Gas — LEL and UEL

Natural gas (primarily methane, CH_4) is flammable within a specific concentration range in air:

- **LEL (Lower Explosive Limit):** 5 vol% methane in air — below LEL, the mixture is too lean to ignite.
- **UEL (Upper Explosive Limit):** 15 vol% methane in air — above UEL, the mixture is too rich to ignite (though it will ignite if diluted back into the flammable range).
- Plant action levels: alarm at 10% LEL (0.5% methane); evacuate at 25% LEL (1.25% methane); hot work prohibited above 10% LEL.

Fire and Gas (F&G) System

The **Fire and Gas (F&G) system** is an independent system that monitors the gas plant for flammable gas releases, toxic gas (H_2S), smoke, heat, and flame. Key detector types:

- **Catalytic bead (pellistor) detector:** Detects flammable gas by oxidation on a heated bead. Measures 0–100% LEL. Susceptible to poisoning (silicones, lead) and water flooding. Requires regular calibration (monthly or per manufacturer spec).
- **Infrared (IR) point detector:** Detects hydrocarbon gas by infrared absorption. Immune to poisoning; functional in inert atmospheres; better choice for most gas plant applications. Also measures 0–100% LEL.
- **Flame detector:** UV, IR, or UV/IR combination. Detects actual flame (fire). UV detectors are sensitive to welding arcs and sunlight — ensure proper positioning and arc flash suppression settings.
- **F&G voting logic:** F&G system uses voting (e.g., 1oo2 or 2oo3 detectors per zone) before initiating emergency actions (ESD, deluge, PA system) to avoid spurious trips from a single faulty detector.

Cause and Effect Matrix

A **Cause and Effect (C&E) Matrix** documents every safety interlock relationship: which initiating causes (sensors, switches, manual ESD) trigger which protective responses (final elements — valves, trips). This is the definitive reference for ESD logic design and testing.

Initiating Cause ↓ / Final Element →	ESD-XV-001 (Plant Inlet Block)	COMP-XV-201 (Comp Suction Block)	SEP-XV-101 (Sep Inlet Block)	COMP Trip (MCC)	PUMP-MTR-101 Trip	Blowdown-XV-010 Open	PA Announcement	Fire Suppression
SEP-PAHH-101 (Sep Hi-Hi Press)	X		X			X		
SEP-LAHH-101 (Sep Hi-Hi Level)			X					

Initiating Cause ↓ / Final Element →	ESD-XV-001 (Plant Inlet Block)	COMP-XV-201 (Comp Suction Block)	SEP-XV-101 (Sep Inlet Block)	COMP Trip (MCC)	PUMP-MTR-101 Trip	Blowdown-XV-010 Open	PA Announcement	Fire Suppression
COMP-PAHH-201 (Comp Disch Hi-Hi Press)	X	X		X		X		
COMP-TAHH-201 (Comp Disch Hi-Hi Temp)		X		X				
COMP-VT-201 Hi-Hi (Vibration)		X		X				
FIRE-GAS-301 (F&G Zone 3 Vote)	X	X	X	X	X		X	X
Manual ESD (Pushbutton / HMI)	X	X	X	X	X	X	X	
AMINE-PAHH-401 (Amine Hi-Hi Press)						X		

[COMPANY: Replace this template C&E matrix with the plant-specific Cause and Effect Matrix. Reference document: C&E Matrix Drawing No. _____. Any discrepancy between this manual and the official C&E matrix — the official C&E matrix governs.]

LOTO Procedure Summary

Lockout/Tagout (LOTO) is the OSHA-mandated procedure (29 CFR 1910.147) for controlling hazardous energy before maintenance. Steps for electrical LOTO:

1. Notify affected employees and operators.
2. Identify all energy sources (electrical, pneumatic, hydraulic, process pressure/chemistry, gravity).
3. Shut down the equipment using normal stopping procedure.
4. Isolate each energy source at its isolation point (disconnect, valve, etc.).
5. Apply the personal lock and tag to each isolation point.
6. Release/restrain stored energy (bleed lines, block valves, vent pressure, support suspended loads).
7. Verify isolation — test for zero energy state (use calibrated voltage tester, check pressure gauges).
8. Perform work. Keep lock applied at all times.
9. Remove tools; ensure no personnel are in the hazard zone.
10. Remove lock and tag — only by the person who applied it.
11. Restore energy; notify operator; confirm equipment is safe to operate.

Work Permit Types

Permit Type	When Required	Key Controls
Hot Work Permit	Any work that generates ignition sources (welding, grinding, cutting, non-intrinsically-safe power tools) in classified hazardous areas	Gas test before and during work; fire watch; fire extinguisher positioned; area cleared of flammables; LEL <10% confirmed
Cold Work Permit	Non-ignition-source work in hazardous areas (bolting, instrument maintenance with IS tools, painting)	Area classified as safe for cold work; gas test; LOTO if required; PPE specified
Confined Space Entry Permit	Entry into any permit-required confined space (vessels, tanks, large pipes, pits)	Atmospheric testing (O_2, LEL, H_2S); ventilation; attendant stationed outside; rescue equipment available; LOTO on all inlets/outlets; rescue plan documented
Line Breaking Permit	Opening any pressurized or process-containing line, flange, fitting, or instrumentation connection	Line depressurized and drained/purged; process isolated and LOTO applied; PPE for process fluid; atmospheric test if toxic/flammable fluid

NOTES

Section 10: Twelve Hands-On Labs

WARNING — All Labs
All labs are performed under the supervision of a qualified technician or engineer until the technician is qualified for the task (Appendix F.4 competency matrix). Obtain all required permits before starting any lab involving live equipment. Never bypass safety interlocks for a lab exercise without written authorization and a supervisor present.

NOTE
Lab review questions and the answer key are in Appendix F.2.

Lab 1: Instrument Loop Check — Pressure Transmitter (Site Form Discipline)

Objective: Perform a complete loop check on a 4–20 mA pressure transmitter from field sensor through junction box, IS barrier, and AI card to the HMI display, using the plant loop-check form and the work-control flow in Section 0.5. Use a training-loop tag assigned by the supervisor; SEP-PT-101 is an example only.

Equipment / Tools: Calibrated loop calibrator (mA source/measure, e.g., Fluke 789), HART communicator (Emerson 475 or Trex), calibrated digital multimeter, laptop with Studio 5000 (online), access to InTouch HMI, loop drawing, instrument datasheet, loop check form.

Safety Precautions: Obtain loop check work order. Notify control room operator — request alarm suppression on affected tags. Confirm area classification — use only IS tools in Class I Div 1 areas. Do not remove transmitter from process without line breaking permit and LOTO.

Background Theory: A loop check verifies the electrical signal path from the field transmitter to the control system, confirming that scaling, tag mapping, and alarm setpoints are correctly configured. Loop checks are required after installation, after any wiring modification, and at defined intervals per the instrument maintenance plan. The 4–20 mA signal represents 0–100% of the instrument's calibrated range.

Procedure

1. Review the loop drawing. Identify transmitter tag (e.g., SEP-PT-101), range (0–300 PSIG), IS barrier type, AI card slot/channel, and HMI tag name.
2. Notify the control room operator: "Performing loop check on SEP-PT-101. Expect artificial readings for approximately 30 minutes. Please suppress alarms on this tag."
3. At the field transmitter, remove the housing cover. Verify wiring matches the loop drawing. Check terminal block connections are tight.
4. Connect the HART communicator across the transmitter loop terminals (or at the junction box). Read transmitter: model, serial number, configured range, current PV, and loop current. Record on loop check form.
5. Using the loop calibrator in mA-source mode, disconnect the transmitter signal lead and inject 4.000 mA. At the control room/HMI, observe SEP-PT-101 reading — should display 0.0 PSIG (±0.5 PSIG acceptable).
6. Inject 8.000 mA — HMI should read 75.0 PSIG.
7. Inject 12.000 mA — HMI should read 150.0 PSIG.
8. Inject 16.000 mA — HMI should read 225.0 PSIG.
9. Inject 20.000 mA — HMI should read 300.0 PSIG.
10. Inject a value above the PAH setpoint (example: PAH = 250 PSIG; inject 17.6 mA ≈ 255 PSIG). Confirm the PAH alarm annunciates on the HMI after the configured on-delay, with the correct tag, description, and priority.
11. Open the loop (disconnect one wire). Confirm the channel under-range/fault status sets, the HMI shows bad quality, and a bad-quality alarm annunciates.
12. Restore transmitter wiring. Re-read HART — confirm transmitter PV matches process (local gauge if available).
13. Request operator to restore alarm suppression. Confirm tag is reading correctly on HMI.
14. Complete and sign the loop check form. File per plant document control procedure.

Expected Results: All five injection points within ±0.5% of full span (±1.5 PSIG for a 300 PSIG span). Alarms trigger and clear at configured setpoints. Fault alarm triggers on open loop.

Troubleshooting Tips: If the HMI reads a fixed value regardless of injection, check whether the AI tag is aliased to the wrong channel. If HART will not communicate, verify 230–600 Ω loop resistance. If the alarm does not trigger, check the alarm suppression status in InTouch.

Close-out: Record as-found mA and EU, as-left mA and EU, who was notified, alarm-suppression start/stop time, forces used (must be none at the end), and supervisor initials. If the assigned tag is a SIF sensor, stop; that work is outside the technician authorization envelope (Section 0.2).

Review Questions: Appendix F.2, Lab 1.

Lab 2: FLEX I/O Module Replacement (Hot-Swap)

Objective: Safely remove and replace a failed 1794-IE8 analog input module while the FLEX I/O bank remains powered, without losing communication to other modules.

Equipment / Tools: Replacement 1794-IE8 module (same hardware rev as installed), flathead screwdriver, ESD wrist strap, laptop with Studio 5000 online, MOC form, loop check form for re-verification.

Safety Precautions: Complete MOC form before proceeding. Notify operator that AI signals on this module (channels 0–7) will go to fault/zero during replacement. Wear ESD wrist strap when handling module. Confirm replacement module is the same catalog number and hardware revision. Do not remove terminal base — field wiring remains connected throughout.

Background Theory: FLEX I/O hot-swap separates the electronics module (which carries the circuitry) from the terminal base (which carries the field wiring). Removing the module stops data exchange for that module only — the adapter and other modules in the bank remain communicating. When the new module is inserted, the adapter automatically reconfigures it and restores data exchange.

Procedure

1. Complete and obtain approval for the MOC form. Document: affected tags, replacement module catalog number, estimated downtime, notification to operators.
2. In Studio 5000, go online. Verify the 1794-AENTR adapter and other modules in the bank are in normal (green) status. Note which channels are active on the failed 1794-IE8.
3. Notify the control room operator: "Replacing IE8 module in slot [X] of FLEX_SEP_01. Tags SEP_PT_101 through SEP_FT_101 will show fault values for approximately 2 minutes."
4. Locate the 1794-IE8 in the bank. Observe the failed module's status LED (likely red or off).
5. Press and hold the module release latch (orange tab on some versions; check product bulletin). Rock the module slightly and lift it vertically from the terminal base. The terminal base (with wiring) remains mounted.
6. Inspect the terminal base connector pins — no damage or corrosion. Insert the new 1794-IE8 module into the terminal base: align the module body with the base, press down firmly until the latch clicks. The module status LED should flash green, then go solid green.
7. In Studio 5000, observe the I/O tree — the replaced module should show normal (green) status within 10–30 seconds.
8. Verify all AI tags on the replaced module are reading correct process values (not clamped at 0 or full scale, and not flagged Bad).
9. Perform abbreviated loop checks on at least two channels: inject 12 mA (50%) and verify HMI reads correctly.
10. Notify the control room operator: "Module replacement complete. All signals restored. Please verify SEP-PT-101 and associated tags."
11. Document completion of MOC, record new module serial number, and update the instrument maintenance record.

Expected Results: Module replacement completed without loss of communication to other modules. All AI channels restored to correct readings within 2 minutes. No new faults in controller.

Troubleshooting Tips: If module shows red after insertion — check for part number mismatch; may require module configuration re-download. If one channel remains bad after replacement — fault is in field wiring or instrument, not module.

Review Questions: Appendix F.2, Lab 2.

Lab 3: Build a Motor Start/Stop Rung in Studio 5000

Objective: Create a motor start/stop rung with seal-in logic on a bench controller, create the required tags, download, and test using bench inputs, including wire-break and mode-change behavior.

Equipment / Tools: Studio 5000 Logix Designer v36; 1769-L33ER bench controller at firmware v36 (or Studio 5000 Logix Emulate); bench DI/DO simulator (NO momentary START switch, NC STOP, OL, and E-STOP switches, pilot light); laptop.

Safety Precautions: Bench controller or emulator only; never download to a production controller. Confirm the bench controller is not connected to live process equipment. Clear all forces before ending the lab.

Background Theory: The motor start/stop with seal-in is the most fundamental discrete control circuit in industrial automation. The seal-in (holding) contact is a contact of the output in parallel with the momentary start pushbutton; it keeps the circuit ON after the start button is released. Only the stop circuit, a protective device, or a trip can break the seal-in. Health inputs follow the fail-safe convention in Section 0.3.

Procedure

1. Create a new Studio 5000 project for the 1769-L33ER at firmware v36 (or an Emulate controller). Use a project name that is visually distinct from production projects (Section 0.7).
2. In the Controller Organizer, expand Tasks → MainTask → MainProgram. Right-click MainProgram → Add → New Routine. Name: "Motor_Control", Type: Ladder Diagram.
3. Set the MainProgram properties: Main Routine = Motor_Control.

4. Wire the bench simulator per the bench drawing: START (NO momentary), STOP (NC), OL (NC), and E-STOP auxiliary (NC) to DI points; pilot light to a DO point.
5. Create alias tags to the bench points: START_PB, STOP_OK (1 = stop not pressed and wire intact), OL_OK (1 = overload healthy), E_STOP_OK (1 = E-stop released), and Motor_Run (DO, pilot light).
6. In Rung 0, enter the run rung in Neutral Text (rung block below).

Rung 0 – Run with seal-in

```
[XIC(START_PB) ,XIC(Motor_Run) ]XIC(E_STOP_OK)XIC(STOP_OK)XIC(OL_OK)OTE(Motor_Run);
```

7. Add the rung comment: "Motor start/stop with seal-in. START_PB = NO momentary. STOP_OK, OL_OK, E_STOP_OK = NC health inputs (1 = healthy). Motor_Run = starter coil."
8. Verify the routine (no errors) and save the project.
9. Download to the bench controller and place it in Run mode.
10. Press START: Motor_Run = 1 and the pilot light turns on.
11. Release START: Motor_Run remains 1 (seal-in holding).
12. Open STOP: Motor_Run = 0. Return STOP to normal: Motor_Run remains 0 until START is pressed again.
13. Open the OL switch and press START: Motor_Run remains 0 while OL_OK = 0. Repeat with the E-STOP switch.
14. Wire-break test: with the motor running, disconnect the STOP wire at the DI terminal. Motor_Run = 0 (fail-safe). Reconnect the wire.
15. Mode-change test: with the motor running, switch the controller to Program and back to Run. Motor_Run = 0 after the transition (OTE cleared at prescan) and does not restart until START is pressed. Confirm the force table is empty and save the final project.

NOTE

Where bench switches are not available, inputs may be simulated by forcing on the bench controller only, and all forces are cleared at the end of the lab. Verify on bench before field use.

Expected Results: Motor_Run energizes on START, holds through the seal-in, and drops on STOP, overload, E-stop, a broken stop wire, or a Program-to-Run transition. It restarts only on a new START. No forces remain at the end of the lab.

Review Questions: Appendix F.2, Lab 3.

Lab 4: Configure and Tune a PID Loop

Objective: Configure a PID instruction in a periodic task, simulate a first-order-plus-dead-time process, tune it with the Ziegler-Nichols closed-loop method, and verify bumpless transfer to manual.

Equipment / Tools: Studio 5000 v36 with the bench controller or Logix Emulate; trending (Studio 5000 trend or plant historian); laptop.

Safety Precautions: Use a simulated process only; do not connect the PID output to live process equipment. Disable all external outputs during this exercise.

Background Theory: Proportional action responds to the current error; integral action eliminates steady-state offset; derivative action responds to the rate of change. Ziegler-Nichols closed-loop tuning uses the ultimate gain (Ku), the gain at which the loop sustains a steady oscillation, and the period of that oscillation (Tu) to calculate starting tuning values. The Logix PID requires a fixed execution interval, so it is placed in a periodic task with .UPD equal to the task period.

Procedure

1. Create a periodic task "PID_Task" with a period of 100 ms and a program "PID_Lab". Create tags: PV_Sim (REAL), CV_Sim (REAL), CV_Delay (REAL[20]), i (DINT), and PIC_LAB (PID).
2. Create a Structured Text routine "Process_Sim" with the code below. The 20-element array at 100 ms gives 2 s of dead time; the last line is a first-order lag with gain 1 and a time constant of about 2 s.

```
(* Process_Sim — first-order-plus-dead-time simulator, 100 ms periodic task *)
FOR i := 19 TO 1 BY -1 DO
    CV_Delay[i] := CV_Delay[i-1];
END_FOR;
CV_Delay[0] := CV_Sim;
PV_Sim := PV_Sim + (CV_Delay[19] - PV_Sim) * 0.05;
```

3. Create a ladder routine "PID_Logic" with the rungs below and set the main routine to call Process_Sim and then PID_Logic. The PID rung is unconditional.

Rung 0 – Output tracking (bumpless entry to manual)
```
XIO(PIC_LAB.SWM)MOVE(PIC_LAB.OUT,PIC_LAB.SO);
```
Rung 1 – PID (unconditional)
```
PID(PIC_LAB,PV_Sim,0,CV_Sim,0,0,0);
```

4. Configure PIC_LAB: .UPD = 0.1; .PE = 0 (independent gains); .MAXI/.MINI = 100.0/0.0; .MAXS/.MINS = 100.0/0.0; .MAXO/.MINO = 100.0/0.0; .CA = 0 (reverse acting, E = SP − PV, because a higher CV raises PV); .SP = 50.0; .KP = 0.1; .KI = 0.0; .KD = 0.0; .SWM = 0 (auto).
5. Download and run. Observe PV_Sim settle near 50 with proportional action only (offset remains).
6. Find the ultimate gain (Ku): with .KI = 0 and .KD = 0, increase .KP in steps (0.5, 1.0, 2.0 …) until PV_Sim sustains a steady oscillation (neither growing nor decaying). Record Ku.
7. Measure the ultimate period (Tu) in seconds from the trend. Record Tu.
8. Calculate Ziegler-Nichols PI values: Kc = 0.45 × Ku; Ti = Tu / 1.2 (seconds). With .PE = 0: .KP = Kc and .KI = Kc / Ti (1/s). With .PE = 1: .KP = Kc and .KI = Ti in minutes per repeat (Ti / 60). Leave .KD = 0.
9. Enter the values. Step .SP from 50 to 70 and observe settling time, overshoot, and offset.
10. Adjust: if oscillatory, reduce .KP by 20%; if slow, increase integral action by 10%.
11. Bumpless transfer test: set .SWM = 1. The output holds at the last automatic value (.SO tracked .OUT). Change .SO by 10% and observe PV. Set .SWM = 0 and confirm the output resumes smoothly from the manual value.
12. Record the final tuning values and trend a setpoint step showing PV tracking.

Expected Results: PV tracks SP with less than about 20% overshoot and settles within a few oscillation periods. Steady-state offset is zero (integral action). Entering manual holds the output without a bump, and returning to auto resumes smoothly.

Review Questions: Appendix F.2, Lab 4.

Lab 5: Add and Configure a FLEX I/O Bank

Objective: Add a 1794-AENTR EtherNet/IP adapter and three I/O modules to the Studio 5000 I/O configuration tree, set RPI values, verify online connection, and map tags to engineering names.

Equipment / Tools: Studio 5000, ControlLogix training controller with EtherNet/IP network, physical 1794-AENTR adapter with 1794-IB16, 1794-IE8, and 1794-OE4 modules (or training rig), network switch, laptop.

Safety Precautions: Confirm no live process I/O is connected to the FLEX bank used for this lab. Confirm IP address assigned to the training adapter does not conflict with production devices.

Background Theory: Adding a FLEX I/O bank to Studio 5000 creates an implicit EtherNet/IP I/O connection between the ControlLogix controller and the 1794-AENTR adapter. The adapter communicates with all I/O modules on its bank and presents their data to the controller in a single I/O image. The RPI governs how frequently this image is refreshed.

Procedure

1. Open the Studio 5000 project. Right-click "I/O Configuration" → "New Module." Search "1794-AENTR." Select and click Create.
2. In Module Properties: Name = "FLEX_TRAIN_01"; IP Address = [training adapter IP, e.g., 192.168.1.20]; set RPI = 20.0 ms. Click OK.
3. Expand FLEX_TRAIN_01 in I/O tree. Right-click → New Module. Add 1794-IB16 in Slot 1; name = "DI_Bank_1"; RPI = 20 ms.
4. Add 1794-IE8 in Slot 2; name = "AI_Bank_1"; set signal type = Current (4–20 mA) for all 8 channels; RPI = 20 ms.
5. Add 1794-OE4 in Slot 3; name = "AO_Bank_1"; set signal type = Current (4–20 mA) for all 4 channels; set the fault state per the loop drawing (hold last value or a defined safe value) and document the rationale; RPI = 20 ms.
6. Go online. Observe I/O tree — all modules should show green status within 30 seconds.
7. In Controller Tags, verify the auto-generated module tags for the DI word, AI channel 0 data, and AO channel 0 data (for example FLEX_TRAIN_01:1:I.Data[0], FLEX_TRAIN_01:2:I.Ch0Data, FLEX_TRAIN_01:3:O.Ch0Data).
8. Create alias tags: DI_Test_Input → FLEX_TRAIN_01:1:I.Data.0 (first DI channel). AI_Test_Raw → FLEX_TRAIN_01:2:I.Ch0Data. AO_Test_CMD → FLEX_TRAIN_01:3:O.Ch0Data.

NOTE
Module tag names come from the module profile and may differ by profile version. Verify on bench before field use.

9. In Studio 5000 online, observe DI_Test_Input; toggle a field switch and confirm the bit changes. Observe AI_Test_Raw; connect a loop calibrator at 12 mA and confirm a raw count of approximately 15420 (4–20 mA mode).
10. Write 15420 to AO_Test_CMD (12 mA). Measure the output at the AO terminals with a milliammeter and confirm 12.0 mA ± 0.1 mA.

11. Disconnect the Ethernet cable to the adapter. Observe the module status change in Studio 5000. Reconnect the cable and confirm automatic recovery within a few seconds.
12. Change RPI to 5 ms. Monitor task execution time — note any increase. Change back to 20 ms.

Expected Results: All three modules communicate normally. Alias tags map correctly. AO output matches commanded mA value within ±0.1 mA. System recovers automatically after cable reconnection.

Review Questions: Appendix F.2, Lab 5.

Lab 6: Create an InTouch HMI Faceplate

Objective: Build a pressure controller faceplate in InTouch WindowMaker with PV bar display, SP entry field, Auto/Manual mode toggle, output bargraph, HiHi alarm indicator, and connect to live PLC tags via OPC.

Equipment / Tools: AVEVA InTouch HMI WindowMaker, RSLinx or KEPServerEX OPC server configured to the training controller, training controller with SEP_PIC_101 PID configured (Lab 4 / Snippet 4), laptop.

Safety Precautions: Connect to the training controller only, not the production system. Do not write values to production PLC tags.

Background Theory: An HMI faceplate (or controller popup) provides operators with a standardized view and interaction interface for a single control loop. Standardized faceplates reduce operator training time, improve situational awareness, and reduce errors during manual intervention. Every analog controller on the gas plant HMI should use a consistent faceplate design.

Procedure

1. Open WindowMaker. Create a new window (fixed popup type, 350×500 pixels). Name: "SEP_PIC_101_FACEPLATE".
2. Add a title bar text object: "SEP-PIC-101 — Separator Pressure Controller". Font: Arial 12 pt bold. Dark text on a light gray background (ISA-101 style).
3. Add a vertical bar graph for the PV. Properties: Min = 0, Max = 300, fill direction upward. Bind to SEP_PT_101 (PLC source). Color: dark gray when normal; alarm color only in alarm. Size: 40×200 px.
4. Add text labels: "PV" above the bargraph, numeric value display below (Expression: SEP_PT_101, format ##0.0 PSIG).
5. Add a vertical bar graph for the output. Bind to SEP_PIC_101.OUT (PLC source). Range: 0–100%. Label: "OUT %".
6. Add an input field for setpoint entry. Bind to SEP_PIC_101.SP (PLC source). Numeric input with range validation 0–300. Label: "SP (PSIG)".
7. Add MAN and AUTO pushbuttons that set the request bits HMI_PIC101_ManReq and HMI_PIC101_AutoReq with a Set action (or the script in Section 6.5). The PLC clears the requests. Add a mode indicator bound to SEP_PIC_101.SWM ("MAN" when 1, "AUTO" when 0) using gray-scale text.

> **NOTE**
> The Set action configuration for discrete pushbuttons differs between InTouch versions. Verify on bench before field use.

8. Add a PAHH indicator (rectangle) bound to the PLC status bit SEP_PT_101.PAHH_Trip (PAHH = 270 PSIG example). It fills red when the bit is 1. Do not recompute the comparison in the HMI.
9. Add a mini trend (if supported): 5-minute trend of SEP_PT_101 and SEP_PIC_101.SP. 2 pens, distinguished by line style or gray shade.
10. Configure OPC I/O server access names in WindowMaker: Access Name = "PLC1", Server = RSLinx or KEPServerEX, Topic = ControlLogix_Training.
11. Switch to WindowViewer. Navigate to the new faceplate. Verify PV updates live. Change SP from keyboard. Toggle Auto/Manual. Confirm output bargraph moves when in Auto.
12. Simulate a PAHH condition on the training controller by injecting a signal above 270 PSIG at the training AI channel with the loop calibrator (no forcing). Confirm the PAHH indicator activates on the faceplate.

Expected Results: The faceplate displays live PV, accepts SP changes within limits, sends mode requests that the PLC executes, and the PAHH indicator activates on the high-pressure simulation.

Review Questions: Appendix F.2, Lab 6.

Lab 7: MSG Read — Inter-PLC Communication

Objective: Configure a MSG instruction to read a REAL tag from a peer controller over EtherNet/IP with periodic polling, stale-data detection, and error capture, and verify the behavior on a cable pull.

Equipment / Tools: Two bench controllers (1769-L33ER at v36) on the same Ethernet network, Studio 5000, laptop.

Safety Precautions: Bench controllers only. Confirm IP addresses do not conflict with the production network. Do not perform this lab on a production system without engineering authorization.

Background Theory: Explicit MSG instructions allow a ControlLogix controller to read from or write to tags in another controller on demand. This is used extensively in gas plants to exchange data between the main process PLC and compressor package OEM PLCs, metering computers, and analyzer systems.

Procedure

1. On the target controller (Controller B, example IP 192.168.1.30), create a controller-scoped REAL tag "Comp_Speed_Fdbk" with External Access = Read Only (or Read/Write). Set its value to 1800.0.
2. On the source controller (Controller A), add Controller B to the I/O configuration under the Ethernet port so that the message path can be browsed.
3. In Controller A, create tags: Comp_MSG_Read (MESSAGE), Comp_Speed_Remote (REAL), Comp_Speed (REAL), Comp_MSG_Poll and Comp_MSG_Stale (TIMER), Comp_Comms_Fault (BOOL), Comp_MSG_ErrCode and Comp_MSG_ExtErr (DINT).
4. Configure Comp_MSG_Read: Message Type = CIP Data Table Read; Source Element = Comp_Speed_Fdbk; Number of Elements = 1; Destination = Comp_Speed_Remote; Path = browse to Controller B in the I/O tree; Connected and Cache Connections enabled.
5. Enter the rungs below in Controller A.

Rung 0 – Poll timer
```
XIO(Comp_MSG_Read.EN)TON(Comp_MSG_Poll,1000,0);
```
Rung 1 – Send
```
XIC(Comp_MSG_Poll.DN)MSG(Comp_MSG_Read);
```
Rung 2 – Stale-data timer
```
XIO(Comp_MSG_Read.DN)TON(Comp_MSG_Stale,5000,0);
```
Rung 3 – Communications fault
```
XIC(Comp_MSG_Stale.DN)OTE(Comp_Comms_Fault);
```
Rung 4 – Capture error codes
```
XIC(Comp_MSG_Read.ER)MOVE(Comp_MSG_Read.ERR,Comp_MSG_ErrCode)MOVE(Comp_MSG_Read.EXERR,Comp_MSG_ExtErr)
;
```
Rung 5 – Use data only when healthy
```
XIO(Comp_Comms_Fault)MOVE(Comp_Speed_Remote,Comp_Speed);
```

6. Download and run. Observe .EN and .DN cycle about once per second and Comp_Speed = 1800.0.
7. Change Comp_Speed_Fdbk in Controller B to 2400.0. Confirm Comp_Speed updates within about 1–2 s.
8. Cable-pull test: disconnect Controller B from the network. Observe .ER, record Comp_MSG_ErrCode and Comp_MSG_ExtErr, and time how long until Comp_Comms_Fault sets. Confirm Comp_Speed holds its last good value.
9. Reconnect Controller B. Confirm reads resume and Comp_Comms_Fault clears.
10. Record the timeout settings: .UnconnectedTimeout (default 30 s) and, for connected messages, .ConnectionRate × .TimeoutMultiplier (default 7.5 s × 4).

NOTE

MSG path configuration and the error codes observed on the 1769-L33ER. Verify on bench before field use.

Expected Results: Comp_Speed follows Comp_Speed_Fdbk in Controller B within about 1–2 s. On a cable pull, the error code is captured and Comp_Comms_Fault sets 5 s after the last successful read; Comp_Speed holds its last good value. The system recovers automatically when the network is restored.

Review Questions: Appendix F.2, Lab 7.

Lab 8: BPCS Trip String and Time-Limited Bypass (Not SIS / Not ESD SIF)

WARNING — BPCS Training Logic

This is BPCS training logic. The output is not an ESD final element, and this bypass pattern is never copied into a SIS. A production SIF bypass requires a bypass permit, safety-engineer approval, a time limit, and an operator display, and is outside the technician authorization envelope (Section 0.2).

Objective: On a training controller only, build a BPCS trip string with PAHH, LAHH, and manual trip inputs; add a maintenance bypass with key-switch authorization and a self-expiring timer; and document the bypass as a note on a training C&E worksheet. This lab does not authorize SIS work.

Equipment / Tools: Studio 5000, bench controller with bench DI switches (NC trip inputs and a key switch), laptop, C&E matrix template (Section 9.4).

Safety Precautions: Training controller only. Use a short bypass duration for training (5 minutes). In production, every BPCS trip bypass requires written authorization (bypass permit), a time limit, an active operator notification, and removal immediately on completion of the maintenance task. SIS bypasses follow the plant functional-safety procedure.

Background Theory: BPCS trip bypasses are sometimes used for testing process interlocks. SIS/SIF bypasses are a different, higher-authority activity and are not performed in this lab. A bypassed trip leaves the process with less protection. Best practice requires (1) authorization, (2) a time-limited bypass that expires automatically, (3) a continuous alarm informing operators that the trip is bypassed, and (4) increased operator vigilance during the bypass period.

Procedure

1. Create a program "BPCS_Trip_Lab". Create tags: SEP_PAHH_OK, SEP_LAHH_OK, MAN_TRIP_OK (BOOL aliases to bench NC switches, 1 = healthy); BYP_KEY (BOOL alias to the bench key switch); SEP_Byp (UDT with members TON (TIMER), Expired, Sup_Reset, Alm (BOOL)); SEP_Lab_Causes_OK, SEP_Trip_OK, SEP_Trip_Alm (BOOL); Area.HMI_Reset, Area.Reset_ONS, Area.Reset_Pulse (BOOL).
2. Enter the rungs below (Pattern A1 reset pulse, bypass timer, trip string, and the Snippet 6 latch and reset).

Rung 0 – Reset pulse
```
XIC(Area.HMI_Reset)ONS(Area.Reset_ONS)OTE(Area.Reset_Pulse);
```
Rung 1 – Clear HMI reset request
```
XIC(Area.HMI_Reset)OTU(Area.HMI_Reset);
```
Rung 2 – Bypass timer (5 min, key on, not expired)
```
XIC(BYP_KEY)XIO(SEP_Byp.Expired)TON(SEP_Byp.TON,300000,0);
```
Rung 3 – Expiry latch
```
XIC(SEP_Byp.TON.DN)OTL(SEP_Byp.Expired);
```
Rung 4 – Re-arm only with key off and supervisor reset
```
XIO(BYP_KEY)XIC(SEP_Byp.Sup_Reset)OTU(SEP_Byp.Expired);
```
Rung 5 – Clear supervisor reset request
```
XIC(SEP_Byp.Sup_Reset)OTU(SEP_Byp.Sup_Reset);
```
Rung 6 – Trip causes (PAHH bypassed only while timing)
```
[XIC(SEP_PAHH_OK) ,XIC(SEP_Byp.TON.TT) ]XIC(SEP_LAHH_OK)XIC(MAN_TRIP_OK)OTE(SEP_Lab_Causes_OK);
```
Rung 7 – Bypass alarm
```
[XIC(BYP_KEY) ,XIC(SEP_Byp.TON.TT) ]OTE(SEP_Byp.Alm);
```
Rung 8 – Trip (and first scan)
```
[XIO(SEP_Lab_Causes_OK) ,XIC(S:FS) ]OTU(SEP_Trip_OK);
```
Rung 9 – Reset with causes healthy
```
XIC(Area.Reset_Pulse)XIC(SEP_Lab_Causes_OK)OTL(SEP_Trip_OK);
```
Rung 10 – Trip alarm
```
XIO(SEP_Trip_OK)OTE(SEP_Trip_Alm);
```

3. Download and run. With all switches healthy, set Area.HMI_Reset: SEP_Trip_OK = 1.
4. Trip test: open the PAHH switch. SEP_Trip_OK = 0 and SEP_Trip_Alm = 1. Close the switch: the trip remains until reset. Reset.
5. Bypass test: turn BYP_KEY on. Confirm SEP_Byp.Alm = 1 and the bypass timer is timing. Open the PAHH switch: no trip. Open the LAHH switch: trip (the bypass covers PAHH only). Close both switches and reset.
6. Expiry test: turn BYP_KEY on and open the PAHH switch. When the timer expires, SEP_Byp.Expired = 1 and the trip occurs. The bypass alarm remains on while the key is on.
7. Re-arm: turn BYP_KEY off and set SEP_Byp.Sup_Reset (supervisor reset): SEP_Byp.Expired clears. Confirm the bypass cannot restart until the key is cycled and the supervisor reset is given.
8. Mode-change test: switch the controller to Program and back to Run. SEP_Trip_OK = 0 after the transition (comes up tripped); a reset is required.
9. Fill in the C&E matrix template (Section 9.4): add the PAHH bypass as a note on the SEP-PAHH-101 row, stating the bypass conditions and time limit.

Expected Results: The trip occurs on any un-bypassed cause. The bypass covers PAHH only, for at most 5 minutes. The bypass alarm is continuous while the key is on or the timer is running. Protection restores automatically on expiry, and the trip string comes up tripped after a mode change.

Review Questions: Appendix F.2, Lab 8.

Lab 9: Network Troubleshooting — Find a Lost FLEX I/O Node

Objective: Simulate a FLEX I/O adapter communication loss, use systematic diagnostics (RSLinx browsing, ping, Studio 5000 I/O diagnostics, LED status) to identify and restore the lost node. Document RPI considerations.

Equipment / Tools: Studio 5000, RSLinx Classic or FactoryTalk Linx, training FLEX I/O bank (1794-AENTR), Ethernet network switch, laptop with command prompt access.

Safety Precautions: Training equipment only. On a production system, any deliberate disconnection of a FLEX I/O bank causes loss of that I/O — notify operator and get authorization before any network cable manipulation.

Background Theory: EtherNet/IP uses a keep-alive mechanism: if the controller does not receive an I/O update from the adapter within a defined connection timeout period, it declares a connection fault. The timeout is typically 4× the RPI value. Understanding LED status codes is faster than launching software diagnostics in an emergency.

Procedure

1. Start with the FLEX bank fully operational and all modules showing green in Studio 5000. Record the normal LED states on the 1794-AENTR: MS (Module Status) = solid green; NS (Network Status) = solid green (connection established).
2. Disconnect the Ethernet cable from the 1794-AENTR adapter. Start a stopwatch.
3. In Studio 5000 (online), observe the I/O tree and note when the adapter shows a fault. The controller declares the connection fault after about 4 × RPI; the software display may take longer to refresh.
4. Observe the AENTR adapter LEDs: MS remains solid green (module powered and healthy); NS flashes red (connection timeout).

> **NOTE**
> AENTR NS LED behavior on connection timeout. Verify on bench before field use.

5. At the Windows command prompt: ping 192.168.1.20 (training adapter IP). Observe "Request timed out" — no response confirms adapter is unreachable.
6. Open RSLinx Classic → Communications → RSWho (Who Active). Browse the EtherNet/IP driver — the adapter should not appear in the browse tree (confirms network-level disconnection).
7. Physically check the cable: inspect both ends (adapter port and switch port) for a seated RJ-45 connector. Check switch port LED — if no LED, port is not detecting link.
8. Reconnect the Ethernet cable. Observe switch port LED lights up (link established).
9. At command prompt: ping 192.168.1.20 -n 10. Confirm all 10 packets receive replies with low latency (<5 ms).
10. In Studio 5000, observe I/O tree — adapter recovers automatically. Record recovery time.
11. RS Who Browse — confirm adapter reappears. All modules show green status.
12. Experiment with RPI: change the adapter RPI to 5 ms. Disconnect the cable again; the connection timeout is now about 20 ms. Reconnect and change the RPI back to 20 ms. Trade-off: lower RPI gives faster fault detection but higher network load.
13. Document findings: fault detection time at 20 ms RPI, recovery time, LED state descriptions.

Expected Results: Connection fault declared after about 4 × RPI (≈ 80 ms at 20 ms RPI); the I/O tree display may lag. Recovery is automatic on cable reconnection. All previous I/O states are restored. LED diagnostics identify the fault before software tools.

Review Questions: Appendix F.2, Lab 9.

Lab 10: Alarm Rationalization Workshop

Objective: Apply ISA-18.2 alarm rationalization methodology to 10 existing HMI alarms, fill out rationalization worksheets, reassign priorities, adjust setpoints and deadbands, and document changes.

Equipment / Tools: ISA-18.2 standard (or facility alarm philosophy document), alarm rationalization worksheet template (Section 6.3), HMI alarm history data (30-day period), laptop.

Safety Precautions: All setpoint changes must be reviewed and approved by the process engineer before implementation. Do not change alarm setpoints on a production system without MOC authorization.

Background Theory: ISA-18.2 requires that every alarm be individually rationalized — meaning the cause, consequence, operator response, priority, setpoint, and deadband are documented and justified. The goal is to ensure alarms demand operator action, are unique and meaningful, and do not overwhelm the operator during upset conditions.

Procedure

1. Review the 30-day alarm history report. Identify the top 10 most frequent alarms (bad actors).
2. For each alarm, fill out the rationalization worksheet (Section 6.3 template): Tag, Description, Cause, Consequence, Existing Safeguards, Proposed Priority (1–4), Proposed Setpoint, Proposed Deadband, Required Operator Action.
3. For each alarm, determine if it meets the ISA-18.2 definition of a true alarm: "An audible and/or visible means of indicating to the operator an equipment malfunction, process deviation, or abnormal condition requiring a timely response."
4. Identify any alarms that are NOT true alarms (e.g., informational only, status flags, equipment position feedback). Recommend removal or reclassification as "advisory" (no sound, no operator action required).

5. Identify any chattering alarms (activating more than once per 10 minutes over the 30-day period). For each: increase deadband, add time delay, or move setpoint further from normal operating point.
6. Assign ISA-18.2 priority to each alarm: Critical (1), High (2), Medium (3), Low (4). Justify each assignment based on consequence severity and required response time.
7. Document setpoint rationalization: compare existing setpoint to normal operating range. Ensure adequate margin between normal operation and alarm setpoint to avoid nuisance alarms during normal transients.
8. Present the 10 rationalized alarms in a completed worksheet table.
9. Identify any alarms that should be state-based (active only in certain operating modes, e.g., active only when compressor is running).
10. Write a brief summary report: total alarms reviewed, number reclassified, priority changes, setpoint changes, expected reduction in alarm rate.

Expected Results: All 10 alarms have completed rationalization worksheets. At least 2–3 alarms are reclassified or have adjusted setpoints/deadbands. Summary report estimates projected alarm rate reduction.

Review Questions: Appendix F.2, Lab 10.

Lab 11: Find and Clear Leftover Forces and Inhibits

Objective: On a training controller, locate planted I/O forces, an inhibited FLEX connection, and a shelved-style bypass bit; clear them using the Section 0.6 checklist; document what would have happened if they had been left in place on a live unit.

Equipment / Tools: Studio 5000 online to the training PLC, Section 0.6 checklist, supervisor to plant the faults.

Safety Precautions: Training controller only. If a production controller is browsed accidentally, disconnect and notify the supervisor immediately. A force on a production controller is never cleared on the basis of a lab procedure.

Background Theory: Forces override program logic. Inhibited modules freeze or fault I/O. Both persist longer than expected, and operators may continue operating on false indications.

Procedure

1. The supervisor plants at least three issues: one digital force, one analog force or test MOVE, and one inhibited module or bypass bit.
2. Without being told which tags, use the controller status, the Force table, I/O tree icons, and a tag search for Bypass / Force / Inhibit.
3. Record tag, type of trap, as-found state, and process implication in one sentence each.
4. Clear only after the supervisor agrees. Confirm force table empty and I/O tree green.
5. Repeat the Section 0.6 checklist aloud as if handing the unit back to operations.

Expected Results: All planted traps found and cleared. Checklist completed with no remaining forces.

Review Questions: Appendix F.2, Lab 11.

Lab 12: First-Out Narrative from Historian / SOE

Objective: Given a recorded training-trip or a replayed event, build a 10-line first-out narrative: what went abnormal first, which alarms followed, which trip fired, and what the as-left state was. Do not guess when the timestamp is missing.

Equipment / Tools: Plant historian or training trend pack, HMI alarm summary export if available, paper timeline template.

Safety Precautions: Read-only. Do not change historian archives, alarm files, or live setpoints.

Background Theory: After a unit trip, the first question is what happened first. HMI snapshots mislead when only the current faceplate is examined. Sequence-of-events records and analog trends are the record.

Procedure

1. Identify the event window (start 15 minutes before the trip, end when the unit is safe).
2. List analog excursions with timestamps (pressure, level, temperature, flow).
3. List discrete trips and operator actions with timestamps (XV closed, motor stop, SEP_Trip_OK dropped; training tags only).
4. Mark the first-out candidate and one sentence why later alarms are consequences.
5. Write a 10-line narrative a night-shift supervisor can read in one minute.
6. Note data gaps (bad quality, inhibited module, missing SOE point).

Expected Results: A timeline that another technician can verify against the same trends.

Review Questions: Appendix F.2, Lab 12.

NOTES

Section 11: Quick Reference

This section is a one-page summary for workstation use. Appendix E is the field pocket card.

Controller Status Indicators — ControlLogix 1756-L8x

Indicator	State	Meaning
RUN	Solid green	Controller in Run mode
RUN	Off	Controller in Program or Test mode, or faulted
FORCE	Solid amber	I/O forces enabled; forced values may be active
FORCE	Flashing amber	Forces present in the project but not enabled
FORCE	Off	No forces present
SD	Flashing green	SD card activity; do not remove the card or power down
OK	Solid green	Controller OK
OK	Flashing red	Recoverable major fault; check the fault log
OK	Solid red	Nonrecoverable fault, or controller completing power-up
Status display	Scrolling text	Controller state, IP address, and fault messages (including I/O faults)
LINK (Ethernet port)	Green, steady or flickering	Link established; flicker indicates traffic
LINK (Ethernet port)	Off	No Ethernet link; check cable and switch port

NOTE

Indicator behavior varies by controller series and firmware; refer to the controller user manual status-indicator tables. Verify on bench before field use.

Common Ladder Instructions — Quick Reference (v36 names)

v36 name (legacy)	Name	One-line description
XIC	Examine If Closed	TRUE when bit = 1; used for health (_OK) bits on the run path
XIO	Examine If Open	TRUE when bit = 0; used for fault or request bits that must be 0 to proceed
OTE	Output Energize	Bit = 1 while rung TRUE, 0 when FALSE; cleared at prescan
OTL	Output Latch	Set bit = 1 and hold (retentive) until OTU
OTU	Output Unlatch	Reset bit = 0 (clears an OTL latch)
ONS	One Shot	TRUE for one scan on a 0→1 transition; storage bit set at prescan
TON	Timer On Delay	Accumulate while rung TRUE; .DN when ACC ≥ PRE
TOF	Timer Off Delay	Time after rung goes FALSE; .DN drops when ACC ≥ PRE
RTO	Retentive Timer On	Accumulate (retained on rung FALSE); reset with RES
CTU	Count Up	Increment ACC on rising edge; .DN when ACC ≥ PRE
CTD	Count Down	Decrement ACC on rising edge; .DN when ACC ≥ PRE
GT (GRT)	Greater Than	TRUE when Source A > Source B
LT (LES)	Less Than	TRUE when Source A < Source B
EQ (EQU)	Equal	TRUE when Source A = Source B (avoid on REAL values)
MOVE (MOV)	Move	Copy Source value to Destination
ADD / SUB	Add / Subtract	Destination = A + B or A − B
PID	PID Controller	PID algorithm; periodic task, .UPD = task period
MSG	Message	Explicit read/write to a remote device; starts on a FALSE→TRUE rung transition
JSR	Jump to Subroutine	Call a named routine with parameters
JMP / LBL	Jump / Label	Skip rungs within a routine; not used in interlocks (Appendix G)

4–20 mA Quick Math

```
% of range  = (mA − 4) ÷ 16 × 100
EU value    = EU_Min + ((mA − 4) ÷ 16) × (EU_Max − EU_Min)
mA output   = ((EU − EU_Min) ÷ (EU_Max − EU_Min)) × 16 + 4

EXAMPLES (range 0–300 PSIG):
   4 mA →   0 PSIG (0%)       12 mA → 150 PSIG (50%)
   8 mA →  75 PSIG (25%)      16 mA → 225 PSIG (75%)
                              20 mA → 300 PSIG (100%)

Fault thresholds (NAMUR NE43):
   ≤ 3.6 mA  = fail-low (open circuit / transmitter fault)
   ≥ 21.0 mA = fail-high
   20.5 mA   = typical over-range (saturation) limit
```

Calibration 5-Point Expected mA Values

Input %	Expected mA	Tolerance (±0.1%FS)
0%	4.000 mA	±0.016 mA
25%	8.000 mA	±0.016 mA
50%	12.000 mA	±0.016 mA
75%	16.000 mA	±0.016 mA
100%	20.000 mA	±0.016 mA

PID Tuning Rules of Thumb

- **Start conservatively:** begin with Kc = 0.5, integral and derivative off. Increase Kc until the loop responds without oscillation before adding integral action.
- **Integral eliminates offset:** Add Ki gradually to remove steady-state error. Too much Ki → oscillation and windup.
- **Derivative on noise:** Avoid Kd for noisy pressure signals. Use only on slow, smooth processes (temperature) and only if loop is stable with P+I first.
- **Ziegler-Nichols (closed loop):** PI: Kc = 0.45 × Ku, Ti = Tu / 1.2. PID: Kc = 0.6 × Ku, Ti = Tu / 2, Td = Tu / 8. Ti and Td are in the time units of Tu; convert to the PID form in use (Appendix B).
- **If oscillating:** reduce Kc by 20% first. If slow with offset: increase integral action by 10%. Kc and integral action are not adjusted at the same time.

I/O Connection Fault Codes (FLEX I/O and 1756 Modules)

Code	Meaning	Action
16#0203	Connection timed out (communications lost after the connection was established)	Check cable, switch port, adapter power, and IP; ping the adapter
16#0204	Connection request timed out (module or adapter not responding)	Check IP address, path, and adapter power
16#0116	Electronic keying mismatch (module type, vendor, or revision)	Verify the catalog number and revision against the I/O configuration
Channel status bits	Under-range, over-range, or channel fault	Check field wiring and device; measure mA at the terminals
Module removed	Connection fault on the affected slot only	Normal during hot-swap; reinsert the module

NOTE

Codes appear on the module properties Connection tab in Studio 5000. Verify on bench before field use.

HMI Alarm Priority Presentation

Priority	HMI presentation	Audible	Response time
1 — Critical	Red; flashing until acknowledged	Yes — distinct urgent tone	< 5 min

Priority	HMI presentation	Audible	Response time
2 — High	Site high-priority color; flashing until acknowledged	Yes — alarm tone	5–15 min
3 — Medium	Site medium-priority color	Yes — lower-priority tone	15–60 min
4 — Low	Site low-priority color	No	Next opportunity
Acknowledged	Same priority color, steady	No (silenced)	Still active — monitor
Return to normal (unacknowledged)	Per site HMI style guide	No	Acknowledge to clear

H_2S Exposure Limits and Actions — Quick Reference

Concentration	Action
< 1 ppm	Normal monitoring (ACGIH TLV 1 ppm, 8-h TWA); note any odor
1–5 ppm	Investigate source; verify ventilation (ACGIH STEL 5 ppm)
≥ plant action level (commonly 10 ppm)	Respiratory protection per the plant H_2S program (SCBA or supplied air); evacuate non-essential personnel (NIOSH REL 10 ppm, 10-min ceiling)
100 ppm	NIOSH IDLH — emergency response; SCBA mandatory; evacuation
> 500 ppm	Rapid incapacitation likely; rescue only by a trained team with SCBA

NOTE
The plant H_2S program governs action levels. OSHA general industry limits are a 20 ppm ceiling and a 50 ppm peak; OSHA does not define an action level for H_2S.

EtherNet/IP Troubleshooting — Top 5

1. **Ping the device first.** If ping fails, the problem is at Layer 1 (cable) or Layer 3 (IP address/routing) — fix network before troubleshooting software.
2. **Check for IP address conflict.** Use arp -a at command prompt to verify MAC addresses. Two devices with the same IP = communication chaos.
3. **Verify RPI settings.** Too-low RPI causes connection timeouts and task overruns. Increase RPI for non-critical remote I/O to 20–50 ms.
4. **Check switch port statistics.** High error counts or discards on a port indicate cable damage, duplex mismatch, or device failure — replace cable or check connector.
5. **Use RSLinx Who-Active for browsing.** If a device appears in RSLinx but not in Studio 5000 I/O tree — check if the I/O connection is inhibited in the module properties.

Studio 5000 Common Commands

Command	Location
Who Active (browse network)	Communications → Who Active
Go Online	Communications → Go Online
Download / Upload	Communications → Download / Upload
Controller mode	Communications → Run Mode / Program Mode (or the online toolbar mode menu)
Find text or tag	Search → Find (Ctrl+F)
Cross Reference	Right-click tag → Go to Cross Reference
Forces	Online toolbar Forces menu → Enable / Disable / Remove All
Instruction help	Select instruction → F1
Properties	Right-click item → Properties

NOTE
Menu locations and keyboard shortcuts vary by Studio 5000 version. Verify on bench before field use.

Key ISA Tag Function Codes

Tag Code	Full Name	Example
PT	Pressure Transmitter	SEP-PT-101
TT	Temperature Transmitter	COMP-TT-201
FT	Flow Transmitter	METER-FT-501
LT	Level Transmitter	SEP-LT-101
AT	Analyzer Transmitter	AMINE-AT-301
XV	On/Off (Block or Shutdown) Valve, discrete	ESD-XV-001
PCV	Pressure Control Valve (modulating)	SEP-PCV-101
LCV	Level Control Valve (modulating)	AMINE-LCV-201
FCV	Flow Control Valve (modulating)	DEHYD-FCV-301
SDV	Shutdown Valve	COMP-SDV-201
MOV	Motor-Operated Valve	SEP-MOV-102
ESD	Emergency Shutdown Device/System	ESD-XV-001

NOTES

Appendix A: Glossary of Terms

Term	Definition
AI	Analog Input — a module channel that receives a variable signal (typically 4-20 mA) from a field sensor and converts it to a digital value for the PLC processor.
AO	Analog Output — a module channel that converts a digital PLC value to an analog signal (typically 4-20 mA or 0-10 VDC) to drive a field actuator or positioner.
AOI	Add-On Instruction — a user-defined, reusable custom instruction in Studio 5000 with its own logic, parameters, and local tags, appearing like a native PLC instruction.
BPCS	Basic Process Control System — the primary PLC/DCS used for process control (maintaining setpoints). Distinguished from the SIS, which handles safety functions independently.
CIP	Common Industrial Protocol — the application-layer protocol used by EtherNet/IP, DeviceNet, and ControlNet for object-based communication between Rockwell Automation devices.
DCS	Distributed Control System — a process control architecture where control functions are distributed across multiple controllers and I/O nodes, typically used in large continuous processes.
DI	Digital Input — a module channel that reads a discrete (on/off) voltage signal from a field device (limit switch, pushbutton, relay contact) as a BOOL value in the PLC.
DLR	Device Level Ring — an EtherNet/IP ring topology that provides fault-tolerant network communication by reconfiguring from ring to linear topology in <3 ms when a cable break is detected.
DO	Digital Output — a module channel that drives a discrete field device (solenoid, relay coil, pilot light, contactor) ON or OFF based on a BOOL value from the PLC.
ESD	Emergency Shutdown — a safety action or system that drives a process to a defined safe state on detection of a hazardous condition. ESD SIFs are implemented in a SIS (or hardwired relay system) independent of the BPCS; BPCS trips are not ESD SIFs.
FBD	Function Block Diagram — an IEC 61131-3 graphical programming language where function blocks (PID, analog I/O, arithmetic) are connected by signal lines, resembling an instrument diagram.
FLEX I/O	Allen-Bradley 1794 FLEX I/O — a distributed, modular I/O system where modules snap onto terminal bases on a DIN rail, connected to the ControlLogix controller via EtherNet/IP or DeviceNet.
FSM	Functional Safety Management — the organizational and procedural framework required by IEC 61511 for managing all activities related to the safety lifecycle of a SIS.
HART	Highway Addressable Remote Transducer — a bi-directional digital communication protocol superimposed on the 4-20 mA signal using FSK at 1,200 baud, allowing remote configuration and diagnostics of field instruments.
HMI	Human-Machine Interface — the operator interface (screen, keyboard, touchscreen) for monitoring and controlling a process. This manual covers AVEVA (Wonderware) InTouch HMI and AVEVA System Platform.
IDLH	Immediately Dangerous to Life or Health — NIOSH-defined concentration of a hazardous substance that poses an immediate threat to life or causes irreversible health effects. H_2S IDLH = 100 ppm.
IEC 61511	International standard: "Functional Safety — Safety Instrumented Systems for the Process Industry Sector." Defines the lifecycle requirements for specification, design, installation, operation, and maintenance of SIS in the process industries. SIS devices are certified to IEC 61508 or justified by prior use.
ISA	International Society of Automation (formerly Instrument Society of America) — the professional organization that publishes instrument standards (ISA-5.1 for symbology, ISA-18.2 for alarms, ISA-84 for SIS) used throughout this manual.
ISA-101	HMI standard (ISA-101.01) for display design; recommends gray-scale normal displays with color reserved for abnormal conditions.
JSR	Jump to Subroutine — a ControlLogix ladder instruction that transfers execution to a named routine, passes parameters, and returns after the routine executes RET.
LOPA	Layers of Protection Analysis — a semi-quantitative risk assessment method that evaluates whether existing and proposed protection layers (BPCS, alarms, SIS, PSV, dikes) provide sufficient risk reduction for a specific accident scenario.
LOTO	Lockout/Tagout — OSHA-mandated procedure (29 CFR 1910.147) for controlling hazardous energy (electrical, pneumatic, chemical) before performing maintenance on equipment to prevent accidental startup or energy release.
LRV	Lower Range Value — the minimum of an instrument's calibrated span, corresponding to 4 mA output (0% of range). Example: LRV = 0 PSIG for a 0–300 PSIG pressure transmitter.
MOC	Management of Change — a formal process (required by OSHA PSM 29 CFR 1910.119) for reviewing, documenting, approving, and communicating changes to process equipment, instrumentation, control systems, or procedures before implementation.

Term	Definition
MSG	Message Instruction — a ControlLogix ladder instruction that performs on-demand, explicit CIP data transfer (read or write) between the controller and another device or controller over EtherNet/IP.
NE43	NAMUR recommendation NE43 — transmitter failure signaling on 4–20 mA loops: ≤ 3.6 mA or ≥ 21 mA indicates failure.
OPC	OLE for Process Control (now Open Platform Communications) — a standardized software interface specification (OPC DA, OPC UA) for data exchange between industrial software applications and hardware devices, enabling HMI-to-PLC communication.
OTE	Output Energize — a ControlLogix coil instruction that sets a BOOL tag to 1 when the rung is TRUE and resets it to 0 when the rung is FALSE (non-retentive).
P&ID	Piping and Instrumentation Diagram — an engineering drawing showing all process piping, vessels, equipment, instruments, and control loops with their interconnections, using ISA symbols and tag numbers.
PFD	Probability of Failure on Demand — the probability that a safety function will fail to respond when a process demand occurs. The key SIS performance metric. SIL 2 requires PFD between 0.01 and 0.001.
PID	Proportional-Integral-Derivative — a control algorithm that calculates a corrective output based on the error between a process variable and its setpoint, plus the accumulated error over time (I) and the rate of error change (D).
PSV	Pressure Safety Valve — a mechanical device that automatically opens to relieve excess pressure at its set pressure (at or below the vessel MAWP), independent of any electronic system.
RPI	Requested Packet Interval — the configurable cyclic update rate (in ms) for an EtherNet/IP I/O connection between a ControlLogix controller and an I/O module or adapter.
RTD	Resistance Temperature Detector — a temperature sensor whose electrical resistance changes predictably with temperature. Pt100 (100Ω at 0°C) is most common in process applications; typically paired with a 4-wire head-mount transmitter for accurate temperature measurement.
RTO	Retentive Timer On — a ControlLogix timer that accumulates time only while the rung is TRUE and retains its accumulated value when the rung goes FALSE. Must be reset with a RES instruction. Used for running-hours meters.
SCADA	Supervisory Control and Data Acquisition — a system for remote monitoring and control of distributed field devices, data acquisition, alarming, and historical logging. AVEVA InTouch HMI / System Platform is the HMI/SCADA platform covered in this manual.
SIF	Safety Instrumented Function — a specific safety function implemented by the SIS (e.g., "high-high pressure shutdown on the inlet separator"). Each SIF has a defined SIL requirement, architecture, and proof test schedule.
SIL	Safety Integrity Level — a discrete performance level (1–4) for a safety function, defined by its Probability of Failure on Demand (PFD). Higher SIL = lower PFD = more reliable safety function.
SIS	Safety Instrumented System — an independent system (logic solver, dedicated sensors and final elements) that implements one or more SIFs, designed and managed per IEC 61511, with devices certified to IEC 61508 or justified by prior use. Operates independently from the BPCS.
ST	Structured Text — an IEC 61131-3 high-level text programming language (PASCAL-like syntax) for PLC programming, used for complex math, array operations, and algorithms not suited for ladder diagram.
TON	Timer On Delay — a ControlLogix timer that accumulates elapsed time while the rung is TRUE; resets to 0 when rung goes FALSE. .DN bit sets when accumulator ≥ preset. Most commonly used timer type.
UDT	User-Defined Type — a custom data structure in Studio 5000 that groups related variables (e.g., PV, SP, OUT, alarm bits) under one tag name, enabling consistent, structured data organization for equipment modules.
URV	Upper Range Value — the maximum of an instrument's calibrated span, corresponding to 20 mA output (100% of range). Example: URV = 300 PSIG for a 0–300 PSIG pressure transmitter.
VFD	Variable Frequency Drive — an electronic device that controls AC motor speed by varying the frequency and voltage of the power supplied to the motor. Used for compressors, pumps, and fans to match process demand without throttling losses.
XIC	Examine If Closed — a ControlLogix contact instruction that is TRUE when the referenced BOOL tag = 1. Equivalent to a normally open contact in a hardwired relay circuit.
XIO	Examine If Open — a ControlLogix contact instruction that is TRUE when the referenced BOOL tag = 0. Equivalent to a normally closed contact. Used for bits that must be 0 to proceed (fault or request bits); field health (_OK) inputs use XIC (Section 0.3).

Appendix B: Useful Formulas

4–20 mA to Engineering Units (Forward and Reverse)

```
(* FORWARD: mA signal → engineering units *)
EU  = EU_Min + ((mA - 4.0) / 16.0) * (EU_Max - EU_Min);

(* REVERSE: engineering units → expected mA *)
mA  = ((EU - EU_Min) / (EU_Max - EU_Min)) * 16.0 + 4.0;

(* Percent of span *)
Pct = (mA - 4.0) / 16.0 * 100.0;
```

Raw Counts to Engineering Units (1794-IE8 and 1756-IF8)

```
(* 1794-IE8 FLEX I/O, 4-20 mA mode: 0 counts = 4 mA, 30840 counts = 20 mA *)
EU = EU_Min + (DINT_TO_REAL(Raw_Counts) / 30840.0) * (EU_Max - EU_Min);
(* Example: Raw = 15420 (12 mA), range 0-300 PSIG → EU = 150.0 PSIG *)

(* 1756-IF8: use float (engineering units) data format with module     *)
(* scaling (Low/High Signal 4.0/20.0 mA → Low/High EU) and read the     *)
(* channel data directly. In integer mode 4 mA reads about -20030       *)
(* counts; scale from the module manual, not from 0-32767.              *)
```

NOTE

IE8 count values from the 1794 analog user manual; IF8 integer-mode counts depend on module configuration. Verify on bench before field use.

Orifice Plate Flow Calculation (Differential Pressure)

```
(* AGA-3 / ISO 5167 simplified — reference only; custody transfer uses the flow computer *)
Q_actual = Cd * (PI/4) * d^2 * SQRT(2 * dP / rho)

  Q_actual = volumetric flow rate (m3/s or ft3/s)
  Cd       = discharge coefficient (about 0.6 for a sharp-edged orifice; per AGA/ISO)
  d        = orifice bore diameter (m or ft)
  dP       = differential pressure across the orifice (Pa or lbf/ft2)
  rho      = fluid density at flowing conditions (kg/m3 or lbm/ft3)

(* For gas: rho = MW * P / (Z * R * T)                                  *)
(*   MW = molecular weight; P = absolute pressure; Z = compressibility;   *)
(*   R = gas constant; T = absolute temperature                         *)

(* In Logix (v36 name SQRT, legacy SQR): *)
IF DP_Value < 0.0 THEN DP_Value := 0.0; END_IF;     (* no MAX function in Logix ST *)
Q_flow := K * SQRT(DP_Value);
```

NOTE

DP_Value is a copy of DP_Transmitter.Value. K is a calibration constant from the flow element data sheet.

PID Output Formula (Independent Form, .PE = 0)

```
(* Independent PID form — each term acts on the error *)
E(t)   = SP - PV                (* .CA = 0, reverse acting; .CA = 1 uses PV - SP *)

OUT(t) = Kp * E(t)
       + Ki * integral(E dt)     (* integral — eliminates offset; Ki in 1/s *)
       + Kd * dE/dt              (* derivative — responds to rate; Kd in s  *)

(* Discrete form, sampled at Ts = .UPD: *)
OUT_n  = Kp * E_n
       + Ki * Ts * sum(E)
       + Kd * (E_n - E_n-1) / Ts

(* Output clamped: .MINO <= OUT <= .MAXO; integral stops accumulating while clamped *)
```

Ziegler-Nichols Tuning Formulas (Closed Loop)

Controller type	Kc	Ti (integral time)	Td (derivative time)
P only	0.5 × Ku	∞ (no integral)	0 (no derivative)
PI	0.45 × Ku	Tu / 1.2	0 (no derivative)
PID (classic)	0.6 × Ku	Tu / 2	Tu / 8
PID (some overshoot)	0.33 × Ku	Tu / 2	Tu / 3

Ku = ultimate gain — the proportional gain at which the loop sustains a steady oscillation with integral and derivative off. Tu = period of that oscillation. Ti and Td are in the time units of Tu. Logix PID conversion: with .PE = 0 (independent), KP = Kc, KI = Kc / Ti (1/s, Ti in seconds), KD = Kc × Td (s). With .PE = 1 (dependent), KP = Kc, KI = Ti (minutes per repeat), KD = Td (minutes).

Appendix C: Revision Log

All revisions to this manual are logged below. The revision log provides traceability of all changes and an audit trail for regulatory inspections.

Rev	Date	Section(s) affected	Description of change	Revised by	Approved by
1.0	September 15, 2026	All	Initial issue	[COMPANY: Author]	[COMPANY: Supervisor]
1.1	September 17, 2026	0, 3.3, 5.2, 10, 12, 13, App C–F	Working rules; BPCS vs SIS relabel; tag/discrete convention; Labs 11–12; competency matrix; pocket card; force checklist; illustrative-calc warning.	Training author	[COMPANY: Supervisor]
1.2	October 27, 2026	All	Reference-manual conversion and technical corrections; see Revision 1.2 detail below.	[COMPANY: Author]	[COMPANY: Supervisor]
1.3					
1.4					
2.0					

Revision 1.2

Changes:

- Removed personal references
- Converted notebook to reference-manual format
- Removed duplicate Lab 3
- Removed duplicate Lab 8
- Standardized formatting
- Added duplex-print layout
- Removed instructor-oriented language
- Improved printer friendliness
- Corrected BPCS/SIS terminology throughout

Additional technical changes:

- Ladder logic: rungs in Logix neutral-text format with v36 instruction names (legacy names in brackets); OTE/OTL prescan and power-up behavior documented; Snippets 1–8 corrected (seal-in with _OK health bits, unconditional PID with .SWM/.SO bumpless transfer, MSG poll/stale-data/error-capture, BPCS trip latch with first-scan trip).
- Ladder patterns added (Section 5.4): reset pulse, alarm on/off delay, first-out, bad-quality handling, time-limited bypass, motor and valve fail-to-respond, hysteresis; E-stop/MCR wiring; MTR_BASIC and XV_BASIC UDTs; JSR parameter passing; state machine with a single commit rung; ST clamps with IF.
- PID: .PE independent/dependent units, .UPD = task period, .CA direction, .MAXI/.MINI versus .MAXS/.MINS corrected; Ziegler-Nichols tables corrected.
- I/O and networks: 1794-IE8 counts (0 / 15420 / 30840), 1756-IF8 data formats, NAMUR NE43 thresholds, HART loop resistance (230–600 Ω) and polling addresses, AENTR NS LED meanings, MSG error codes, I/O connection fault codes, STRING length.
- Process and safety data: separator alarm/trip setpoints made consistent with the PSV/MAWP (PAH < PAHH < PSV ≤ MAWP); H_2S exposure limits; ISA-18.2 alarm-rate benchmarks; IEC 61511 / IEC 61508 relationship; ISA-101 gray-scale HMI guidance.
- Labs: Lab 3 rebuilt for the bench L33ER at v36 with wire-break and mode-change tests; Lab 4 dead-time simulator and Ziegler-Nichols PI; Lab 7 rebuilt with stale-data detection; Lab 8 retitled "BPCS Trip String and Time-Limited Bypass"; Labs 11–12 placed after Lab 10; review questions and answers moved to Appendix F.2.
- Training material (review cards, study plan, competency matrix, assessment notes) moved to Appendix F; low-use instructions moved to Appendix G; sources collected in Appendix H.
- Items not yet confirmed on the bench are marked "NOTE: Verify on bench before field use."
- Ladder logic peer review by Kirk.

Appendix D: Company-Specific Placeholders Checklist

The checklist below tracks all [COMPANY: ...] placeholders in this manual that require plant-specific information. Each item is checked off when filled in, and the change is logged in Appendix C (Revision Log).

☐	Section	Placeholder	Date Filled	Initials
☐	Title Page	[COMPANY: Insert company name, logo, and document control number]		
☐	Section 1.1	[COMPANY: Insert site-specific P&ID reference numbers for each stage]		
☐	Section 3.3	[COMPANY: Insert the plant's official tag naming standard document reference]		
☐	Section 3.4	[COMPANY: All IP addresses in Network Address Layout Template] (9 entries)		
☐	Section 4.3	[COMPANY: I/O Mapping Table — replace example rows with actual plant I/O; reference drawing number]		
☐	Section 4.3	[COMPANY: Add additional I/O rows] — fill all blank rows at bottom of I/O table		
☐	Section 7.1	All terminal block group numbers — verify against plant cable schedule and panel drawings		
☐	Section 7.2	[DRAWING: Elementary Wiring Diagram — Motor Control Circuit] — insert drawing number		
☐	Section 7.2	[DRAWING: Loop Wiring Diagram — 4-20mA Pressure Transmitter to PLC AI Card] — insert drawing number		
☐	Section 7.2	[DRAWING: Digital Input Wiring — Proximity Switch to DI Card] — insert drawing number		
☐	Section 7.2	[DRAWING: Digital Output Wiring — PLC DO Card to Solenoid Valve] — insert drawing number		
☐	Section 7.2	[DRAWING: ESD Loop Wiring — SIS Solenoid Valve] — insert drawing number		
☐	Section 9.4	[COMPANY: Replace C&E Matrix template with plant-specific C&E Matrix] — insert official document number		
☐	Appendix C	[COMPANY: Author name] in Rev 1.0 entry		
☐	Appendix C	[COMPANY: Supervisor name] in Rev 1.0 approval entry		
☐	Appendix E	Who-to-call table (names, radio, phone)		

CAUTION — Before This Manual Is Used as a Plant Reference

All items in the checklist above are completed and verified by a qualified plant engineer or instrumentation supervisor. Until all placeholders are replaced with verified, plant-specific information, this manual serves as a training reference only and is not the authoritative source for plant configuration data. Official plant documents (P&IDs, loop drawings, I/O schedules, C&E matrix, calibration records) always govern over this manual in case of any discrepancy.

NOTES

Appendix E: Field Pocket Card

This appendix prints on one double-sided sheet for field use and may be laminated.

> **CAUTION — Plant documents govern**
> If a live drawing or procedure disagrees with this card, the plant document governs. Training tags are examples.

4–20 mA math

```
% = (mA − 4) / 16 × 100
EU = EU_min + (mA − 4) / 16 × (EU_max − EU_min)
mA = (EU − EU_min) / (EU_max − EU_min) × 16 + 4
```

Faults (NAMUR NE43): ≤ 3.6 mA fail-low / open; ≥ 21 mA fail-high; 20.5 mA typical over-range. 4.00 / 8.00 / 12.00 / 16.00 / 20.00 mA = 0 / 25 / 50 / 75 / 100%.

Work-control reminder

Drawing → MOC? → permits/LOTO → notify board → do only the scoped work → restore covers/forces → operator confirm → close permits.

BPCS vs SIS

BPCS = process control. SIS = independent safety. PSV = mechanical last line. The SIS is not bypassed by technicians. BPCS trips are not ESD SIFs.

Fail-safe discrete (plant convention, Section 0.3)

Health bits (STOP_OK, OL_OK, E_STOP_OK) are 1 when healthy and are examined with XIC on the run path. A wire break or stop press makes the bit 0 and the output drops.

H_2S / LEL (confirm plant numbers)

Condition	Action
H_2S ≥ plant action level (commonly 10 ppm)	Respiratory protection per plant H_2S program; odor is not a reliable indicator
H_2S 100 ppm (NIOSH IDLH)	Emergency response only
LEL ≥ 10% of LEL	No hot work
Personal monitor	Bump test before the shift; know wind direction and muster point

End-of-job software checks

Force table empty. No unexpected inhibits. No leftover bypass timer. Official project saved. Operator sees real data.

Who to call (fill in)

Need	Name / radio / phone
Board operator	[COMPANY:]
Automation supervisor	[COMPANY:]
Process safety / SIS owner	[COMPANY:]
Emergency / H_2S	[COMPANY:]

NOTES

Appendix F: Training Program Materials

This appendix holds the training-program material for the plant technician qualification program: review cards, lab review questions and answers, the eight-week program schedule, the competency matrix, and assessment notes. The Training Coordinator or designated assessor administers this material.

F.1 Review Cards

The following question-and-answer cards summarize key facts from Sections 1–9.

#	Question (Front)	Answer (Back)
1	What are the six main stages of a natural gas processing plant in order?	1) Inlet Separation, 2) Compression, 3) Dehydration (Glycol), 4) Amine Treating (H_2S/CO_2 removal), 5) NGL Fractionation, 6) Custody Transfer Metering.
2	What is the primary purpose of glycol dehydration in a gas plant?	To remove water vapor from the gas stream to prevent hydrate formation and meet pipeline water dew-point specifications, typically <4°C or <7 lb/MMSCFD using tri-ethylene glycol (TEG) as the absorbent.
3	What pipeline specification limits does sweet gas typically need to meet for H_2S and CO_2?	H_2S ≤4 ppm (some pipelines ≤0.25 grain/100 SCF); CO_2 ≤2 mol%. Amine treating (MDEA, DEA, MEA) removes both acid gases to meet these specs.
4	What is the difference between PAHH and PSV as protective devices?	PAHH (Pressure Alarm High-High) is an instrumented function that trips the process (valve closure, equipment trip) through BPCS trip logic or, where it is a SIF, through the SIS, as defined in the C&E matrix. A PSV (Pressure Safety Valve) is a purely mechanical device that opens at its set pressure regardless of any electronic system, providing the last line of pressure protection.
5	What is custody transfer metering and why is it highly accurate?	Custody transfer metering measures the volume and energy content of natural gas sold to a downstream operator — it determines billing. It must achieve ±0.25% or better accuracy using AGA-7 ultrasonic or AGA-3 orifice meter standards, with temperature and pressure compensation and chromatographic BTU measurement.
6	What does "4-20 mA live zero" mean and what is its advantage?	"Live zero" means 4 mA represents 0% of range — not 0 mA. This allows the control system to distinguish between a true zero reading (4 mA) and a broken wire or transmitter failure (0 mA or ≤ 3.6 mA). Any signal below ~3.6 mA indicates a fault, triggering a "below live zero" alarm.
7	Given a pressure transmitter with range 0–500 PSIG reading 14 mA, what is the process pressure?	EU = ((14-4)/16) × 500 = (10/16) × 500 = 0.625 × 500 = 312.5 PSIG.
8	What is the expected mA output for a level transmitter (0–100%) when the vessel is at 75% full?	mA = ((75/100) × 16) + 4 = 12 + 4 = 16.0 mA.
9	What is HART and what is its primary benefit for a field technician?	HART (Highway Addressable Remote Transducer) is a bi-directional digital communication protocol superimposed on the 4-20 mA analog signal using FSK at 1,200 baud. Primary benefit: a technician can use a HART communicator to read transmitter configuration, perform remote calibration, read secondary variables, and run diagnostics without interrupting the analog signal or disconnecting wiring.
10	What is the difference between a Zener barrier and a galvanic isolator, and when is each used?	Zener barrier: passive IS protection using Zener diodes and resistors; requires a certified ≤1Ω IS earth ground; low cost. Galvanic isolator: active IS protection using optical or transformer isolation; no ground bond required; preferred when ground loop issues exist or reliable IS earth is unavailable. Use galvanic isolators in modern installations for reliability and ease of installation.
11	In a 5-point calibration, what are the five input % levels and their corresponding mA values?	0% = 4.000 mA; 25% = 8.000 mA; 50% = 12.000 mA; 75% = 16.000 mA; 100% = 20.000 mA.
12	What symptom indicates an open circuit (broken wire) in a 4-20 mA instrument loop?	The signal reads 0 mA (or ≤ 3.6 mA — "below live zero"). The PLC/DCS annunciates a "bad quality" or low fault alarm, and the HMI tag shows ??? or a minimum value. A calibrated milliammeter in series will read 0 mA, confirming open circuit.
13	What does "intrinsically safe" mean and in what hazardous area classification is IS wiring required?	Intrinsically safe (IS) means the circuit cannot release enough electrical energy (voltage, current, capacitance, or inductance) to ignite a flammable atmosphere, even under single-fault conditions. IS wiring is required in Class I, Division 1 (NEC) or ATEX/IECEx Zone 0 and Zone 1 locations — areas where flammable gas is continuously or occasionally present.

#	Question (Front)	Answer (Back)
14	What is the ControlLogix 1756 chassis slot numbering convention?	The controller commonly occupies Slot 0, but a ControlLogix controller can be placed in any slot; the slot is defined in the project. The power supply mounts at the left end of the chassis and is not a numbered slot. I/O and communication modules (for example 1756-EN2T or 1756-EN2TR) can occupy any remaining slot.
15	What does RPI stand for and what happens if it is set too low for a remote FLEX I/O bank?	RPI = Requested Packet Interval — the cyclic update rate (in ms) for an I/O connection. If set too low for network capacity, the controller cannot receive I/O updates in time, causing connection timeouts, I/O module faults in Studio 5000, and potential task overruns in the controller program.
16	What are the three ControlLogix task types and when is each used?	Continuous: one per controller; executes all programs repeatedly at background priority; for main process control logic. Periodic: executes at a fixed time interval (e.g., 10 ms, 100 ms); for time-critical loops like PID or high-speed monitoring. Event: triggered by a specific hardware or software event; for time-critical responses to events (module input change, motion complete).
17	What is the difference between controller-scoped and program-scoped tags in Studio 5000?	Controller-scoped tags: accessible from any program, routine, or external device (HMI, MSG, OPC) — used for I/O tags, shared variables, and HMI-visible tags. Program-scoped tags: local to one program only — used for internal calculation variables, timers, and counters not needed elsewhere. Program-scoped tags cannot be directly accessed by the HMI or MSG instructions.
18	What is a UDT (User-Defined Type) and why is it useful in a gas plant PLC program?	A UDT is a custom data structure grouping related variables under one tag name. In a gas plant, a UDT for a pressure controller would contain PV, SP, OUT, Kp, Ki, alarm bits, and mode — all accessed as SEP_PIC_101.PV, SEP_PIC_101.SP, etc. UDTs promote consistency, simplify HMI binding, and allow creating identical configurations for multiple similar loops by instantiating the same type.
19	What is an AOI (Add-On Instruction) in Studio 5000?	An Add-On Instruction is a reusable custom instruction created once, with its own logic, input/output parameters, and internal tags — it appears and behaves like a native ControlLogix instruction. AOIs are used for standardized valve control, motor starters, PID loops, and alarm management blocks that are used on multiple pieces of equipment, ensuring consistent behavior and reducing programming errors.
20	What does EtherNet/IP implicit messaging (I/O messaging) do, and how is it different from explicit messaging?	Implicit (I/O) messaging: cyclic, automatic data exchange between controller and I/O device at the RPI rate — established once and runs continuously; used for real-time I/O data. Explicit messaging (MSG instruction): on-demand, triggered by program logic; request-response; used for configuration, diagnostics, and data transfer between controllers or to non-I/O devices.
21	What is DLR (Device Level Ring) and what is its key advantage over a standard star network?	DLR is a ring-topology EtherNet/IP network where devices are daisy-chained in a ring. A ring supervisor monitors the ring with beacon frames every 400 µs. Upon detecting a single cable break, the supervisor reconfigures the network to a linear topology in <3 ms — maintaining I/O communication without any managed switches in the ring. Key advantage: fault-tolerant network connectivity with low-cost infrastructure.
22	What is the maximum number of 1794 FLEX I/O modules per adapter bank?	Eight (8) I/O modules per bank. The adapter (1794-AENTR or 1794-ADN) is the first module (adapter position), followed by up to 8 I/O modules in slots 1–8. A terminator cap (1794-END) is required on the last terminal base's expansion port.
23	What is hot-swap capability on FLEX I/O and how is it implemented?	Hot-swap allows removing and replacing a FLEX I/O module while the adapter and bank remain powered and communicating. It is implemented by physically separating the module (electronics) from the terminal base (field wiring): the module lifts off the terminal base while the terminal base — with all wiring — stays mounted. Only the replaced module's data is interrupted during the swap.
24	What raw count value does the 1794-IE8 output for a 4 mA input (0% of 4-20 mA range)?	0 counts. The 1794-IE8 in 4–20 mA mode reads 0 counts at 4 mA and 30840 counts at 20 mA (the 12-bit value is left-justified in the 16-bit word; 12 mA = 15420). The 1756-IF8 in integer mode reads about −20030 at 4 mA; float (engineering units) mode is commonly used instead. The module data format is verified in the product manual before scaling logic is written.

#	Question (Front)	Answer (Back)
25	What does the 1794-AENTR NS LED indicate when it is flashing green?	Flashing green indicates the IP address is set but no EtherNet/IP connection is established. Solid green indicates at least one established connection, the normal state when connected to a ControlLogix controller.
26	What does the XIC instruction do in ladder logic?	XIC (Examine If Closed): evaluates the referenced BOOL tag — the instruction is TRUE (passes power) when the tag = 1 (bit is SET). Equivalent to a normally open contact. Used for: start pushbuttons, run permissives, timer done bits, positive logic conditions.
27	What does the XIO instruction do and when is it used instead of XIC?	XIO (Examine If Open) is TRUE when the referenced BOOL tag = 0 (bit clear); it is equivalent to a normally closed contact. It is used for bits that must be 0 to proceed, for example XIO(Trip_Alm) or XIO(HMI_Stop). Field stop, E-stop, overload, and trip inputs are _OK health bits (1 = healthy) examined with XIC (Section 0.3); XIO of an _OK bit appears only in trip-detect or first-out rungs.
28	What is the difference between OTE, OTL, and OTU?	OTE (Output Energize): non-retentive — bit is 1 when the rung is TRUE and 0 when FALSE; cleared at prescan. OTL (Output Latch): retentive set — the bit stays 1 after the rung goes FALSE until an OTU executes. OTU (Output Unlatch): resets the bit to 0 when the rung is TRUE. OTL/OTU pairs are used for latched alarms and BPCS trip latches, never to drive a motor output directly.
29	What does the TON instruction do and what are its key bits?	TON (Timer On Delay): accumulates elapsed time (.ACC in ms) while the rung is TRUE. Key bits: .EN (Enable — rung is TRUE), .TT (Timing — ACC < PRE), .DN (Done — ACC ≥ PRE). When rung goes FALSE, ACC resets to 0 (non-retentive). Used for: time delays before starting equipment, alarm delay filtering, sequence step timers.
30	What is the difference between TON and RTO timers?	TON (Timer On Delay): resets accumulator to 0 when rung goes FALSE — non-retentive. RTO (Retentive Timer On): retains its accumulator value when rung goes FALSE; continues accumulating each time rung returns to TRUE. Must be reset explicitly with a RES instruction. Used for: running-hours meters, total runtime accumulation.
31	What does the ONS (One Shot) instruction do and why is it important for MSG triggers?	ONS (One Shot) passes power for exactly one scan when the rung input transitions from FALSE to TRUE. It requires a dedicated storage BOOL, which is set at prescan, so a condition already TRUE at power-up does not produce a pulse. ONS converts a sustained condition into a single-scan action, for example a reset pulse (Pattern A1). A MSG instruction itself starts only on a FALSE-to-TRUE rung transition; periodic messages are triggered from a poll timer conditioned on .EN (Snippet 8).
32	What does the JSR instruction do and how does it differ from JMP?	JSR (Jump to Subroutine): transfers execution to a named routine in the same program, passes input parameters, and returns to the JSR rung after the subroutine executes RET. Scanned subroutine executes completely each call. JMP (Jump): branches to a LBL label within the same routine, skipping all rungs between — no parameters, no return. JSR creates modular, reusable code; JMP creates conditional execution paths within one routine.
33	What are the PID instruction's timing requirements in Studio 5000?	The PID instruction is placed in a periodic task, and its update time (.UPD) is set equal to the task period (for example 0.1 s for a 100 ms task). In a continuous task with variable scan time, the integral and derivative terms are calculated incorrectly because they depend on a fixed sample interval. The PID executes only while its rung is TRUE, so the rung is unconditional; manual mode uses .SWM and .SO.
34	What does the GT (legacy GRT) instruction do? Give a gas plant example.	GT (Greater Than; legacy GRT) is TRUE when Source A > Source B. Gas plant example: GT(SEP_PT_101.Value,250.0)OTE(SEP_PAH_101); — the PAH bit sets when separator pressure exceeds 250 PSIG (example) and clears when it drops below. A deadband (Pattern A9) prevents chatter near the setpoint.
35	What does the MSG instruction's .ER bit indicate and what is the first thing to check?	MSG.ER is TRUE when the message failed to complete. First check: read .ERR and .EXERR directly. Common codes: 16#0001 = connection failure (extended 16#0204 = unconnected timeout; 16#0203 = connection timeout); 16#0004 = path or tag-name (IOI) syntax error; 16#0005 = destination unknown (tag not found); 16#0006 = insufficient packet space. Verify the target controller is online and the path is correct.

#	Question (Front)	Answer (Back)
36	How does a state machine pattern in ladder logic work?	A DINT tag holds the current state number (for example 0 = Idle, 1 = Purge, 2 = Running). Transition rungs use EQ (legacy EQU) to check the current state and a transition condition, then MOVE (legacy MOV) a value into a next-state tag; one final rung commits the next state, so only one transition occurs per scan. Action rungs use EQ to check the state and drive outputs (Section 5.3).
37	What is Structured Text (ST) and when should it be used instead of Ladder Diagram?	Structured Text is a high-level, PASCAL-like text language for PLC programming (IEC 61131-3). It should be used when the logic involves: complex math calculations (dew point, flow compensation, efficiency calculations), array operations, string handling, or algorithms that would require dozens of math rungs in ladder. Ladder is preferred for discrete control (on/off, interlocks) because it is more readable for field technicians.
38	What is Wonderware InTouch and what are its two main applications?	Wonderware InTouch (AVEVA InTouch HMI) is a SCADA/HMI software platform for industrial process visualization. Two main applications: WindowMaker — the development environment for creating graphic displays, configuring tags, writing scripts, and setting up animations. WindowViewer — the runtime environment where operators interact with the process: monitoring values, acknowledging alarms, entering setpoints, and navigating screens.
39	What does the HMI tag showing "???" indicate and what is the most common cause?	"???" in InTouch indicates the tag has lost communication with its data source — the I/O server (OPC server, RSLinx) has disconnected from the PLC, or the tag name is not found in the I/O server's tag database. Most common cause: OPC/SuiteLink I/O server crashed or the PLC went offline. Check if the I/O server service is running and the PLC is accessible on the network.
40	What is ISA-18.2, and what are its benchmark alarm rates?	ISA-18.2 is the standard for alarm management in the process industries. Benchmark rates per operator console: about 6 alarms per hour (1 per 10 minutes) is very likely acceptable; about 12 per hour (2 per 10 minutes) is the maximum manageable. More than 10 alarms in 10 minutes is an alarm flood.
41	What are the four ISA-18.2 alarm priority levels and their response times?	Priority 1 (Critical): immediate hazard — respond within <5 minutes; red flashing, audible. Priority 2 (High): significant deviation — respond within 5–15 minutes; red steady. Priority 3 (Medium): process deviation — respond within 15–60 minutes; yellow. Priority 4 (Low): advisory — respond at next opportunity; no audible, displayed in alarm list only.
42	What is alarm "rationalization" and what does it produce?	Alarm rationalization is the process of reviewing every alarm against ISA-18.2 criteria to ensure it requires operator action, has a documented cause and consequence, has a justified setpoint and priority, and has a defined operator response. The output is a completed alarm rationalization worksheet for each alarm — the official record justifying why each alarm exists at its configured priority, setpoint, and deadband.
43	When troubleshooting a control loop that won't respond, what is the most common place to find the root cause?	In field instrumentation and wiring: the majority of control loop problems (estimated 60-70%) originate in the field — broken wires, corroded terminals, failed sensors, plugged impulse lines, or loss of instrument air to actuators. Check physical components before suspecting PLC software errors. The second most common cause is the final element (valve) — mechanical failure, loss of air supply, or positioner malfunction.
44	What does a "task overrun" mean in ControlLogix and what are its main causes?	A task overrun occurs when the PLC program takes longer to execute than the configured task period — the periodic task period expires before the previous scan completes. Causes: too many I/O modules with short RPI creating network poll overhead, MSG instructions adding latency, computationally intensive loops (large arrays, complex math), or program logic added without adjusting task period. Solution: increase task period, optimize code, or move low-priority logic to a slower periodic task.
45	What are the seven steps of systematic troubleshooting?	1) Define the problem precisely. 2) Observe all symptoms. 3) Identify possible causes. 4) Test hypotheses — least invasive first. 5) Isolate the fault to a specific component. 6) Repair or replace. 7) Verify normal operation is restored and document the resolution.

#	Question (Front)	Answer (Back)
46	What is the difference between SIS and BPCS?	BPCS (Basic Process Control System): the PLC/DCS that maintains normal operating conditions; it may provide BPCS protection layers (alarms, trips) but is not SIL-rated. SIS (Safety Instrumented System): independent logic solver with dedicated sensors and final elements, designed and managed per IEC 61511 (devices certified to IEC 61508 or justified by prior use) to detect unsafe conditions and drive the process to a safe state. The SIS does not share critical sensors, logic, or final elements with the BPCS.
47	What is SIL 2 and what does it mean in terms of PFD and risk reduction?	SIL 2 (Safety Integrity Level 2): the safety function fails to respond on demand with a probability of 0.01 to 0.001 (1 in 100 to 1 in 1,000 demands). Risk Reduction Factor = 100 to 1,000. This means the SIF reduces the frequency of the hazardous event by a factor of 100 to 1,000 compared to having no safety function. Typical SIL 2 applications: compressor ESD, absorber high-level shutdown.
48	What is the H_2S IDLH, and what respiratory protection applies above the plant action level?	H_2S IDLH (NIOSH) = 100 ppm. Respiratory protection follows the plant H_2S program; plants commonly set an action level (often 10 ppm) at which SCBA or supplied air with an escape bottle is required. OSHA does not define an action level. Exposure limits: ACGIH TLV 1 ppm, STEL 5 ppm; NIOSH REL 10 ppm (10-min ceiling); OSHA general industry ceiling 20 ppm, peak 50 ppm. Also required: buddy system, personal H_2S monitor, awareness of wind direction, and knowledge of the muster points and H_2S response plan.
49	What is a proof test for a Safety Instrumented Function, and why is it required?	A proof test is a periodic functional test of the complete SIF (sensor → logic solver → final element) to detect dangerous undetected (DU) failures — failures that would prevent the SIF from responding when a process demand occurs. Required by IEC 61511 because DU failures accumulate over time, increasing the PFD. The proof test interval (typically annual or biennial) is a key variable in the SIL verification — longer intervals mean higher average PFD.
50	What is the LEL of methane and at what % LEL should hot work be stopped?	Methane LEL = 5 vol% in air. Hot work must NEVER be performed if the atmosphere exceeds 10% LEL (0.5% methane). Plant action levels: alarm at 10% LEL, evacuate non-essential personnel at 25% LEL, emergency response above 25% LEL. The UEL of methane is 15 vol% — above UEL, the mixture is too rich to ignite but will ignite if diluted back into the 5–15% flammable range.

F.2 Lab Review Questions and Answers

Copy rule: lab review answers are written on a separate sheet before the answer key is consulted. The supervisor initials a lab only after the practical steps are demonstrated. Labs use the training rack only unless the procedure states otherwise.

Lab 1: Instrument Loop Check — Pressure Transmitter (Site Form Discipline)

#	Question	Answer
1	What is the expected HMI reading when 10 mA is injected into a 0–300 PSIG transmitter?	(10–4)/16 × 300 = 112.5 PSIG.
2	What does a signal at or below 3.6 mA indicate, and how does the PLC detect it as a fault?	A transmitter low fault (NAMUR NE43: ≤ 3.6 mA). The PLC detects it through the AI channel under-range/fault status. A 1794-IE8 reads 0 counts at or below 4 mA, so the count value alone cannot detect it.
3	What loop resistance does HART communication require, and why?	230–600 Ω (typically 250 Ω). HART uses FSK at 1,200 baud superimposed on the 4–20 mA loop; without sufficient impedance, the HART modem cannot develop an adequate voltage swing.
4	What is the purpose of suppressing alarms before a loop check?	To prevent operators from taking unnecessary action on artificial readings generated during testing.
5	If the HMI reads 150 PSIG but the field gauge reads 220 PSIG, what are three possible causes?	Wrong AI channel addressed by tag; incorrect scaling in PLC (wrong EU min/max); transmitter not re-zeroed after calibration or installation.

Lab 2: FLEX I/O Module Replacement (Hot-Swap)

#	Question	Answer
1	What happens to the other modules in the FLEX bank when one module is removed?	Other modules continue operating normally — only the removed module's data is interrupted.

#	Question	Answer
2	Why must the terminal base NOT be removed during hot-swap?	The terminal base holds all field wiring; removing it would disconnect all field devices in that bank and risk an uncontrolled process response.
3	What information should always be recorded when replacing any I/O module?	Old module serial number, new module serial number, catalog number, hardware revision, date, technician name, and reason for replacement.
4	What does the module's status LED indicate immediately after proper insertion of a healthy replacement?	Green flashing then solid — indicates healthy module has been recognized by the adapter.
5	What is an MOC and why is it required before replacing a module in a live system?	Management of Change is a process safety regulation requirement ensuring all planned changes to instrumentation and control systems are reviewed, documented, and approved before implementation, preventing unintended consequences to safety, environment, or production.

Lab 3: Build a Motor Start/Stop Rung in Studio 5000

#	Question	Answer
1	What is the purpose of the seal-in contact, and what would happen if it were omitted?	The seal-in contact maintains the motor output after the momentary start button is released; without it, the motor would run only while the start button is held down.
2	Why does the stop circuit use XIC(STOP_OK) rather than XIO?	Per Section 0.3, name the bit STOP_OK and wire the field NC contact so that healthy/not-pressed = 24 VDC = 1. Use XIC(STOP_OK) in series. Pressing stop or breaking the wire makes the bit 0, the XIC opens, and the OTE drops. XIO(STOP_OK) would be wrong under this convention because it would run the motor when the stop circuit is unhealthy. If a plant drawing uses a different contact and bit meaning, follow the drawing — not habit.
3	What is the difference between an OTE and an OTL instruction? When is each used for motor control?	OTE de-energizes when the rung goes false and is cleared at prescan (non-retentive); OTL retains its state until an OTU executes. For motor control, OTE with seal-in is used so that the output re-evaluates every permissive every scan and drops on a mode change. OTL is used for latched status bits (for example a trip latch or alarm) that must persist until explicitly reset, and never to drive an output directly.
4	Why must all forced I/O be cleared at the end of a session?	Leaving forced I/O can cause unintended equipment operation on the next startup — it is a safety and production risk.
5	In a real installation, what additional permissive contacts might be added in series with the start circuit?	Gate permissives (vessel pressure in safe range), line-up permissives (isolation valves open), lube oil pressure present, not-in-ESD-state, VFD ready signal, and any pre-start sequence confirmations.

Lab 4: Configure and Tune a PID Loop

#	Question	Answer
1	What is the role of the integral term in a PID controller?	The integral term eliminates steady-state offset by accumulating the error over time and driving the output until error = 0.
2	What is "integral windup" and how is it prevented in Studio 5000?	Integral windup occurs when the controller output saturates (0% or 100%) while the integral keeps accumulating, so a large opposite error is needed to unwind it before the output responds. The Logix PID limits the output at .MAXO/.MINO and stops integral accumulation while the output is clamped; set realistic output limits.
3	Why is derivative action often disabled (Kd = 0) for pressure control in gas plants?	Derivative acts on the rate of change of error; pressure signals are often noisy, and derivative amplifies noise into the output signal, causing valve chatter and wear.
4	What is the difference between parallel PID form and series (interacting) PID form?	Parallel (independent) form: each term acts on the error with its own gain. Series or dependent forms: the controller gain multiplies all terms, so changing Kc changes the effective integral and derivative action. The Logix PID instruction offers both independent (.PE = 0) and dependent (.PE = 1) forms.
5	If a loop exhibits sustained oscillation after tuning, which parameter is reduced first?	Reduce Kp (proportional gain) first — it is the most common cause of oscillation.

Lab 5: Add and Configure a FLEX I/O Bank

#	Question	Answer
1	What is the maximum number of I/O modules that can follow a 1794-AENTR adapter on one bank?	Eight I/O modules per bank (slots 1–8, with adapter in slot 0).
2	What does RPI stand for, and what is the effect of setting it too low?	Requested Packet Interval — the cyclic update rate between controller and I/O module; setting it too low for the available network capacity causes communication faults, connection timeouts, and potential task overrun in the controller.
3	What is the difference between an alias tag and a controller tag?	An alias tag is a pointer to another tag — changing the underlying hardware mapping requires only changing the alias definition, not every rung that uses it. A controller tag is an actual variable with memory allocated.
4	What raw count values are expected for 4 mA and 20 mA inputs on a 1794-IE8 in 4–20 mA mode?	0 counts at 4 mA and 30840 counts at 20 mA (16#7878). The 12-bit converter value is left-justified in the 16-bit data word.
5	What does the NS (Network Status) LED on the 1794-AENTR indicate when it is flashing green?	Flashing green indicates the IP address is set but no connection is established. Solid green indicates at least one established EtherNet/IP connection, the normal state when connected to the controller.

Lab 6: Create an InTouch HMI Faceplate

#	Question	Answer
1	What is the purpose of input validation on the SP entry field?	Prevents operator entry errors — e.g., entering 3000 instead of 300 PSIG — which could cause severe process upset or equipment damage.
2	What is bumpless transfer, and where is it implemented in this design?	Bumpless transfer means no step in controller output when switching modes. It is implemented in the PLC: .SO tracks .OUT while in auto (Snippet 4), so entering manual holds the output, and returning to auto resumes from the current output. The HMI only sends the mode request.
3	Why should operator console HMI faceplates be standardized across all controllers at the plant?	Standardization reduces training time, reduces errors during emergencies (operators can interpret any faceplate without learning a new layout), and simplifies maintenance.
4	What is an Access Name in InTouch and what does it define?	An Access Name defines a logical name for a connection to an external data source (OPC server, DDE server, SuiteLink server); it specifies the server application, topic name, and communication protocol.
5	If the PV bar graph shows ??? instead of a numeric value, what is checked first?	Check if the OPC/SuiteLink I/O server is running and connected to the PLC — "???" always indicates a loss of communication between InTouch and the data source.

Lab 7: MSG Read — Inter-PLC Communication

#	Question	Answer
1	What is the difference between implicit (I/O) messaging and explicit (MSG) messaging in EtherNet/IP?	Implicit messaging is cyclic/periodic I/O data exchange (controller to field I/O module at RPI rate, always running once connected); explicit messaging is on-demand request-response communication triggered by program logic (MSG instruction).
2	Why must the target tag have "External Access = Read" set in Studio 5000?	External access is controlled per tag. External Access must be set to Read Only or Read/Write for a tag to be accessible to other controllers through MSG or to OPC clients.
3	What does MSG error 16#0001 with extended error 16#0204 indicate?	A connection failure (16#0001) with extended code 16#0204, an unconnected timeout: the target did not respond, for example because it is unreachable, powered off, or the path is wrong.
4	Why is "Cache Connections" recommended for MSG instructions that execute frequently?	Caching keeps the CIP connection open between MSG executions, avoiding the overhead of establishing a new connection every time; without caching, frequent MSG triggers waste bandwidth on connection setup/teardown.

#	Question	Answer
5	Why does this design use a stale-data timer rather than relying on a short MSG timeout?	The stale-data timer declares a fault a fixed time (5 s) after the last successful read, regardless of which failure occurred (timeout, error, or the MSG rung not executing), and it gates every use of the data. Shortening the MSG timeouts alone causes nuisance errors on a busy network and still leaves stale data ungated.

Lab 8: BPCS Trip String and Time-Limited Bypass (Not SIS / Not ESD SIF)

#	Question	Answer
1	Why must BPCS trip bypasses be time-limited and self-expiring?	A bypass left in place indefinitely removes protection; automatic expiry restores protection even if the bypass is forgotten or the technician is called away.
2	What should an operator do if a BPCS trip bypass alarm appears unexpectedly on the HMI?	Immediately investigate whether a bypass was authorized. If not, alert the supervisor and safety team; an unauthorized bypass of a trip is a potentially serious safety violation.
3	Why is a key-switch authorization used rather than only a software pushbutton?	A physical key switch provides a tangible, auditable control — only authorized personnel with the physical key can enable bypass, reducing accidental or unauthorized activation.
4	What is the difference between bypassing a BPCS trip (in the PLC) and bypassing the SIS?	Bypassing a BPCS trip leaves the SIS (an independent system) still protecting the process; the SIS is unaffected by BPCS changes. Bypassing the SIS removes the SIL-rated protection; this is a much more critical action requiring a higher level of authorization under the plant functional-safety procedure.
5	What documentation must be completed before and after a BPCS trip bypass on a production system?	Before: MOC form, bypass permit signed by safety engineer and area supervisor, operator notified with time limit agreed. After: bypass removed, permit closed, process returned to normal protection, bypass duration and reason recorded in maintenance management system.

Lab 9: Network Troubleshooting — Find a Lost FLEX I/O Node

#	Question	Answer
1	What is the typical connection timeout for an EtherNet/IP I/O connection as a multiple of RPI?	Four times the RPI (4×RPI) is the standard EtherNet/IP connection timeout for I/O messaging — at RPI=20ms, the timeout ≈80ms.
2	What does a solid red NS LED on the 1794-AENTR indicate?	Solid red NS indicates a duplicate IP address: the adapter has detected that its IP address is already in use on the network. Flashing red indicates a connection timeout.
3	Why is the ping test a useful first step in network troubleshooting?	Ping verifies basic Layer 3 (IP-level) network connectivity at the target device's IP address; if ping fails, all higher-level protocols (CIP, MSG, RSLinx) will also fail — confirming a physical or network layer issue before spending time in software diagnostics.
4	What advantage does DLR (Device Level Ring) topology provide in a scenario like this lab?	DLR would maintain communication on the remaining ring path even with the cable disconnected — I/O data would continue flowing in <3 ms, with only an alert that the ring is broken. The lab failure scenario (star topology) would not occur in a DLR ring with a single cable break.
5	What information should be recorded in the maintenance logbook after resolving a network fault?	Date/time of fault, affected tags and I/O bank, symptoms observed, root cause identified (e.g., cable pulled from port by maintenance crew), corrective action taken (re-seated cable), time to restoration, any process impact, and initials of technician who resolved the issue.

Lab 10: Alarm Rationalization Workshop

#	Question	Answer
1	According to ISA-18.2, what is the maximum acceptable alarm rate for a well-rationalized alarm system (in steady state)?	ISA-18.2 benchmarks per operator console: about 6 alarms per hour (1 per 10 minutes) is very likely acceptable; about 12 per hour (2 per 10 minutes) is the maximum manageable. More than 10 alarms in 10 minutes is an alarm flood.

#	Question	Answer
2	What is "alarm deadband" and why is it critical for chattering alarms?	Alarm deadband is a hysteresis band around the alarm setpoint; the alarm activates when the PV exceeds the setpoint but does not clear until the PV returns past (setpoint minus deadband). Without deadband, a PV oscillating exactly at the setpoint will generate continuous alarm chatter.
3	What is the difference between alarm "suppression" and alarm "shelving"?	Suppression is a system-driven, rules-based automatic masking of a specific alarm when a defined condition exists (e.g., suppress low-flow alarm when pump is stopped); shelving is an operator-initiated, time-limited masking of an alarm the operator is aware of and has decided to temporarily silence (with automatic restoration after the shelf period).
4	What is a "nuisance alarm" and what are three strategies to eliminate it?	A nuisance alarm is one that activates frequently but requires no meaningful operator action. Strategies: increase deadband; move setpoint further from normal operation; add time delay (alarm must persist for N seconds before activating); implement state-based alarming (active only in relevant operating modes).
5	Why should alarm setpoints not be set at the same value as the process control setpoint?	If the alarm setpoint equals the control setpoint, the controller will be fighting to maintain the process at exactly the alarm boundary — any normal control variability will continuously trigger and clear the alarm; there must be adequate margin between the normal operating range, the alarm setpoint, and the ESD/safety setpoint.

Lab 11: Find and Clear Leftover Forces and Inhibits

#	Question	Answer
1	Why can a force make a loop check look healthy while the field device is dead?	The CPU writes the forced value to logic and HMI regardless of the card.
2	What action is required when an unbriefed force is found?	Stop, do not clear it blindly, and notify the board operator and supervisor; it may be a permitted force.
3	How does an inhibited remote adapter differ from a single forced BOOL?	An inhibit can freeze an entire bank of PVs/DOs; a single force affects one tag.

Lab 12: First-Out Narrative from Historian / SOE

#	Question	Answer
1	Why might the loudest HMI alarm not be the first-out?	Consequence alarms flood after the initiating bad actor.
2	What is the risk of using only the current HMI value after the trip?	Post-trip values are the safe state, not the initiating state.
3	If a tag shows ??? through the event, what is written in the narrative?	Record 'data gap / bad quality from hh:mm to hh:mm — cannot prove this bit' instead of inventing a value.

F.3 Eight-Week Program Schedule

Overall Weekly Schedule

Week	Focus Area	Key Topics	Labs to Complete	Review Cards	Milestone / Goal
1	Gas Plant Orientation & Instrumentation Basics	Process flow (Sec 1); 4-20mA signal standard; instrument types; HART; loop check procedure; IS basics	Lab 1 (Loop Check)	Cards 1–13	Perform a supervised loop check on one field instrument. Pass a verbal check on process stages.
2	ControlLogix 1756 Hardware & Studio 5000 Navigation	Chassis & modules (Sec 3.1); Studio 5000 project tree; task types; tag database; data types; UDT; AOI; EtherNet/IP basics	Lab 3 (first half — create project and tags)	Cards 14–21	Navigate Studio 5000 without assistance. Identify every module in the training chassis. Create a new project from scratch.
3	Ladder Logic Fundamentals — Basic Instructions	XIC, XIO, OTE, OTL, OTU, ONS, TON, TOF, RTO, CTU, CTD; Snippets 1–3 (Sec 5.2); tag naming convention (Sec 3.3)	Lab 3 (complete); start Lab 4	Cards 26–35	Write a motor start/stop rung with timer delay from memory. Test with bench inputs without referring to notes.

Week	Focus Area	Key Topics	Labs to Complete	Review Cards	Milestone / Goal
4	FLEX I/O Configuration & I/O Mapping	FLEX I/O hardware (Sec 4.1–4.3); hot-swap; module configuration in Studio 5000; I/O mapping table; DLR topology	Lab 2 (Hot-Swap); Lab 5 (Configure FLEX Bank)	Cards 22–25	Configure a FLEX I/O bank from scratch in Studio 5000. Complete a blank I/O mapping table for a 20-point bank. Perform hot-swap under supervision.
5	Advanced Ladder Logic — PID, MSG, State Machines	PID instruction (Sec 5.2 Snippet 4); Ziegler-Nichols tuning; MSG instruction (Snippet 8); state machine pattern; Structured Text (Sec 5.3)	Lab 4 (complete PID tuning); Lab 7 (MSG Read)	Cards 32–37	Configure and tune a PID loop to achieve <20% overshoot on a SP step change. Successfully execute an inter-PLC MSG read.
6	Wonderware HMI — Tags, Alarms, Screens	InTouch vs. System Platform (Sec 6.1); HMI tag list; alarm management (ISA-18.2); screen design guidelines; scripts	Lab 6 (HMI Faceplate); Lab 10 (Alarm Rationalization)	Cards 38–42	Build a complete PID controller faceplate in InTouch without assistance. Complete alarm rationalization for 5 tags with justified priorities and setpoints.
7	Safety, SIS, BPCS Trips and Bypass Discipline, Troubleshooting	HAZOP; LOPA; SIS vs. BPCS; SIL levels; voting logic; proof tests; H_2S hazard; F&G system; C&E matrix; BPCS trip latch (Snippet 6); troubleshooting methodology (Sec 8)	Lab 8 (BPCS Trip String and Bypass); Lab 9 (Network Troubleshooting)	Cards 43–50	Build a BPCS trip string with time-limited bypass on the trainer under supervision. Diagnose and restore a simulated FLEX I/O network fault. Explain SIL 2 requirements verbally to the supervisor.
8	Review, Practice, Lab Completion, Assessment	All topics — focus on weak areas. Complete the Quick Reference (Section 11) from memory. All 50 review cards. Work all troubleshooting tables (Sec 8.2)	Complete any remaining labs. Repeat Lab 1 independently (no notes).	All 50 review cards — aim for 100% correct without prompts	Pass plant competency assessment (written and practical). Perform unsupervised loop check and documented MOC. Sign-off on the technician qualification checklist.

Week 1 — Daily Breakdown

Day	Morning (4 hrs)	Afternoon (4 hrs)	Review Question (closed-book)
Monday	Plant orientation tour with supervisor. Review Section 1 (Gas Plant Process Flow). Study inlet separation and compression stages in detail.	Study Section 2.1–2.2 (4-20mA standard; instrument types). Study 4-20mA formula. Practice formula calculations: given mA → EU and EU → mA for three different ranges.	What does a 14 mA reading mean for a 0–200 PSIG transmitter?
Tuesday	Study Section 2.3 (HART protocol). HART communicator demonstration (video or senior technician in the field). Study Section 2.4 (calibration procedures).	Shadow a senior technician performing calibration on at least one field instrument. Fill in a practice calibration record template. Review the 5-point calibration expected values.	What is the expected mA output at 75% of a 0–500 PSIG transmitter? What does a reading of 3.5 mA indicate?
Wednesday	Study Section 2.5 (Loop Check Procedure) — all 11 steps. Study Section 2.6 (IS basics — Zener barrier vs. galvanic isolator). Obtain a loop drawing for a real plant instrument.	Perform Lab 1 (Loop Check) — supervised. Focus on HART communicator operation and 5-point injection. Document results on loop check form. Supervisor reviews.	List the 11 steps of a loop check in order. What is the purpose of requesting alarm suppression before starting?
Thursday	Study Section 2.7 (Instrument Faults table). For each fault in the table, state the symptom and corrective action aloud, closed-book. Study Section 1.4 (ESD, PAHH, PALL descriptions).	Review plant P&IDs for the inlet separation area. Identify five instrumented loops on the P&ID. For each loop, write down: tag number, instrument type, range, and the control action it enables. Compare with Section 1.3 table.	What is the difference between PAH, PAHH, and PSV? What does each one do when it activates?

Day	Morning (4 hrs)	Afternoon (4 hrs)	Review Question (closed-book)
Friday	Review cards: Cards 1–13. Each card is studied until it can be answered closed-book. Repeat any missed cards. Study Section 11 (Quick Reference) — focus on 4-20mA quick math section.	Week 1 review (closed-book): (1) Draw and label the 6 stages of the gas plant process. (2) Calculate EU values for 5 different mA inputs on a 0–300 PSIG transmitter. (3) List the 11 loop check steps. (4) Explain IS to a non-technical person. Review any weak areas with supervisor before end of week.	Explain the difference between a Zener barrier and a galvanic isolator and when each is used.

Weeks 2–8 — Daily Breakdown Summary

Week	Monday	Tuesday	Wednesday	Thursday	Friday
2	Studio 5000 install & navigation; Section 3.1 hardware	Section 3.2 project tree; create first project	Section 3.3 tag naming; Section 3.4 networking; create tags	EtherNet/IP; DLR; network address table exercise	Review cards 14–21; Week 2 review: identify all chassis modules by sight
3	Section 5.1 instructions table; Snippet 1 (motor start/stop)	Snippet 2 (TON timer); Snippet 3 (CTU counter); start Lab 3	Complete Lab 3; test all rungs with bench inputs	Snippet 5 (alarm logic); Snippet 6 (BPCS trip latch); tag naming practice	Review cards 26–35; review: write a TON + motor start rung from memory
4	Section 4.1 FLEX I/O hardware; adapter types; terminal bases	Section 4.2 module table; Section 4.3 I/O mapping; fill in blank I/O table	Lab 2 (hot-swap) supervised; Section 4.4 configuration steps	Lab 5 (configure FLEX bank); verify online connection	Review cards 22–25; review: identify FLEX module by LED status
5	Section 5.2 Snippet 4 (PID); PID instruction configuration in Studio 5000	Lab 4 (PID tuning); Ziegler-Nichols method practice	Section 5.2 Snippet 8 (MSG); Lab 7 (MSG inter-PLC read)	Section 5.3 state machine; write state machine for simple 3-state compressor sequencer	Review cards 32–37; review: explain Ziegler-Nichols method verbally
6	Section 6.1 InTouch overview; WindowMaker navigation; create first window	Section 6.2 HMI tag list; bind 5 tags to OPC server; verify live data	Lab 6 (HMI faceplate — build and test)	Section 6.3 alarm management; ISA-18.2; Lab 10 (alarm rationalization — first 5 alarms)	Review cards 38–42; complete Lab 10 remaining 5 alarms; review: explain 4 alarm priorities
7	Section 9.1 HAZOP and LOPA; review C&E matrix template	Section 9.2–9.3 SIL, voting logic; Section 9.4 H_2S hazard; LEL/UEL; work permits	Lab 8 (BPCS trip string and bypass); document in C&E matrix	Section 8.1–8.2 troubleshooting methodology; Tables A–D; Lab 9 (network troubleshooting)	Review cards 43–50; review: explain the BPCS trip and SIS boundary to the supervisor
8	Full review: Quick Reference (Section 11) from memory. Repeat weak review cards.	Practice all troubleshooting scenarios (Tables A–D) verbally with supervisor as proctor	Repeat Lab 1 independently — no notes; document results correctly	Written competency assessment (all sections); review results	Practical assessment with supervisor; sign-off on the technician qualification checklist

NOTE — Supplementary Resources

Rockwell Automation ControlLogix training; AVEVA InTouch HMI tutorials; ISA CCST (Certified Control Systems Technician) study materials; and the plant approved document list (ADL) for plant-specific P&IDs, cause and effect matrices, and loop drawings. Regular field observation of instrumentation during normal operations (at least 2 hours per week) is recommended.

F.4 Competency Matrix (Supervisor Sign-Off)

Skill	W1	W2	W3	W4	W5	W6	W7	W8	Sign-off
Plant process walk / stage ID	O							I	
4–20 mA math + live-zero fault	S							I	
Loop check + site form (Lab 1)	S							I	
Studio 5000 navigation		O/S	S					I	

Skill	W1	W2	W3	W4	W5	W6	W7	W8	Sign-off
Seal-in + timer rungs (Lab 3)			S					I	
FLEX config + hot-swap (Labs 2, 5)				S					
PID on simulator only (Lab 4)					S			—	
MSG read-only (Lab 7)					S				
HMI faceplate on training node (Lab 6)						S			
Alarm rationalization on paper (Lab 10)						S			
BPCS trip string on trainer (Lab 8)							S	—	
Network restore on trainer (Lab 9)							S		
Forces/inhibits cleared (Lab 11)							S	I	
SOE first-out narrative (Lab 12)							S	I	
SIS proof test							O	O	
Production download / SIS bypass / fiscal config	—	—	—	—	—	—	—	Never	

Key: O = Observe, S = Supervised practical, I = Independent, — = not required this week, Never = outside the technician authorization envelope (Section 0.2).

Gate rule: the next week does not start until the supervisor has initialed that week's Independent or Supervised column for the required skills. A blank cell means the skill is not expected yet.

F.5 Assessment Notes

The Training Coordinator or designated assessor keeps a marked assessor copy of Appendices F.1–F.2. Oral checks are closed-book. The Week 8 independent loop check is not signed if Lab 1 was performed only on a simulator.

Week 8 assessment blueprint

Part	Method	Pass bar
Written	Closed-book, 25 questions from review cards + Section 0	≥ 80% and zero missed safety-boundary questions
Practical A	Unsupervised loop check on assigned training or approved spare loop	Form complete; 5-point within tolerance; forces clear
Practical B	Planted network or force fault on trainer (Lab 9 or 11 style)	Correct isolation in ≤ 20 minutes; no production browse
Oral	Explain one plant SIF without opening the SIS cabinet or changing anything	Correct layer (SIS vs BPCS vs PSV) and a correct statement that the technician does not operate it

Safety-boundary questions (a miss fails the written assessment)

1. Which document governs if this manual disagrees with a P&ID?
2. May a newly assigned technician bypass a SIF?
3. What action is required for an unexplained force?
4. Is Snippet 6 SIS logic?
5. When is a production download allowed in the first eight weeks?

Appendix G: Additional Instructions (Not Used in Interlocks)

The instructions below appear in existing plant programs and vendor examples. They are not used in interlock or trip logic in this manual.

Instruction	Name	Description	Reason not used in interlocks
CMP	Compare	Evaluates an expression (for example A > B AND C < D); TRUE when the expression is true	Expression hides individual conditions from cross reference and troubleshooting
OSR	One Shot Rising	Output-format one shot: OSR(Storage_Bit,Output_Bit) sets the output bit for one scan on a FALSE→TRUE transition	ONS on an input-side condition is used instead (Pattern A1)
OSF	One Shot Falling	Sets the output bit for one scan on a TRUE→FALSE transition	Falling-edge actions are easily missed after prescan
FAL	File Arithmetic and Logic	Applies an expression across array elements (All, Incremental, or Numerical mode) using a CONTROL tag	Multi-scan modes spread results over several scans
FSC	File Search and Compare	Searches array elements for a condition; sets .FD and .POS when found	Multi-scan behavior and CONTROL handling
JMP / LBL	Jump / Label	Skips rungs between the JMP and the matching LBL within one routine	Outputs in skipped rungs hold their last state

NOTE

Operand formats follow the Logix 5000 General Instructions Reference Manual. Verify on bench before field use.

Appendix H: References

Rockwell Automation Literature

Document	Publication	Location
Logix 5000 Controllers General Instructions Reference Manual (v36 mnemonic changes; ONS, timers, counters, PID)	1756-RM003 (served as 1756-RM018A-EN-P, Sep 2025)	literature.rockwellautomation.com/idc/groups/literature/documents/rm/1756-rm003_-en-p.pdf
Logix 5000 Advanced Process Control and Drives Instructions (PIDE, SCL, ALMA, FGEN)	1756-RM006	literature.rockwellautomation.com/idc/groups/literature/documents/rm/1756-rm006_-en-p.pdf
Logix 5000 Controller Messages Programming Manual (MSG configuration, error codes, timeouts)	1756-PM012	literature.rockwellautomation.com/idc/groups/literature/documents/pm/1756-pm012_-en-p.pdf
ControlLogix Analog I/O Modules User Manual (1756-IF8 data formats)	1756-UM009	literature.rockwellautomation.com/idc/groups/literature/documents/um/1756-um009_-en-p.pdf
FLEX I/O Analog Modules User Manual (1794-IE8 / OE4 counts)	1794-UM002 (1794-6.5.2)	literature.rockwellautomation.com/idc/groups/literature/documents/um/1794-um002_-en-p.pdf
Product Compatibility and Download Center (firmware versions, 1769-L33ER v36)	PCDC	compatibility.rockwellautomation.com

Regulatory and Health Sources

Topic	Source
H_2S standards (OSHA general industry ceiling 20 ppm, peak 50 ppm; construction 10 ppm)	OSHA Hydrogen Sulfide — Standards, osha.gov/hydrogen-sulfide/standards
H_2S IDLH 100 ppm; REL 10 ppm ceiling	NIOSH Pocket Guide / IDLH documentation, cdc.gov/niosh
H_2S TLV 1 ppm, STEL 5 ppm	ACGIH TLVs and BEIs (current edition)
Lockout/Tagout; Process Safety Management (MOC)	29 CFR 1910.147; 29 CFR 1910.119

Industry Standards

Standard	Subject
ANSI/ISA-18.2 (IEC 62682)	Management of alarm systems for the process industries
ANSI/ISA-101.01	Human-machine interfaces for process automation systems
IEC 61511 (ANSI/ISA-61511)	Functional safety — SIS for the process industry sector (lifecycle)
IEC 61508	Functional safety of E/E/PE safety-related systems (device certification)
ANSI/ISA-5.1	Instrumentation symbols and identification
NAMUR NE43	Signal level for failure information of digital transmitters (4–20 mA)
HART Communication Protocol (FieldComm Group)	HART 5/6/7 physical layer and polling addresses
IEC 61131-3	PLC programming languages (LD, ST, FBD)
AGA Report No. 3 / ISO 5167	Orifice metering of natural gas

Secondary Sources

Vendor application notes and the internal ladder-logic peer review (Kirk) were used to cross-check examples. Plant documents (P&IDs, loop drawings, C&E matrix, I/O schedules) govern over all sources listed here.

[COMPANY: Insert company name, document control number, and classification footer here]